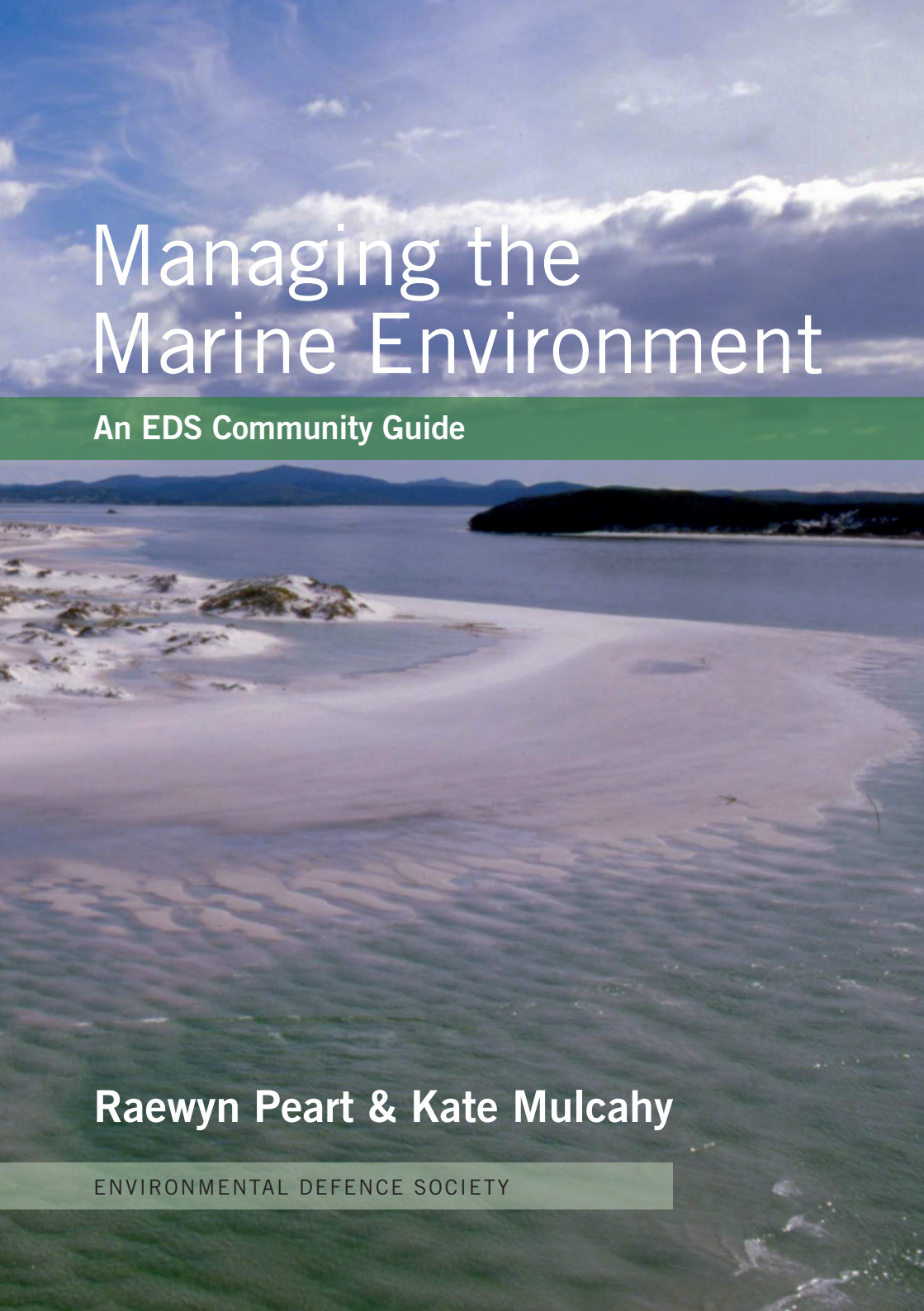




Managing the Marine Environment

An EDS Community Guide

Raewyn Peart & Kate Mulcahy

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Acronyms

ACE	Annual catch entitlement
AMA	Aquaculture management area
B_{MSY}	The size of a fish stock that can produce the maximum sustainable yield
DoC	Department of Conservation
EEZ	Exclusive economic zone
HGMPA	Hauraki Gulf Marine Park Act 2000
ITQ	Individual transferable quota
MFE	Ministry for the Environment
MPA	Marine protected areas
MSY	Maximum sustainable yield
NZHPT	New Zealand Historic Places Trust
NZCPS	New Zealand coastal policy statement
QMA	Quota management area
QMS	Quota management system
RMA	Resource Management Act 1991
SLED	Sea lion exclusion device
SMS	Seafloor massive sulphides
TAC	Total allowable catch
TACC	Total allowable commercial catch
UNCLOS	United Nations Convention on the Law of the Sea

Glossary of Māori Terms

Hapū	Sub-tribe
Hui	Meeting or gathering
Iwi	Tribe
Kaimoana	Seafood
Kaitiaki	Guardian
Kaitiakitanga	The exercise of guardianship by the tangata whenua of an area in accordance with tikanga Māori in relation to natural and physical resources; and includes the ethic of stewardship (RMA) The exercise of guardianship; and, in relation to any fisheries resources, includes the ethic of stewardship based on the nature of the resources, as exercised by the appropriate tangata whenua in accordance with tikanga Māori (Fisheries Act 1996)
Mana	Prestige
Mana whenua	Customary authority exercised by an iwi or hapū in an identified area (RMA and Fisheries Act 1996)
Mātaitai	Food resources from the sea (RMA)
Rāhui	A custom that involves placing restrictions of access and use over an area for cultural reasons
Rangatiratanga	Sovereignty
Rohe Moana	Territorial waters
Taiā pure rāhui	Sea reserve
Tangata Kaitiaki/Tiaki	Person who can authorise customary fishing in their territory
Tangata whenua	In relation to a particular area, means the hapū, or iwi, that is Māori and holds mana whenua over that area (Fisheries Act 1996) In relation to a particular area, means the iwi, or hapū, that holds mana whenua over that area (RMA)
Tikanga Māori	Maori customary values and practices (RMA)
Wāhi tapu	Sacred place(s)
Whakapapa	Geneology

Preface

This Guide has been written for people who want to know more about how to promote effective management of New Zealand's marine environment. It provides information for those who would like to obtain better environmental outcomes for the marine area. It is also designed to assist people engaged in activities impacting on the marine area and in designing marine-related projects. The Guide focuses on what 'good practice' marine management might look like rather than what the minimum environmental requirements are under the relevant legislation.

Content of guide

This Guide has been prepared in two parts. **Part One** provides an overview of New Zealand's marine area and how it is currently managed. It also identifies opportunities for public involvement in seeking better environmental outcomes. It addresses:

- The characteristics of New Zealand's marine environment and the species which live within it
- How the marine environment is currently managed
- Kaitiakitanga
- Opportunities for public involvement in marine decision making
- Statutory mechanisms which can protect areas of the marine environment

Part Two of the Guide provides more detailed information on a range of activities which can potentially impact on the marine environment and the elements of best practice that can be applied to help minimise negative environmental effects. The activities include:

- Fishing
- Aquaculture
- Coastal sand mining
- Deep sea exploration and mining
- Major marine developments
- Vessels and offshore installations
- Catchment-based activities

Supporting resources are provided in the **Appendices** to the Guide, including a summary of useful cases relating to seabed mining and aquaculture, the

provisions of the New Zealand Coastal Policy Statement (NZCPS), and the additional management objectives which apply specifically to the Hauraki Gulf. The Guide will be regularly updated to reflect changes in marine law.

Companion guides

There are three companion volumes to this Guide:

- *The New Zealanders' Guide to the Resource Management Act 1991* which provides more detailed information about participating in processes under the RMA
- *The Community Guide to Coastal Development under the Resource Management Act 1991* which provides more detailed information on managing coastal development under the RMA
- *The Community Guide to Landscape Protection under the Resource Management Act 1991* which provides more detailed information on addressing landscape management issues under the RMA

Web-based versions of all guides can be found at www.rmaguide.org.nz. Hard copies can be obtained from the Environment Defence Society Incorporated, PO Box 95 152 Swanson, Waitakere. See www.eds.org.nz for more details.

Disclaimer

The material in this guide is provided in good faith, but with no guarantee as to it being comprehensive or accurate. The Guide is not a substitute for proper professional advice where that is needed.

Environmental Defence Society

The Environmental Defence Society is a non-profit environmental advocacy organisation consisting largely of resource management professionals. Coastal and marine issues are a major focus of the Society's work. If you would like to find out more about the Society, or support its environmental programmes, please visit the society's website at www.eds.org.nz.

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The final contents of the Guide remain the responsibility of the authors.

PART ONE

OVERVIEW OF MARINE MANAGEMENT



one

New Zealand's Marine Environment

Introduction

New Zealand is fortunate to have a very extensive and diverse marine area. It is home to many different plant, invertebrate, fish, marine mammal and seabird species. The identity of many of these species, and the interactions between them, are still not fully understood. This makes the effective management of the marine environment a considerable challenge.

Physical Marine Area

Size

For a relatively small country, New Zealand has a long coastline and jurisdiction over a very large area of sea. Some 19,000 kilometres of coastline borders a marine area extending over 4.2 million square kilometres (see Figure 1.1). This is fifteen times larger than New Zealand's total land area.

But the marine environment is considerably larger than this two-dimensional figure indicates. It also has a substantial third dimension, extending down from the surface of the sea, to the seabed. Although the sea is relatively shallow in coastal areas, where the land abuts the continental shelf, seventy-five per cent of New Zealand's marine area has depths of more than one kilometre. The deepest place is within the Kermadec Trench, some ten kilometres from the surface.

Topography

New Zealand's seabed has a very diverse topography. This is because the country sits astride two colliding tectonic plates. The movement of the plates has resulted in arcs of volcanic activity and the uplift and depression of large areas of coastal land and seabed.

Many coastal features have been formed from the remnants of volcanic activity including cones, sunken craters and lahars. New Zealand's seabed features chains of underwater volcanoes and deep ocean trenches and ridges. The mountains which are entirely under the sea are called 'seamounts'.

Closer to land, the New Zealand coastline is punctuated by many estuaries and harbours. These shallow, sheltered areas were formed after the last ice age, when the rising seas flooded river valleys and other low-lying areas. They have continued to infill with sediment washed off the land.

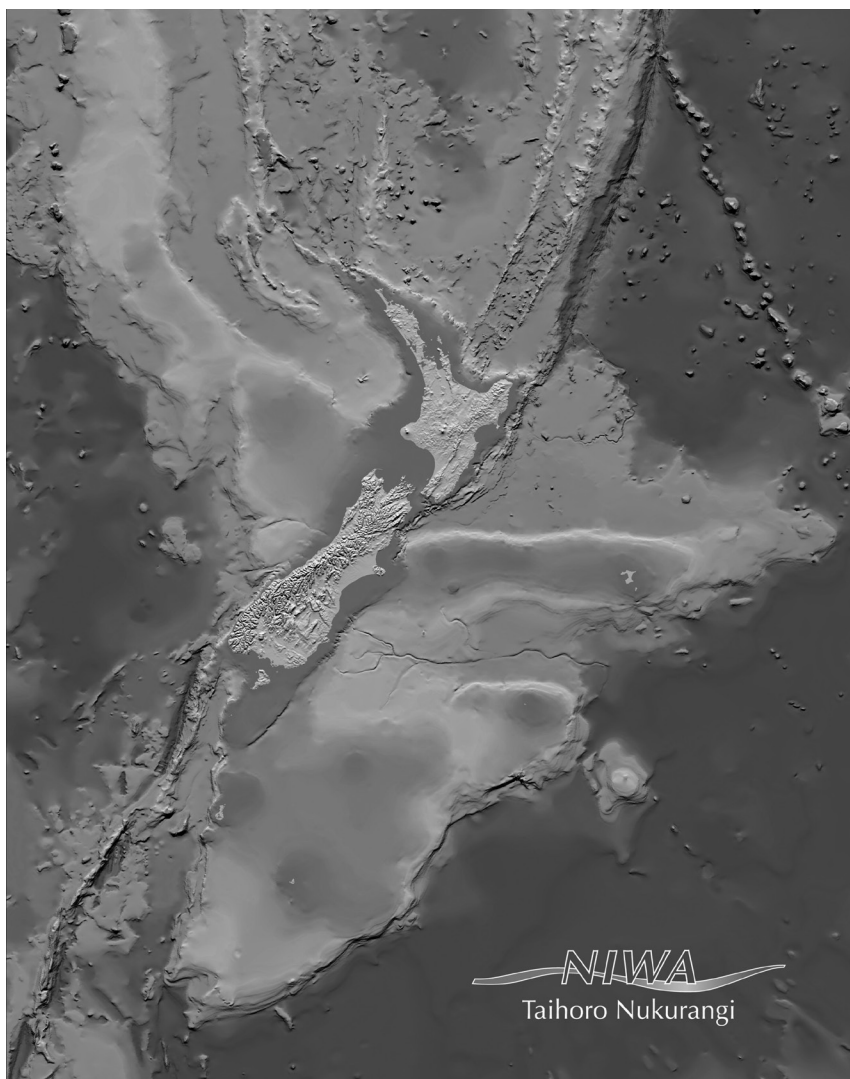


Figure 1.1: New Zealand's marine area (NIWA)

Rocky reefs have formed around the more exposed parts of the coast, with soft rock being eroded away by waves to form bays, and harder rock remaining as headlands and reefs. Sand spits and barrier islands are also common around

the coast, forming across the mouths of harbours and bays, where rivers have dropped their sediment load.

Movement of water

In terms of the sea itself, New Zealand's marine area is influenced by two major bodies of oceanic water: subtropical surface water which travels from the central Pacific Ocean via the eastern coast of Australia; and subantarctic surface water which originates from the Southern Ocean.

Both of these water bodies move in an easterly direction and they meet along a subtropical front which bisects the southern part of New Zealand's marine area. This oceanic meeting point is located along the same latitude as Fiordland on the west coast and along the Chatham Rise to the east (see Figure 1.2). It is a very productive area for marine life, particularly over the Chatham Rise. More localised currents move the subtropical surface water around the New Zealand coast.

Wind blowing over the surface of the sea can create surface currents which move seawater towards or away from the coast depending on the direction of the wind. When surface currents move away from the coast, deeper oceanic water is drawn up closer to the surface. This water usually has more nutrients than surface water, and such upwellings can help support highly productive fisheries. Upwellings commonly occur along the northeast coast of the North Island and along the entire west coast of the South Island.¹

Marine Habitats

New Zealand has a high diversity of marine habitats, some of which are particularly important to the ecological health and productivity of the marine area. As much as 80 per cent of New Zealand's total biodiversity lives in the sea. It is estimated that there might be up to 65,000 different marine species, of which only twenty-five per cent have been described. The greatest biodiversity of fish and invertebrates is found along the northeast coast of the North Island.

Estuaries

Estuaries are located in the brackish water at river mouths and are one of the most ecologically productive marine habitats. New Zealand has around 300 estuarine systems situated along the coastline, and they cover over 100,000 hectares.

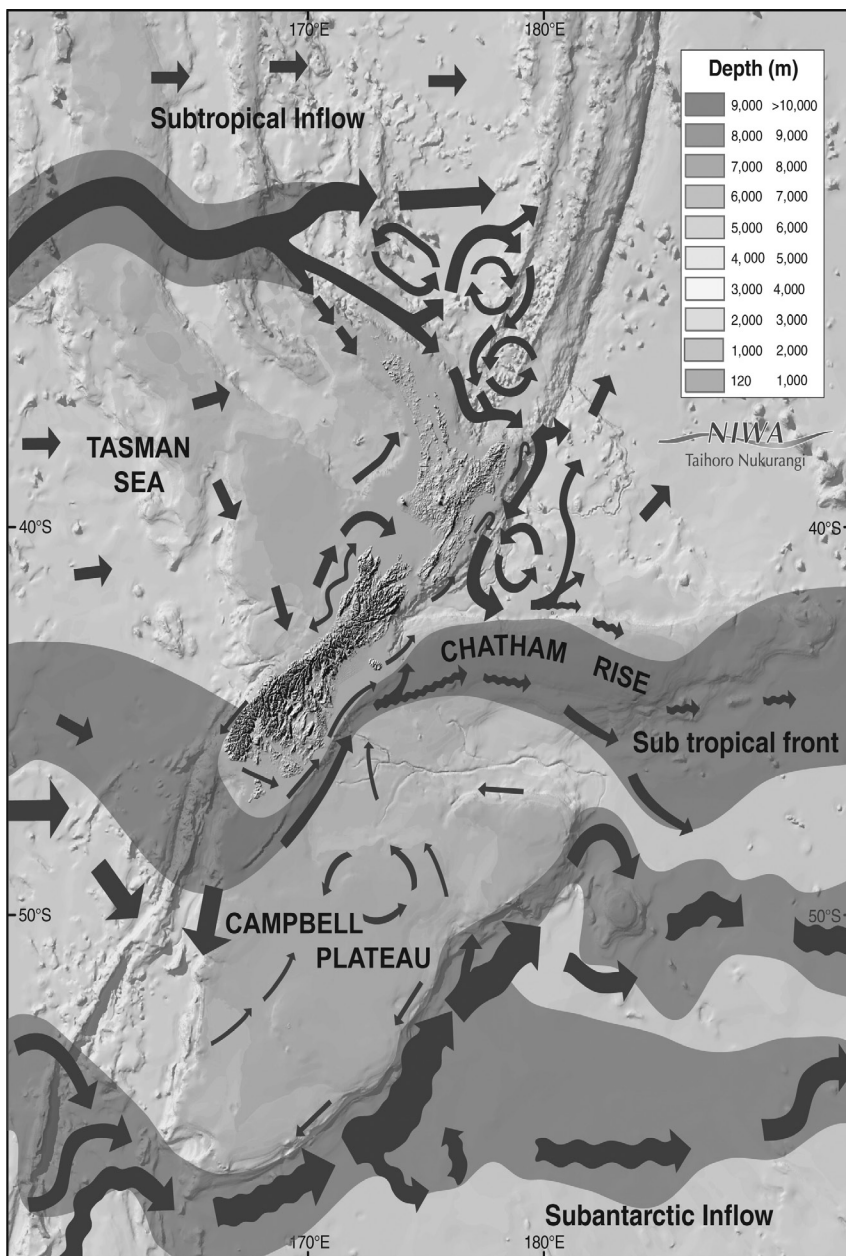


Figure 1.2: Ocean currents around New Zealand (NIWA)

Estuaries provide habitat for mangroves, seagrasses, reeds, worms, algae and phytoplankton. They provide spawning and nursery areas for a wide range of fish and shell-fish species. They also provide critical habitat and food for migratory wading birds, some which travel to New Zealand from as far afield as Alaska each year.

Rocky reefs

Rocky reefs are common in nearshore areas around New Zealand's coast. Many reefs are covered in rich forests of seaweed, which can contain several large kelp species and smaller red and green seaweeds. They provide important habitats for a diverse range of sea creatures.

Rocky reefs are home to over 2,000 species of invertebrates, which are permanently attached to the rocks. Many of these species are only found in New Zealand. These 'sessile' animals harvest plankton and organic particles from passing seawater. They include sponges, corals, anemones, tube worms and barnacles.

Bryozoans are fascinating creatures, which have been variously referred to as lace corals, moss animals or sea mats. These animals form coral-like groupings on rocky and gravel areas, which in turn provide shelter for other invertebrates and juvenile fish. Nearly 1,000 different species of bryozoans are found within New Zealand's marine area and most of these are not found anywhere else in the world.

Another group of invertebrates move around the reef to feed. These include herbivores such as kina and top shells, which graze on seaweed, and carnivores such as octopus, starfish, crayfish and crabs. The reef is also home to brightly coloured nudibranchs, shellfish-type animals which have evolved to live without shells.

Rocky reefs support some 250 species of reef fish. Some of these fish, including parore, silver drummer and butterfish, graze on seaweed. Others including maomao, trevally and demoiselle sift plankton from the seawater. Other fish targeted by fishers, including snapper, blue cod, kahawai, John Dory and moki, predate on smaller fish and invertebrates living in the rocky reef environment.

Some unusual species also make New Zealand's rocky reefs their home including sea horses and sea dragons. The sea horse species found in New Zealand is one of the largest in the world. Sea horses anchor themselves to the reef with their prehensile tails and ambush their food, mainly small crustaceans. Male sea horses incubate fertilised eggs in a specialised pouch.

Soft sediment habitats

Much of the seabed with New Zealand's marine area consists of soft sediments, which are particularly common in harbours, estuaries and along the open coast. The sediment is often rich in organic matter which supports algal growth and provides food for bottom-dwelling and burrowing creatures which live in the soft sand or mud. Common residents of soft sediments include a variety of worms, crabs, shellfish, starfish, sea urchins and sea cucumbers. In coastal areas these are eaten by bottom-feeding fish including flounder, tarakihi, snapper, red cod and warehou. Some fish, such as the stargazer, completely bury themselves in the sand and pounce on unsuspecting passers-by.²

Calcareous gravels

One of the most biodiverse marine habitats in New Zealand is rare, and occurs where gravel lying on the seabed is intermixed with the hard calcium remains of shellfish and bryozoans. These areas are called 'calcareous gravels' and are home to rich thickets of bryozoans and sponges. A very ecologically important area of these gravels occurs around the northern tip of the North Island near Spirits Bay, where more than 330 species of bryozoans and 220 species of sponges have been found. They also occur near the southern coast of the South Island, off the Otago coast, in Foveaux Strait and in Tasman Bay.

Fjords

New Zealand's fjords provide an unusual marine habitat, because copious rainwater leaches through the rich humus of the beech forest and results in a yellow-coloured layer of freshwater overlaying the sea. This reduces the light levels within the fjords and has resulted in normally deep water species living closer to the surface. Spectacular large tree-like black corals, normally only present in deep waters, can be found growing in Fiordland.

Seamounts

Seamounts, which are located within deep parts of New Zealand's marine area, can support a very diverse range of life. They provide a hard stable surface for sessile plankton feeders to attach to, and act as oases within large plains of seabed covered in soft mud. They provide a food-rich environment by causing localised turbulence and upwellings in seawater and through physically trapping plankton as it moves with the ocean currents.

Seamounts can support very large deep-water sponges and corals that can live for hundreds of years. Sponges can reach several metres in height and corals have been found growing up to 10 metres tall. These communities provide a

refuge for deep-water fish species including orange roughy, black oreo and black cardinalfish.

Hydrothermal vents

Scientists have recently discovered new forms of life deep under the sea within hydrothermal vents. Here, creatures are no longer dependent on the sun as a source of energy for the food chain, but are supported by sulphur compounds rising up from the Earth's crust within superheated seawater. Despite the often very high temperatures, hydrothermal vents are home to enormous worms, mussels and blind shrimps.

Migratory fish species

As well as having a wide range of resident marine life, New Zealand waters are visited by many migratory fish species. Species such as the southern bluefin tuna and striped marlin migrate seasonally across vast ocean basins.

Marine mammals

New Zealand's seas are a marine mammal 'hotspot'. At least 38 species of dolphin and whale are found here, which is just under half of the world's total number of species.

Nine different species of dolphin can be sighted in New Zealand's coastal waters including the common dolphin, dusky dolphin, bottlenose dolphin and orca.

Although there are numerous populations of orca around the world, the New Zealand population numbers only around 200 individuals and is classified as nationally critical.

New Zealand is also home to one of the smallest dolphins in the world, the Hector's dolphin. This species is threatened with just over 7,000 remaining. A sub-species, the Maui's dolphin, is only found along the west coast of the North Island. There are as few as 111 of this critically endangered species remaining.

Whale populations are slowly recovering from being hunted to near extinction during the late nineteenth and early twentieth century. Species such as the humpback and blue whale migrate through New Zealand waters. Other species such as the Bryde's whale and southern right whale breed here. A population of Bryde's whales resides in the Hauraki Gulf and breeds there in summer. The species is classified as nationally critical in New Zealand, although there are secure populations overseas. A group of young sperm whales live for

most of the year in the vicinity of the Kaikoura Canyon and is the focus of a whale watching tourism industry in the area.

Seals are the other main group of marine mammals found in New Zealand's waters. It is estimated that up to two million fur seals lived around the New Zealand coastline before humans arrived. They were reduced to very small numbers as a result of harvesting, but the population is now recovering. The New Zealand sea lion can be seen around our southern coasts, and southern elephant seals and leopard seals are occasional visitors to New Zealand beaches, travelling up from Subantarctic and Antarctic waters.

Seabirds

New Zealand has been called the 'seabird capital of the world'. This is because 84 species of seabirds breed here, more than in any other country. This is also greater than the number of New Zealand's landbird species. Thirty-five species of seabird are endemic to New Zealand and breed nowhere else.

New Zealand's seabirds range from the large southern royal albatross, which can have a wingspan of over three metres and which spends most of its time at sea, to the small and critically endangered New Zealand fairy tern, which nests on Northland beaches.

Seven species of penguin can be found in New Zealand, with four of these (yellow-eyed, white-flippered, Fiordland-crested and blue) breeding on the mainland. The blue penguin is the smallest penguin species in the world and can be found along the coasts of the North and South Islands. The largest penguin in New Zealand is the distinctive yellow-eyed penguin which nests along the south-east coast of the South Island and on sub-Antarctic islands further south.

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1 Andrew N and M Francis (eds), 2003, 30

2 Westerskov K and K Probert, 1981, 74-75



two

Management of the Marine Environment

Introduction

New Zealand’s marine environment is managed by numerous statutory agencies under a myriad of different pieces of legislation. This chapter provides an overview of the management regimes that apply to the marine environment.

Maritime areas

Management of the sea is governed internationally by multilateral treaties, the prime one being the United Nations Convention on the Law of the Sea (UNCLOS). Under UNCLOS, New Zealand exercises different jurisdictional powers over four defined maritime areas – the territorial sea, the contiguous zone, the exclusive economic zone (EEZ) and the continental shelf (see Figure 2.1).

The *territorial* sea is part of New Zealand’s sovereign territory. This means that the government’s power to govern this marine area is the same as on land. The territorial sea comprises the marine area within 12 nautical miles of low-water mark. It encompasses 167,650 square kilometers of sea.¹

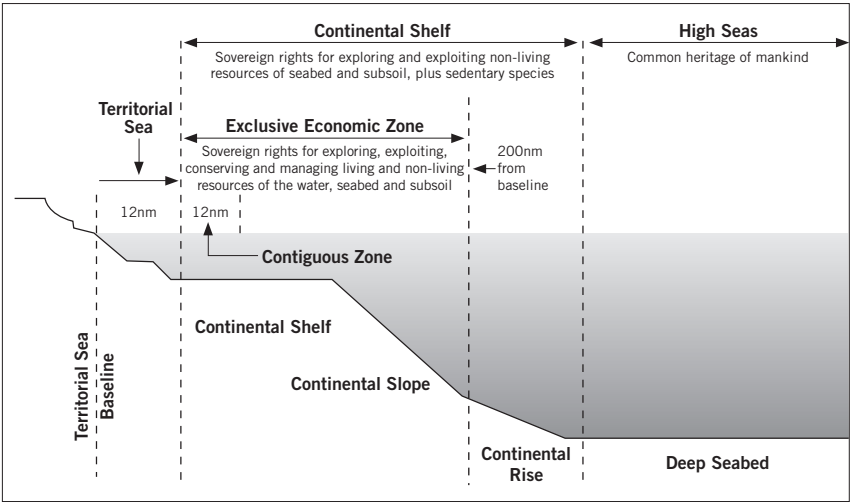


Figure 2.1: Maritime jurisdictions under UNCLOS

The *contiguous zone* lies on the seaward side of the territorial sea, extending from 12 to 24 nautical miles from the low-water mark. This is not part of New Zealand's territory, but under UNCLOS, New Zealand has the right to police the area to prevent or punish infringements (of customs, fiscal, immigration or sanitary laws and regulations) in its territory or territorial sea.

The *Exclusive Economic Zone* is the largest of New Zealand's marine areas, covering an area which extends from the outer edge of the territorial sea to 200 nautical miles from the low-water mark. It overlaps the contiguous zone. In some places, the EEZ extends over the edge of the continental shelf into deep ocean areas. New Zealand's EEZ covers 3.93 million square kilometers of ocean and is the fourth largest EEZ in the world.²

The EEZ is not strictly speaking part of New Zealand's territory, but under UNCLOS New Zealand has sovereign rights within the EEZ for the purpose of exploring, exploiting, conserving and managing all of its living and non-living natural resources. These resources include fish, other marine life, oil, gas, minerals, and energy produced from water, currents and winds. Other countries retain the freedom of navigation and over-flight within New Zealand's EEZ and can lay submarine cables and pipelines within it.

Where the outer edge of the *continental shelf* (which includes the continental slope and rise) extends beyond the EEZ, New Zealand has more limited rights which only cover the exploitation of minerals, gas and other non-living resources within the seabed and subsoil as well as sedentary species on the seabed, but excluding mobile fish. When taking non-living resources such as minerals and petroleum from this part of the continental shelf, payments must be made to the International Seabed Authority. This authority was established under the Law of Sea to control activities within areas of the seabed outside national jurisdiction.³ The United Nations Commission on the Limits of the Continental Shelf confirmed the boundary of New Zealand's continental shelf in 2008 (see Figure 2.2).

Because some of the boundaries of New Zealand's EEZ and continental shelf overlap with areas claimed by other countries, the boundaries need to be agreed by treaty. New Zealand has signed a treaty with Australia to settle joint maritime boundaries but has yet to conclude maritime treaties with Fiji, Tonga and potentially France in respect of New Caledonia.

The area outside the territorial sea, EEZ and continental shelf areas claimed by New Zealand and other countries is effectively the high seas and is outside the national jurisdiction of any one country. This area is managed under international agreements including UNCLOS.

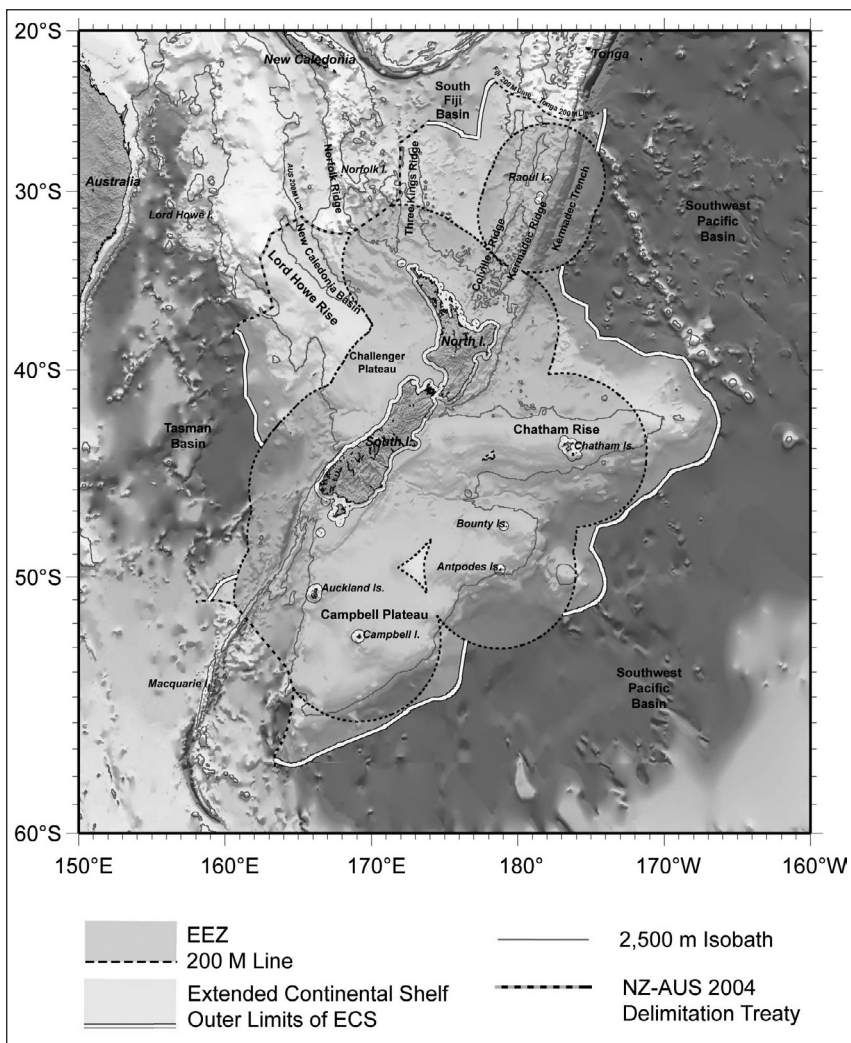


Figure 2.2: New Zealand's maritime areas

Sourced from the LINZ New Zealand Continental Shelf Project. Crown Copyright Reserved.

Marine management agencies

Marine management in New Zealand is largely split among nine types of statutory management agencies with overlapping responsibilities (see Figure 2.3).

Figure 2.3: Key New Zealand coastal and marine management agencies

Management Agency	Areas of management focus	Key legislation
Minister and Department of Conservation	Conservation of natural and historic resources; protected species; marine protection; coastal management	Conservation Act 1987 Foreshore and Seabed Act 2004 Marine Mammals Protection Act 1978 Marine Reserves Act 1971 Resource Management Act 1991 Wildlife Act 1952
Minister and Ministry of Fisheries	Fisheries resources	Fisheries Act 1996 Fisheries (Quota Operations Validation) Act 1997 Māori Commercial Aquaculture Claims Settlement Act 2004 Māori Fisheries Act 2004 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992
Minister of Energy and Ministry of Economic Development	Allocation of minerals and petroleum resources	Continental Shelf Act 1964 Crown Minerals Act 1991
Biosecurity New Zealand	Invasive non-indigenous species	Biosecurity Act 1993
Maritime New Zealand	Shipping and marine pollution	Maritime Transport Act 1994
New Zealand Historic Places Trust - Pouhere Taonga	Protection of historic heritage places	Historic Places Act 1993
Regional Councils	Catchment and marine management (excluding fisheries allocation)	Local Government Act 2002 Reserves Act 1977 Resource Management Act 1991
Territorial authorities	Land subdivision and use	Local Government Act 2002 Reserves Act 1977 Resource Management Act 1991
Iwi and hapū	All coastal and marine resources	All the above legislation Co-management agreements

Minister and Department of Conservation

The Department of Conservation (DoC) is a central government department established by statute – the Conservation Act 1987. Its prime role is to manage Crown-owned conservation land and to protect and rehabilitate endangered indigenous species and habitats.

The Department is also tasked with advocating the conservation of natural and historic resources generally, amongst other things. The definition of ‘natural resources’ in the Conservation Act is very wide and includes living organisms, habitats, ecosystems, landscapes, landform and geological features.

The Department is headquartered in Wellington and is structured around twelve conservancies that are managed by regional conservators and overseen by independent conservation boards. The work of the conservancies is guided by conservation management strategies prepared for each region. The New Zealand Conservation Authority provides oversight of DoC’s work and gives advice to the Minister and the Director-General of Conservation.

The Minister of Conservation and DoC play a significant role in the protection of marine habitats and species. The Minister approves the establishment of marine reserves (with the concurrence of the Ministers of Fisheries and Transport) under the Marine Reserves Act 1971, and marine mammal sanctuaries under the Marine Mammals Protection Act 1978. Once created, these areas are managed by DoC. The Department administers the Wildlife Act 1952 which provides protection for seabirds and some marine species. The Department is also involved, in conjunction with the Ministry of Fisheries, in managing the impacts of fishery activity on protected marine species such as dolphins, sea lions, seals and seabirds.

The Minister of Conservation has a specific role in the management of coastal land and the territorial sea under the Resource Management Act 1991 (RMA). The Minister develops the New Zealand Coastal Policy Statement (NZCPS) which guides, and in some cases directs, the planning and decision making functions of councils within the ‘coastal environment’. The extent of this area is not defined in the RMA, but it includes the coastal marine area which extends from mean high water springs⁴ to the outer edge of the territorial sea, and land near the coast that is influenced by the sea.

The Minister approves regional coastal plans to the extent that they apply to the coastal marine area. The Minister also nominates a person to the hearings panel to consider resource consent applications for restricted coastal activities. Restricted coastal activities include works such as large reclamations, structures and seawalls which are likely to have significant and irreversible adverse effects on the marine environment. The Department of Conservation becomes involved

in making submissions on plans and resource consents under the RMA as part of its support for the Minister's wider coastal management responsibilities.

Under the Foreshore and Seabed Act 2004 the Minister also has ownership responsibilities on behalf of the Crown for public foreshore and seabed. This legislation is currently under review.

Minister and Ministry of Fisheries

The Minister of Fisheries and associated Ministry manage fisheries resources under the Fisheries Act 1996, which includes fish, shellfish, seaweed and other aquatic life. The Minister determines a wide range of matters including what quantity of fish can be caught and what controls are placed on where, when and how fishing is to be undertaken. The Ministry supports the Minister's decision-making role by commissioning research, managing stakeholder inputs and providing advice. The Ministry also enforces fisheries regulations.

The Minister of Fisheries plays a role in the allocation of marine areas for aquaculture under the RMA, through determining whether a proposed Aquaculture Management Area (AMA) in a regional coastal plan will have an undue adverse effect on fishing. The Ministry also maintains a register of fish farms.

With respect to marine reserves under the Marine Reserves Act 1971, the Minister of Fisheries (along with the Minister of Transport) has a concurrence role in deciding whether or not proposed marine reserves should be established. In doing so, the Minister applies the decision-making criteria in section 5 of the Marine Reserves Act 1971. For fisheries, this includes determining whether declaring the area a marine reserve would interfere unduly with commercial fishing or recreational usage in the area. If the concurrence is not provided, the proposed marine reserve is not created.

The Ministry of Fisheries is headquartered in Wellington. It has regional offices in Auckland, Wellington, Christchurch and Dunedin as well as a number of Field Operations (Compliance) offices located around the country.

Minister of Energy and Ministry of Economic Development

The Minister of Energy manages the allocation of minerals resources within New Zealand's marine area. The Minister is supported by the Crown Minerals Division of the Ministry of Economic Development which processes prospecting, exploration and mining permits, collects royalties and enforces mining regulations.

Biosecurity New Zealand

Biosecurity New Zealand is a division of the Ministry of Agriculture and Forestry which manages biosecurity issues under the Biosecurity Act 1993. This includes preventing new organisms from entering the country and eradicating or controlling them once they have arrived. Marine biosecurity is a small part of Biosecurity New Zealand's activities, which are more terrestrially focused, but capacity is gradually being built up in this area.

Maritime New Zealand

Maritime New Zealand is governed by an independent board appointed by the Governor-General on the recommendation of the Minister of Transport. It manages shipping and marine pollution under the Maritime Transport Act 1994. This includes controlling discharges from ships and offshore installations. The agency also controls the dumping of waste within the EEZ and continental shelf region beyond.

Maritime New Zealand, develops on behalf of the Minister of Transport, and enforces marine protection rules aimed at preventing pollution incidents. It maintains the country's marine oil spill response strategy. It also supports New Zealand's engagement with the International Maritime Organisation.

New Zealand Historic Places Trust - Pouhere Taonga

The New Zealand Historic Places Trust (NZHPT) is an independent body established under the Historic Places Act 1993. The Act also establishes a Māori Heritage Council, which advocates for and advises on Māori heritage issues.

The Trust is charged with identifying, recording and protecting historic heritage areas throughout New Zealand including sites within the marine area. The Trust maintains a register of historic and wāhi tapu places and has a direct consenting role in relation to archaeological sites. The Trust also directly manages some historic places and buildings.

Regional councils

Regional councils are democratically elected bodies constituted under the Local Government Act 2002. There are 12 regional councils throughout the country whose boundaries broadly coincide with water catchment areas. In addition, there are four unitary authorities, which carry out the role of both regional authority and territorial authority. Restructuring of local government within the Auckland region will result in one unitary council, the Auckland Council, undertaking the current roles of the Auckland Regional Council and seven city and district councils.

Regional councils undertake a broad role in managing the marine environment through their jurisdiction under the RMA over both catchments and activities in the coastal marine area. The scope of the RMA is broad and it encompasses the environmental impacts of all activities on land and within the coastal marine area, except where they are specifically excluded, as is the case with the allocation of fisheries under the Fisheries Act 1996. Council management of the coastal environment under the RMA is overseen by the Minister of Conservation.

Regional councils prepare regional policy statements and plans under the RMA, which can have a major impact on the way the marine environment is managed. Regional policy statements provide an overview of regionally significant issues. Regional coastal plans apply to the coastal marine area and sometimes coastal land. Regional plans can address catchment issues such as run-off into the marine area.

Regional councils are also consent authorities for coastal permits which apply to activities within the marine area, and water and discharge permits within the catchments which address run-off from land amongst other things.

Regional councils play a role in biosecurity management under the Biosecurity Act 1993, but to date this has been primarily on land with less involvement in the marine area. Some regional councils also manage regional parks which border the sea under the Reserves Act 1977 and other legislation. Regional council activities and expenditure are guided by long term council community plans prepared under the Local Government Act 2002.

Territorial authorities

Territorial authorities, which include both city and district councils, are also democratically elected bodies constituted under the Local Government Act 2002. There are 69 territorial authorities operating throughout the country. These bodies have a more limited role in coastal management under the RMA, being primarily focused on land subdivision, use and development. This role is significant, however, as land development can have major impacts on the coastal environment. Territorial authorities prepare district plans under the RMA and make first-instance decisions on subdivision and land use consents.

Many territorial authorities manage coastal reserve land and esplanade reserves under the Reserves Act 1977. Similar to regional councils, the activities and expenditure of territorial authorities are guided by long term council community plans prepared under the Local Government Act 2002. These plans can have a significant impact on the marine environment as they determine the levels of expenditure on infrastructure such as stormwater and wastewater treatment facilities. The Local Government Act 2002 also empowers territorial

authorities to implement bylaws to regulate the management of stormwater and wastewater systems.

Iwi and hapū

Iwi and hapū are increasingly applying contemporary forms of kaitiakitanga in the coastal environment through involvement in management decision-making and potentially in governance. The meaning and implementation of kaitiakitanga is described more fully in chapter 3. As well as engaging in kaitiakitanga through statutory processes, iwi and hapū are becoming major marine resource managers in their own right through the fisheries settlement, the aquaculture settlement and individual iwi settlements.

Marine management legislation

New Zealand's marine environment is managed within a somewhat complex and overlapping legislative framework which has developed over the past fifty or so years. The key pieces of legislation are shown in Figure 2.4 and described in the following sections. Provisions, which enable the establishment of marine protected areas, are described in chapter five.

Resource Management

Most activities that have an environmental impact on the territorial sea, with the major exception of the allocation of fishing resources, are managed under the RMA. These include land-based activities which impact on the marine area such as farming, forestry and urban development; activities within the marine area such as boating and aquaculture; and major marine works such as reclamation, dredging and the construction of wharves, jetties, sea walls and outfalls.

Part 2 of the RMA provides a purpose and set of principles to govern all management under the Act. The purpose of the RMA is to promote 'sustainable management'. This is defined in the legislation to mean (section 5(2)):

... managing the use, development and protection of natural and physical resources in a way, or at a rate, which enables people and communities to provide for their social, economic, and cultural wellbeing and for their health and safety while –

- (a) Sustaining the potential of natural and physical resources (excluding minerals) to meet the reasonable foreseeable needs of future generations;

Figure 2.4: Marine management legislation

Management area	Legislation	Spatial area	Managing authorities
Resource management	Resource Management Act 1991	Catchments, islands and territorial sea	Minister of Conservation Regional councils Territorial authorities
	Hauraki Gulf Marine Park Act 2000	Hauraki Gulf territorial sea, islands and catchments	Those represented on the Hauraki Gulf Forum
Fisheries	Fisheries Act 1996 Fisheries (Quota Operations Validation) Act 1997 Māori Commercial Aquaculture Claims Settlement Act 2004 Māori Fisheries Act 2004 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992	Freshwater, territorial sea and exclusive economic zone	Minister of Fisheries Supported by Ministry of Fisheries
Mining	Crown Minerals Act 1991 Continental Shelf Act 1964	Land, territorial sea, exclusive economic zone and continental shelf	Minister of Energy Supported by Ministry of Economic Development
Marine pollution	Maritime Transport Act 1994	Territorial sea and exclusive economic zone	Maritime New Zealand
Biosecurity	Biosecurity Act 1993	Land and territorial sea	Biosecurity New Zealand
Marine protection	Marine Reserves Act 1971 Marine Mammals Protection Act 1978 Wildlife Act 1952	Territorial sea Territorial sea and exclusive economic zone Land, territorial sea and exclusive economic zone	Minister of Conservation Supported by Department of Conservation
Historic heritage	Historic Places Act 1993	Land and territorial sea	New Zealand Historic Places Trust - Pouhere Taonga
Coastal reserves	Reserves Act 1977	Land	Department of Conservation Regional councils Territorial authorities
Foreshore and seabed	Foreshore and Seabed Act 2004	Foreshore and territorial sea	Ministry of Justice responsible for administration of Act Minister of Conservation exercising ownership responsibilities Foreshore and seabed reserve boards

(b) Safeguarding the life-supporting capacity of air, water, soil, and ecosystems; and

(c) Avoiding, remedying or mitigating any adverse effects of activities on the environment.

Matters of national importance identified in section 6 of the RMA, which decision-makers are required to recognise and provide for, and which are relevant to marine management include:

- The preservation of the natural character of the coastal environment (including the coastal marine area), wetlands, and lakes and rivers and their margins, and the protection of them from inappropriate subdivision, use and development (section 6(a))
- The protection of outstanding natural features and landscapes from inappropriate subdivision, use and development (section 6(b))
- The protection of areas of significant indigenous vegetation and significant habitats of indigenous fauna (section 6(c))
- The maintenance and enhancement of public access to and along the coastal marine area, lakes, and rivers (section 6(d))
- The relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wāhi tapu, and other taonga (section 6(e))
- The protection of historic heritage from inappropriate subdivision, use, and development (section 6(f))
- The protection of recognised customary activities (section 6(g))

The Māori-related provisions in Part 2 of the RMA, which have been utilised extensively, are section 6(e) which relates to the relationship of Māori with the marine area as shown above; section 7(a) which requires decision-makers to have particular regard to kaitiakitanga; and section 8 which requires decision-makers to take into account the principles of the Treaty of Waitangi.

Management of the coastal environment at a regional and local level is carried out within the framework established by the NZCPS, which is prepared by the DoC and approved by the Minister of Conservation, and other policy and planning documents within a planning hierarchy established under the RMA (see Figure 2.5).

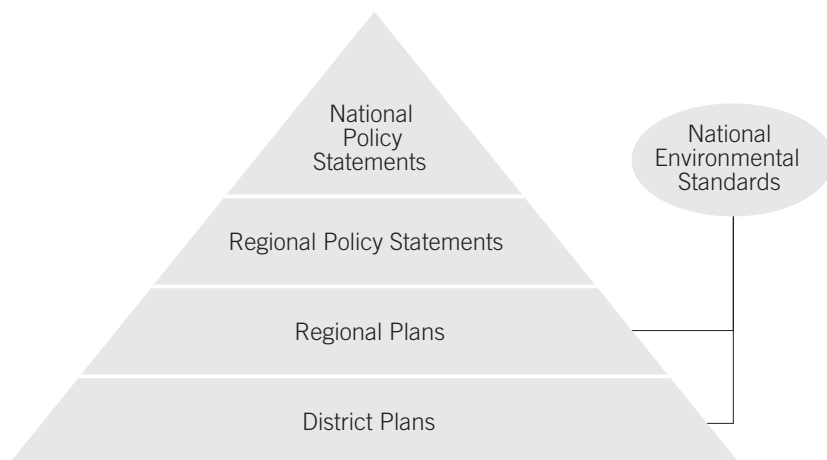


Figure 2.5: Hierarchy between RMA policies and plans

The NZCPS, which came into force in 1994 and which is currently under review, sets out a number of policies to be applied to the coastal environment. These relate to protecting the natural character of the coastal environment, protecting the characteristics of the coastal environment of special value to tangata whenua, and maintaining public access on the coast. They include protecting landscapes and seascapes, significant habitats of indigenous fauna and the integrity, functioning and resilience of the coastal environment. A copy of the 1994 NZCPS is reproduced in Appendix 3.

Regional policy statements, prepared by regional councils, are designed to provide an overview of resource management issues throughout the region and to identify policies and methods to achieve the integrated management of natural and physical resources (section 59).

Activities, which will potentially impact on the marine environment, are controlled by objectives, policies and rules in regional and district plans. These determine whether or not a consent is required before an activity can commence and the criteria that will apply when considering resource consent applications. Unlike the case with the use of land, most activities within the coastal marine area cannot be undertaken unless expressly authorised by a provision of a regional coastal plan or a resource consent.

In most cases applications for activities within the marine area are considered by the regional council in the first instance, and this decision can be appealed to the Environment Court.

In some cases, such applications will be referred directly to the Environment Court for determination. Applications of national significance may be called in and decided by a board of inquiry or Environment Court.

If an activity is classified as a restricted coastal activity in the regional coastal plan, which includes works such as large reclamations, structures and seawalls which are likely to have significant and irreversible adverse effects on the marine environment, the Minister of Conservation appoints a member to the hearing panel.

Regional plans are required to assist a regional council to carry out its functions under the RMA. Regional plans must state the objectives for the region and give effect to those objectives through policies and relevant rules. They can address issues such as soil erosion, marine water quality and the maintenance of indigenous biological diversity. Plans, which apply to the land catchment area, are optional.

Regional councils are required to prepare a regional coastal plan that applies to activities in the marine area below mean high water springs. These usually include the identification and 'zoning' of some marine areas for protection as well as for specific activities such as ports, wharves, marinas, moorings and marine farms. Other activities, not specifically provided for, are usually assessed against a set of criteria that are laid out in the plan. These include activities such as mining, energy generation and aquaculture. Regional coastal plans must be approved by the Minister of Conservation before becoming operative.

Several regional councils have prepared regional coastal environment plans that extend over the land and water portions of the coast. They typically contain objectives and policies which apply to the whole area but rules which apply only to the coastal marine area. This leaves district councils to develop rules to control land-based development within the guidance provided by the regional plan.

Territorial authorities are charged with controlling the 'effects' of the use, development or protection of land (section 31(b)), including effects within the land portion of the coastal environment but not in the marine area. As well as containing objectives and policies, district plans usually include rules, which control activities such as subdivision and the building of houses and structures. Such rules determine the location, density and design of development that is likely to be permitted within the catchment and on the coast. Some district plans identify special coastal zones or identify tighter coastal development controls in recognition of the sensitivity of the particular environment.

Hauraki Gulf

As well as being managed under the RMA and other legislation, the Hauraki Gulf has its own dedicated piece of legislation - the Hauraki Gulf Marine Park Act 2000 (HGMPA). This legislation seeks to improve the environmental management of the Gulf through achieving better integration of the numerous statutory authorities which impact on the area and providing better recognition of the relationships of tangata whenua with the Gulf (section 3 HGMPA).

The HGMPA establishes a Hauraki Gulf Forum. Members of the Forum include representatives of the Ministers of Conservation, Fisheries and Māori Affairs, 12 local authorities, six representatives from the tangata whenua of the Hauraki Gulf and its islands appointed by the Minister of Conservation and two further representatives appointed by the Auckland Regional Council.

Functions of the Forum include integrating the management of the Gulf, facilitating communication, co-operation and co-ordination of members' statutory functions exercised within the Gulf, preparing a list of strategic issues, determining priorities for action on each issue and publishing a report on the state of the environment every three years.

When undertaking its functions, the Forum must have particular regard to the historic, traditional, cultural, and spiritual relationship of tangata whenua with the Gulf. Day-to-day management of the Gulf continues to be carried out by individual statutory agencies.

The HGMPA sets out matters of national significance and management objectives which are to be incorporated into decision-making affecting the Gulf under other legislation (sections 7 and 8 HGMPA). Under the RMA, these must be given 'effect to' by all regional policy statements, regional plans and district plans which apply within the Gulf area, as though they were a national policy statement. They must also be treated as a NZCPS within the coastal environment. In addition, consent authorities must have regard to the objectives when considering an application for a resource consent within that area.

Sections 7 and 8 of the HGMPA are reproduced in Appendix 2. They seek to promote the protection and enhancement of:

- The life-supporting capacity of the Gulf's environment
- The historic, traditional, cultural and spiritual relationship of tangata whenua with the Gulf
- The Gulf's natural, historic and physical resources
- People's relationship to, use and enjoyment of, these resources.

When setting or varying any ‘sustainability measure’ within the Gulf under the Fisheries Act 1986, the Minister of Fisheries must ‘have regard to’ sections 7 and 8 of the HGMPA (section 11(2)(c), Fisheries Act). Statutory bodies undertaking powers and functions affecting the Gulf under a range of legislation must also ‘have particular regard to’ the provisions of section 7 and 8 of the HGMPA (section 13, HGMPA). This includes the Fisheries Act 1996, the Conservation Act 1987, the Biosecurity Act 1993, the Historic Places Act 1993, the Local Government Act 2002, the Marine Mammals Protection Act 1978, the Marine Reserves Act 1971, the Wildlife Act 1953 and the Reserves Act 1977.

The HGMPA also establishes the Hauraki Gulf Marine Park which is described in chapter five.

Fisheries Management

The Fisheries Act 1996 governs fisheries management throughout New Zealand’s territorial sea and EEZ. The purpose of the Act is ‘to provide for the utilisation of fisheries resources while ensuring sustainability’ (section 8). ‘Ensuring sustainability’ is defined as meaning:

- (a) Maintaining the potential of the fisheries resources to meet the reasonably foreseeable needs of future generations: and
- (b) Avoiding, remedying, or mitigating any adverse effects of fishing on the aquatic environment.

‘Utilisation’ is defined as meaning ‘conserving, using, enhancing, and developing fisheries resources to enable people to provide for their social, economic and cultural wellbeing.’

Three environmental principles are to be taken into account when decisions are made under the legislation (section 9):

- Associated or dependent species should be maintained above a level that ensures their long-term viability
- Biological diversity of the aquatic environment should be maintained
- Habitat of particular significance for fisheries management should be protected

Four principles are to apply to information used in fisheries decision making (section 10):

- Decisions should be based on the best available information

- Decision-makers should consider any uncertainty in the information available in any case
- Decision-makers should be cautious when information is uncertain, unreliable, or inadequate
- The absence of, or any uncertainty in, any information should not be used as a reason for postponing or failing to take any measure to achieve the purpose of the Fisheries Act

There are three main categories of fishing activity which are managed under the legislation – commercial, customary (non-commercial) and recreational.

Commercial fishing

Fishing activity is treated as commercial if the fish are taken for the purpose of sale (section 89). Most commercial fisheries are managed under the quota management system (QMS) which first requires the identification of a ‘fish stock’. A fish stock may include a single species such as snapper, or occasionally several species as is the case with the flatfish stock which includes eight different species. A stock comprises the population of such a species or species group within a defined area called a quota management area (QMA). A single species may be managed as a number of stocks. There are generally between one and 10 QMAs per species.

Quota management areas

The boundaries of QMAs differ between species and are designed to facilitate the management of each stock. For example the quota management area for flatfish 1 extends over the entire northern half of the North Island whereas the quota management area for Snapper 1 extends only over the northern east coast (see Figure 2.6). Once a quota management area is established, it can only be changed with the agreement of the owners of at least 75 per cent of the affected quota (section 25), or if the Minister is satisfied that it is necessary to ensure sustainability (section 25B).

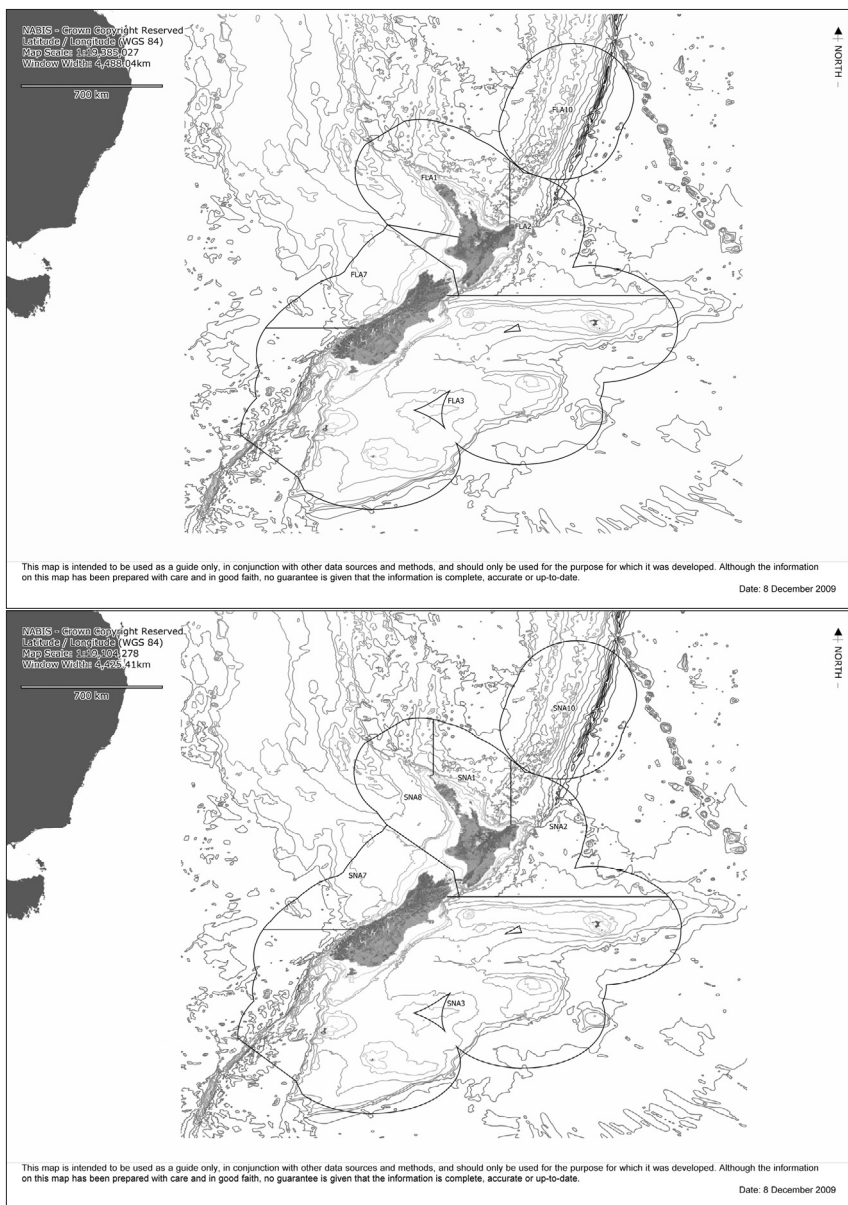


Figure 2.6: Quota management areas for flatfish and snapper (MFish)

Total allowable catch

Under section 13 of the Act, the Minister is required to set a total allowable catch (TAC) that maintains each fish stock at or above a level that can produce the 'maximum sustainable yield' (B_{MSY}). Maximum sustainable yield (MSY) is defined in the Fisheries Act as (section 2):

the greatest yield that can be achieved over time while maintaining the stock's productive capacity, having regard to population dynamics of the stock and any environmental factors that influence the stock.

The total allowable catch indicates the maximum amount of fish which can be taken by the combined commercial, customary and recreational fishing effort each year from specific stocks, as well as making an allowance for other sources of fishing-related mortality including illegal fishing.

Where a fish stock is below its estimated B_{MSY} the Minister is required to set a TAC which will enable the stock to increase to a level at or above B_{MSY} . The time period within which the stock should be able to rebuild to this level is undefined, but is to be a 'period appropriate to the stock and its biological characteristics' having regard to relevant 'social, cultural and economic factors' (section 13).

The Ministry of Fisheries *Harvest Strategy Standard 2008* is a Minister-approved policy statement of best practice for the setting of fishery and stock targets and limits. This serves as a statement of intent as to how stock management will be carried out within the discretion provided by the Act. The standard provides reference points for the size of stock biomass that trigger requirements for management action, such as the development and implementation of a stock rebuilding plan.

The TAC is the primary sustainability measure for a stock, but the Minister may also establish other sustainability measures to control the effects of commercial and other fishing on a fish stock or on the marine environment more generally. These measures may include restrictions on fishing methods, the size of fish taken and where and when fishing may be undertaken (section 11).

Total allowable commercial catch

The Minister is required to set a total allowable commercial catch (TACC), which specifies how much of the TAC can be harvested by commercial fishers, and may adjust it from time to time (section 20). The TACC must not exceed the TAC and the Minister must 'allow for' Māori customary non-commercial fishing interests and recreational interests before setting or adjusting the TACC. Where fish stocks are shared between commercial, recreational and/or customary fishers, the Act provides no guidance as to what proportion of the

TAC should be allocated to each sector. This is a matter which is left up to the discretion of the Minister of Fisheries.⁵

Individual transferable quota

The main means of allocating commercial rights to fishing is through the creation of individual transferable quotas (ITQ) in individual fish stocks. Individual transferable quota are expressed as 'quota shares' and they provide a right in perpetuity to harvest a proportional share of the TACC for a fish stock. ITQ is traded freely.

Annual catch entitlement

Each quota share generates an Annual Catch Entitlement (ACE) which is the right to harvest that share of the TACC during one fishing year. For instance, if someone owns 10 per cent of the quota shares for snapper in the QMA known as SNA1, the ACE generated by their quota shares gives them the right to harvest 10 per cent of whatever the TACC for TAC1 is in that year (expressed in kilograms). The TACC may change from year to year, but unless the quota owner buys or sells their shares, they will continue to have rights to 10 per cent of the TACC for SNA1. ACE is also freely traded.

Deemed values

A system of 'deemed values' encourages commercial fishers to ensure that they have sufficient ACEs to cover the species and amount of fish caught. When fishers have insufficient ACEs to cover their catch, they are required to pay to the Ministry of Fisheries the 'deemed value' of the excess fish. The level at which deemed values are set is important. If they are too high, they will encourage fishers to (illegally) discard excess fish. However, if they are too low, they will fail to provide an incentive for fishers to acquire sufficient ACEs or to keep their catch within their quota.

Māori fisheries

The issue of Māori fishing rights was brought to a head during the late 1980s. The implementation of the quota management system, in particular, led to concern by Māori that rights to fisheries guaranteed under the Treaty of Waitangi were being alienated by the Crown. In 1987 the High Court granted claimants' relief through an injunction on the further implementation of the quota management system pending settlement of the claims.

At that time, the Waitangi Tribunal was already considering claims in respect of fisheries resources and the 1988 Muriwhenua report found that the Crown

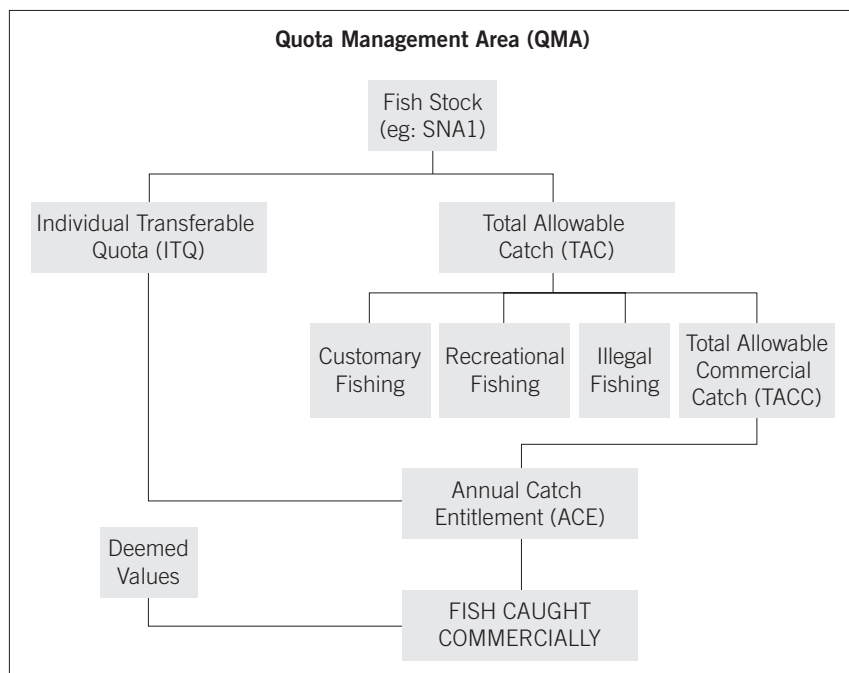


Figure 2.7: Management framework for commercial fishing

was in breach of Treaty obligations. Significantly, the Tribunal found that the quota management system was a fit and proper system for the management of fisheries, but that allocation of rights had not recognised the Māori interest.

This finding led to the 1989 interim settlement of claims legislated in the Māori Fisheries Act 1989, which provided for 10 per cent of all existing quota to be provided to Māori by the Crown. A further report by the Tribunal on claims by Ngāi Tahu found that a more extensive ‘development interest’ was inherent in the Treaty provisions and history of Māori fishing. This led to further negotiations between Māori and the Crown, culminating in the full and final settlement of commercial claims to fisheries in 1992. In addition to the provision of 10 per cent of existing quota, 20 per cent of any new quota being brought into the system was to be allocated to Māori.

Non-commercial Māori fishing rights and interests continue to give rise to Treaty obligations on the Crown, but have no legal effect except to the extent that they are provided for in regulations made under the Fisheries Act.

Customary fishing is managed under the Fisheries (Kaimoana Customary Fishing) Regulations 1998 and the Fisheries (South Island Customary Fishing)

Regulations 1999. These provide for iwi and hapū to manage their non-commercial fishing through the appointment of kaitiaki (guardians) who can issue permits to authorise the harvest of fish in their area. Implementation of these regulations, which arose from the Crown obligations under the 1992 settlement, is incomplete around the country. Many Māori also fish under the general provisions of the amateur fishing regulations.

Special management tools – taiāpure management areas and mātaihai reserves – were also developed for implementation of Treaty settlement obligations. These are described more fully in chapter 3.

Recreational fishing

Recreational fishing includes all non-commercial fishing apart from Māori customary fishing. There is no licensing of recreational fishers and anyone can legally fish for non-commercial purposes so long as he or she abides by the applicable regulations. Regulations limit the number of fish that each fisher can catch per day (bag limits), impose minimum size limits on fish caught, restrict fishing methods and close some areas to fishing. The marine area has been divided into five regions for the purpose of managing recreational fishing activity and different regulations apply to each area.

Fisheries plans

The Fisheries Act 1996 enables the Minister to approve fisheries plans which must be taken into account by the Minister when setting sustainability measures (section 11(2A)). The Act is silent on who may prepare a fisheries plan, but in recent years the Ministry of Fisheries has taken on this function. The plans are not mandatory and may relate to fish stocks, fishing years, or areas. They may include fisheries management objectives, strategies to achieve the objectives, and/or performance criteria to measure the achievement of the objectives amongst other things (section 11A). The Minister of Fisheries is currently developing a set of fisheries plans to cover the most significant fisheries as part of a shift to an objectives-based management framework.

Mineral exploration and development

Inside the territorial sea, the environmental effects of prospecting, exploration and mining are managed under the RMA and mineral resources are allocated under the Crown Minerals Act 1991.

On the continental shelf (including the area of the shelf within the EEZ), licences or permits issued by the Minister of Energy are required in order to exploit minerals or petroleum respectively under the Continental Shelf Act

1964. In the case of petroleum within the EEZ, applications are dealt with through procedures under the Crown Minerals Act 1991. Neither Act has any explicit environmental provisions as they are primarily concerned with allocation issues. There is a proposal to develop new legislation which would apply an environmental management regime to the EEZ and continental shelf.

Marine pollution

Marine pollution from ships and offshore installations such as oil rigs is managed by Maritime New Zealand under the Maritime Transport Act 1994. The Act provides that ‘harmful substances shall not be discharged or escape’ into the sea or seabed within the EEZ or continental shelf except in accordance with marine protection rules (section 226). This applies to discharges from ships, offshore installations, pipelines and any marine operations.

Maritime New Zealand has developed on behalf of the Minister of Transport a series of Marine Protection Rules, which govern:

- Marine pollution by ships
- Oil pollution and discharges from offshore installations
- The discharge of garbage and the dumping of waste or other matter at sea.

Biosecurity

The transport of invasive species into New Zealand waters by shipping is primarily controlled under the Biosecurity Act 1993. The jurisdiction of the Biosecurity Act is currently limited to the territorial sea.

Under this Act, an Import Health Standard has been established which requires all ships to exchange their ballast with mid-ocean water. The International Convention for the Control and Management of Ships’ Ballast Water and Sediments, adopted in February 2004 but yet to come into force, will provide internationally consistent controls over shipping ballast water.

The responsibility for the eradication or management of invasive marine organisms, once they have been introduced into New Zealand, is shared between Biosecurity New Zealand and regional councils.

Regional councils are able to address the marine biosecurity risks of activities when carrying out their resource management responsibilities. The proposed NZCPS 2008 reinforces this by providing that (policy 9):

Regional coastal plans shall control activities in the coastal marine area that could, because of associated biosecurity risks, have adverse effects on the coastal environment. Relevant activities include, but are not limited to:

- (a) the movement of structures likely to be contaminated with harmful organisms;
- (b) the disposal of organic material from vessel maintenance;
- (c) the provision of moorings, marina berths, jetties and wharves;
- (d) the establishment and movement of equipment and stock required for or associated with aquaculture activities.

Coastal permits, where relevant, shall include conditions requiring monitoring for biosecurity risks.

Wildlife protection

Marine mammals are managed by DoC under the Marine Mammals Protection Act 1978 and a permit is required to hold a marine mammal in captivity or to remove one from its natural habitat (section 4). Any fishing operation using a purse seine net is required to incorporate an escape panel or aperture in the net which allows any dolphin or porpoise to readily escape (section 17). When a marine mammal is accidentally injured or killed the legislation provides a defence against prosecution so long as the incident is reported to the relevant authority.

The legislation makes provision for the preparation of population management plans for marine mammal species (section 3E). Where a species is threatened, such plans can determine the maximum allowable level of fishing-related mortality for the species that would 'allow the species to achieve non-threatened status as soon as reasonably practicable, and in any event within a period not exceeding 20 years' (section 3F). Such plans require the approval of the Minister of Conservation and the concurrence of the Minister of Fisheries before they take effect. The Minister of Fisheries considers the impact of the plan on commercial fishing (section 3H).

No population management plans have been approved since these provisions were inserted into the legislation in 1996. In their absence, provisions in sections 15(2) and 11 of the Fisheries Act have been used to manage the effect of fishing on protected species, including Hector's and Maui's dolphins, New Zealand sea lions and seabirds. For example, under s15 (2) the effect of fishing on sea lions is mitigated by setting an annual maximum limit on fishing-related mortality. If this limit is reached, the squid fishery is closed for the season. Set net and trawl prohibitions to protect Hector's and Maui's dolphins were also put in place under the authority of s15 (2).

Marine mammal sanctuaries can be established by the Minister of Conservation under the Act for the purpose of protecting, conserving and managing marine mammals and these are managed by DoC.

The Wildlife Act 1952 identifies protected species. It is an offence to hunt or kill such species (section 63A). However, protected species may be captured, injured or killed inadvertently or as part of a fishing operation and the legislation provides a defence against prosecution when this happens, so long as the incident is reported to the relevant authority.

Currently, marine species protected within the territorial sea and EEZ under the Act include black coral (all species in the order of Antipatharia), red coral (all species), the spotted black groper, the white pointed shark and marine reptiles such as turtles and sea snakes.

All seabirds are protected under the Act except for the southern black-backed gull. Sooty shearwaters and grey-faced petrels may be killed subject to conditions set by the Minister of Conservation and their chicks can be legally harvested by iwi and mana whenua (land trustees) at identified sites (section 6 and schedule 3, Wildlife Act).

Historic heritage

Historic marine sites and places can be protected either under the Historic Places Act 1993 or the RMA. Some sites and places are protected under both pieces of legislation. There are numerous marine heritage sites throughout New Zealand including wāhi tapu, submerged sites and structures, shipwrecks and historic anchorages. Most marine heritage sites have yet to be identified. For example, there are over 2000 known shipwrecks around the coast of which only 150 have been located.⁶

Archaeological sites are protected under the Historic Places Act and can only be modified or destroyed if an authority (permit) is first obtained from the NZHPT. Archaeological sites are defined in the legislation as (section 2):

... any place in New Zealand that

(a) Either –

(i) Was associated with human activity that occurred before 1900; or

(ii) Is the site of the wreck of any vessel where that wreck occurred before 1900; and

(b) Is or may be able through investigation by archaeological methods to provide evidence relating to the history of New Zealand.

Other marine heritage sites can be protected under the RMA, through listing them in the regional coastal plan and including protective rules. Few plans, however, provide protection for marine cultural heritage. District plans can provide protection for land-based marine heritage through scheduling and relevant protection rules. Heritage orders only take effect thorough district plans, which apply above mean high water springs, so such orders are not applicable to marine heritage.

Coastal reserves

Reserves can be established on the coast (and inland) under the Reserves Act 1977. Reserve land is categorised on the basis of its primary purpose as either recreation, historic, scenic, nature, scientific, government purpose or local purpose reserves (see Figure 2.8). Each reserve is required to have a reserve management plan and to be managed in accordance with the purpose for which it is classified.

The Reserves Act also provides mechanisms for the protection of privately-owned land including the declaration of protected private land, the establishment of conservation covenants and the establishment of Nga Whenua Rahui kawenata on Māori land (sections 76, 77 and 77A, Reserves Act).

Figure 2.8 Classification of reserves	
Type of reserve	Primary purpose
Recreation	Providing areas for the recreation and sporting activities and the physical welfare and enjoyment of the public, and for the protection of the natural environment and beauty of the countryside, with emphasis on the retention of open spaces and on outdoor recreational activities, including recreational tracks in the countryside
Historic	Protecting and preserving in perpetuity such places, objects, and natural features, and such things thereon or therein contained as are of historic, archaeological, cultural, educational, and other special interest
Scenic	Protecting and preserving in perpetuity for their intrinsic worth and for the benefit, enjoyment, and use of the public, suitable areas possessing such qualities of scenic interest, beauty, or natural features or landscape that their protection and preservation are desirable in the public interest; or Providing, in appropriate circumstances, suitable areas which by development and the introduction of flora, whether indigenous or exotic, will become of such scenic interest or beauty that their development, protection, and preservation are desirable in the public interest
Nature	Protecting and preserving in perpetuity indigenous flora or fauna or natural features that are of such rarity, scientific interest or importance, or so unique that their protection and preservation are in the public interest

Scientific	Protecting and preserving in perpetuity for scientific study, research, education, and the benefit of the country, ecological associations, plant or animal communities, types of soil, geomorphological phenomena, and like matters of special interest
Government purpose	Providing and retaining areas for such Government purpose or purposes as are specified in any classification of the reserve
Local purpose	Providing and retaining areas for such local purpose or purposes as are specified in any classification of the reserve.

Foreshore and seabed

The Foreshore and Seabed Act 2004 vests ownership of the foreshore and seabed in the Crown, excluding areas subject to a freehold interest, and provides that every natural person has access rights to this ‘public’ foreshore and seabed (section 7). In addition, the legislation confers rights of navigation over the foreshore and seabed to all persons (section 8).

The legislation makes provision for groups to apply to the High Court for a finding that they would have held territorial customary rights to an area of the public foreshore and seabed in common law, if the Foreshore and Seabed legislation had not been enacted (section 33). If the group is successful it has two options. First, it can seek a referral to the Attorney-General and the Minister of Māori Affairs to enable the group to negotiate an agreement for redress. Alternatively, the group can seek the establishment of a foreshore and seabed reserve.

The purpose of foreshore and seabed reserves is to acknowledge the exercise of kaitiakitanga (guardianship) over a specified foreshore and seabed area (section 40). The area is managed by a board comprised of members proposed by the group that holds territorial customary rights over the reserve. The board has the function of preparing a management plan for the reserve.

Alternatively a whānau, hapū or iwi can apply to the Māori Land Court for a customary rights order relating to an area of public foreshore and seabed (section 48). This enables the group to carry out a recognised customary activity and the activity is given specific protection under the RMA.

The Foreshore and Seabed Act 2004 is currently under review. At the time of writing no foreshore and seabed reserves have been established.

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1 WWF-New Zealand, 2009, 11

2 WWF-New Zealand, 2009, 11

3 See <http://www.isa.org.jm/en/about>

4 The term 'mean high water springs' is not defined in the RMA. The New Zealand Nautical Almanac defines it as 'The average of the levels of each pair of successive high waters during that period of about 24 hours in each semi-lunation (approximately every 14 days), when the range of the tide is greatest (Spring Range).

5 See *New Zealand Recreational Fishing Council Inc and other v Sanford Limited and others*, SC 40/2008 [2009] NZSC 54

6 Oceans Policy Secretariat, 2003,7



three

Kaitiakitanga

Iwi as resource owners

Iwi nationally are involved in a range of commercial activities including tourism, property development and forestry. Māori own an estimated 50 per cent of the aquaculture industry and over a third of the fishing industry. Some have interests in other marine ventures such as marinas.

There are 57 iwi or groups of iwi¹ recognised in the Māori Fisheries Act 2004. Commercial fisheries assets held centrally by Te Ohu Kai Moana are being transferred to those iwi. Separate aquaculture settlement legislation (Māori Commercial Aquaculture Claims Settlement Act 2004) requires 20 per cent of marine farm space created since 1992 to be allocated to iwi. Alternatively settlement of a cash equivalent for space created between 1992 and 2004 can be negotiated.

Prior to the allocation of fisheries assets the recognised iwi have been able to lease ACE from Te Ohu Kai Moana. Most of those iwi have established commercial fishing operations which, for many, provide their single most significant income stream.

Tensions can arise within iwi regarding their own development proposals. Whereas a mainstream developer will need to meet RMA and other constraints, an iwi will need to ensure that its development aspirations meet its beneficiaries' expectations of commercial gain as well as the requirements of kaitiakitanga. This can produce a far more stringent set of conditions than other developers confront.

Scope of Kaitiakitanga

'Kaitiakitanga' is now part of the vocabulary of environmental managers in New Zealand, but understanding of its full significance is usually limited. Although there are statutory definitions of kaitiakitanga in the RMA and the Fisheries Act 1996, there is little case law or published best practice, to guide practitioners in its application.

The Aquaculture Steering Group of Te Ohu Kai Moana has developed a useful description of kaitiakitanga which acknowledges the statutory definitions, but provides an understanding within a broader cultural context (see Figure 3.1).

There are a number of key messages in this description. Kaitiakitanga is:

- Integrated with the spiritual, cultural and social life of tangata whenua
- Holistic across land and sea

- Inclusive of people within the concept of environment
- Locally defined and exercised
- Not focused on ownership, but on authority and responsibility
- Concerned with both sustainability of the environment and the utilisation of its benefits.

This concept of kaitiakitanga contains the purpose of the RMA (sustainable management of resources); and that of the Fisheries Act (sustainability and utilisation); but is broader than each of these concepts.

Traditionally kaitiaki had a guardian and stewardship role for natural resources. While individuals may have had specific roles, they were exercised as part of a collective responsibility determined through whakapapa and tikanga. The allocation and distribution of the benefits from those resources were further kaitiaki responsibilities. Tribal mana has, in most aspects of natural resources, been supplanted by legal ownership and statutory regulation. How traditional kaitakitanga can adapt to today's changed circumstances is a major challenge for tangata whenua.

In practice, management of natural resources is exercised either by those having formal legal ownership, or through the exercise of statutory authority. The Māori proportion of land and resources ownership is still relatively small when compared with that of non-Māori private ownership and the Crown Estate. Exercise of kaitiakitanga today must often be implemented through opportunities within the legal framework.

Kaitiakitanga

Kaitiakitanga is already acknowledged in legislation and is defined as follows:

...the exercise of guardianship by the tangata whenua of an area in accordance with tikanga Māori in relation to natural and physical resources, and includes the ethic of stewardship” (section 2, RMA);

and

...the exercise of guardianship; and, in relation to any fisheries resources, includes the ethic of stewardship based on the nature of the resources, as exercised by the appropriate tangata whenua in accordance with tikanga Māori (section 2, Fisheries Act 1992)

Concern has often been expressed however that present legal definitions do not fully express what kaitiakitanga is about, and that any attempt to define it in anything other than te reo Māori will always be insufficient.

The Aquaculture Steering Group offers the following ideas in order to generate discussion on the concept of kaitiakitanga and what it means for the aquaculture reform.

Kaitiakitanga contains many elements that can be described as:

- Mahi tapu – god given and handed down through our tipuna
- Founded in whakapapa - the relationship between everything and everybody in the natural world – there is no distinction between people and their environment
- Exercised on behalf of, and for the benefit of, all who are related through whakapapa
- A set of inalienable responsibilities, duties and obligations that are not able to be delegated or abrogated
- A web of obligations: to the taonga, to the atua and to ourselves and our uri. Kaitiaki have a responsibility to provide for everyone and ensure everyone benefits
- Independent of ‘ownership’ in a European sense. As on land, kaitiaki responsibilities are independent of others who hold ‘ownership’ or use rights under the law. For example, although as kaitiaki, iwi/hapū may ‘own’ only a percentage of the total marine farming space in a region under existing law, they still hold kaitiaki responsibilities over the whole area in accordance with tikanga
- Seamless and all encompassing – making no distinction between moana and whenua
- Given effect at whanau and hapu level
- Expressed in ways that are appropriate to the place and to the circumstances, according to tikanga
- Wider and more complex than existing legal definitions
- Given practical effect by exercising control over access to resources and sharing the benefits of the use of those resources
- Enabled through rangatiratanga, which includes the authority that is needed to control access to and use of resources, and to determine how the benefits will be shared. This means that it can be expressed in part through the concepts of ‘ownership’, ‘property’, ‘title’ or ‘stewardship’ - however it is much wider than any of these.

Kaitiakitanga has been exercised since before the Treaty. Article II of the Treaty guaranteed that iwi/hapū would retain the authority they needed – that is rangatiratanga - to continue to exercise kaitiakitanga.

While the Crown gained the right to govern and to make laws (including for the purpose of resource conservation) under Article I of the Treaty, the Crown must heed the guarantees it made under Article II when designing and implementing its policies and laws.

Figure 3.1 Description of kaitiakitanga

Relevant legislative provisions

Many pieces of legislation, which apply to the marine area, have general provisions of relevance to the exercise of kaitiakitanga.

Resource Management Act 1991

The RMA makes general provisions for recognition of kaitiakitanga through the following provisions:

- All persons exercising functions and powers under the Act must ‘take into account the principles of the Treaty of Waitangi (Te Tiriti o Waitangi)’ (section 8)
- ‘The relationship of Māori and their culture and traditions with their ancestral land, water, sites, wāhi tapu, and other taonga’ is a matter of national importance which must be recognised and provided for by decision makers (section 6(e))
- ‘The protection of historic heritage’ which includes ‘sites of significance to Māori, including wāhi tapu’ from inappropriate subdivision, use, and development is a matter of national importance which must be recognised and provided for by decision makers (section 6(f))
- ‘Kaitiakitanga’ is a matter which decision makers must have particular regard to (section 7(a)).

The RMA also creates opportunities for the practical application of kaitiakitanga through:

- Provision for the transfer of local authority functions, powers or duties under the RMA to iwi authorities (section 33, RMA)
- Provision for the development of joint management agreements between local authorities, other public authorities, iwi authorities and groups that represent hapū (sections 36B-E, RMA)

Fisheries Act 1996

The Fisheries Act makes general provisions for recognition of kaitiakitanga through the following provisions:

- All persons exercising functions, duties, or powers under the Act must do so in a manner consistent with the provisions of the Treaty of

Waitangi (Fisheries Claims) Settlement Act 1992 – which implements the settlement of claims relating to Māori fisheries rights (section 5)

- The provisions relating to taiāpure and customary fishing have the object of making better provision for the recognition of rangatiratanga and of the right secured in relation to fisheries by Article II of the Treaty of Waitangi (section 174).

The Fisheries Act also creates opportunities for the practical application of kaitiakitanga through:

- The identification of rohe moana and appointment of Tangata Kaitiaki/Tiaki to authorise customary food gathering (Fisheries (Kaimoana Customary Fishing) Regulations 1998 and Fisheries (South Island Customary Fishing) Regulations 1999)
- The establishment of mātaihai reserves under the Fisheries (Kaimoana Customary Fishing) Regulations 1998 over traditional fishing grounds of special significance to tangata whenua
- The establishment of taiāpure under the Fisheries Act 1996 (section 186) over estuarine or littoral coastal waters that have customarily been of special significance to iwi or hapū
- The temporary closure of fishing areas or restriction on fishing methods to recognise and make provision for the use and management practices of tangata whenua (section 186A, Fisheries Act)

Local Government Act 2002

- A local authority is required to provide opportunities for Māori to contribute to its decision-making processes (section 14(1)(d))
- Where a local authority is making a significant decision in relation to land or a body of water, it must take into account the relationship of Māori and their culture and traditions with their ancestral land, water, sites, wāhi tapu, valued flora and fauna, and other taonga (section 77(1)(c))
- A local authority must establish and maintain processes to provide opportunities for Māori to contribute to the decision-making processes of the local authority; consider ways in which it may foster the development of Māori capacity to contribute to those decision-making processes; and provide relevant information to Māori for those purposes (section 81)

Other relevant legislation

- The Conservation Act 1987 is to be interpreted and administered to give effect to the principles of the Treaty of Waitangi (section 4)
- All persons exercising functions and powers under the Historic Places Act 1993 shall recognise the relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wahi tāpu, and other taonga (section 4). The Act also makes provision for the registration of wāhi tapu and wāhi tapu areas (part 2)
- The establishment of foreshore and seabed reserves under the Foreshore and Seabed Act 2004 acknowledges the exercise of kaitiakitanga over a specified area

Spatial marine management

Three spatial marine management tools make specific provision for kaitiakitanga – mātaihai reserves, taiāpure and seabed and foreshore reserves. These tools are summarised in Figure 3.2 and described in more detail below.

Figure 3.2 Spatial marine management tools to implement kaitiakitanga	
Mataitai reserves <i>Fisheries (Kaimoana Customary Fishing) Regulations 1998</i> <i>Fisheries (South Island Customary Fishing) Regulations 1999</i>	Recognise and provide for the special relationship between tangata whenua and places of importance for customary food gathering Only tangata whenua can apply to create a mātaihai reserve Created by the Minister of Fisheries through Gazette notice Commercial fishing excluded unless re-instated by regulation Managed by Tangata Kaitiaki/Tiaki, who are chosen by tangata whenua, through bylaws
Taiāpure <i>Fisheries Act 1996 (sections 174 - 185)</i>	Cover estuarine or littoral coastal waters that have customarily been of special significance to iwi or hapū either as a source of food or for spiritual or cultural reasons Anyone can apply to create a taiāpure Created by the Governor General through an Order-in-Council (meaning it must be approved by cabinet) Commercial fishing not usually excluded, but may be constrained Managed by a local committee representative of Māori appointed by the Minister of Fisheries. The committee advises the Ministry of Fisheries on what fisheries regulations should be applied to the area

Foreshore and Seabed Reserves	Purpose is to acknowledge the exercise of kaitiakitanga over a specified foreshore and seabed area
<i>Foreshore and Seabed Act 2004</i>	Applicant group needs to establish its prior territorial customary rights over the area
	Created by the High Court
	Managed by board comprising members of the applicant group which prepares a management plan

Mātaítai reserves

Mātaítai reserves can be declared under the Fisheries (Kaimoana Customary Fishing) Regulations 1998 or the Fisheries (South Island Customary Fishing) Regulations 1999 on application of tangata whenua. These are areas which are recognised traditional fishing grounds with which tangata whenua have a special relationship. The reserve is managed by Tangata Tiaki/Kaitiaki chosen by the tangata whenua and confirmed by the Ministry of Fisheries. The Tangata Tiaki/Kaitiaki can restrict or prohibit fishing in the mātaítai reserve by recommending bylaws. Although the bylaws apply to any person fishing in the mātaítai reserve, fishing for the purposes of sustaining the functions of a marae may continue if authorised by the Tangata Tiaki/Kaitiaki.

The process for creating a mātaítai reserve is as follows:

- The tangata whenua or Tangata Kaitiaki/Tiaki (who have been confirmed by the Minister) lodge an application with the Ministry of Fisheries
- The Minister publishes details of the application twice in a newspaper circulating in the area of the proposed reserve inviting written submissions from the 'local community' in the area. At least 20 working days must be given for submissions to be lodged
- As soon as reasonably practical, and no later than 20 days after the closing date for written submissions, the Minister must publish twice in the local newspaper notice of a local community meeting where the Minister and tangata whenua consult with the 'local community'. The 'local community' is defined in the regulations to include people who own land in the vicinity of the proposed reserve or have been resident in the area for at least three months in the preceding three years. The application may be amended by tangata whenua as a result of the consultation.
- The Minister then advertises the application and calls for written submissions from persons having a fishing interest in the area. At least 20 working days must be given for the submissions to be lodged.
- As soon as practicable after the closing date for submissions, the Minister must advise the tangata whenua of any submissions, and discuss any

conditions that the Minister considers may be necessary to address issues raised by those submissions.

- The Minister must then approve the application if the regulatory criteria are met and a notice will be published in the *Gazette*.
- The Minister will confirm the appointment of Tangata Kaitiaki/Tiaki chosen by tangata whenua to manage the reserve.

Before approving the application the Minister must be satisfied that (clause 23):

- A special relationship exists between the tangata whenua and the proposed reserve
- The general management aims are consistent with the sustainable use of fisheries resources in the area
- The area is an identified traditional fishing ground and is of a size which can be effectively managed by tangata whenua
- The Minister and tangata whenua are able to agree on suitable conditions to address issues raised by submissions
- The area is not a marine reserve

The Minister must also be satisfied that the reserve will not:

- Unreasonably affect the ability of the local community to take fish for non-commercial purposes
- Prevent persons with a commercial interest in a species from taking their quota or annual catch entitlement within the quota management area for that species
- Unreasonably prevent persons with a commercial fishing permit for a non-quota species from taking fish within the fisheries management areas for which that permit has been issued
- Unreasonably prevent non-commercial fishers from fishing

Once a mātaihai reserve is established, commercial fishing is not allowed unless it is re-instated by regulation. The regulations must be recommended by the Tangata Tiaki/Kaitiaki and apply to specified species, restricted by quantity or time period. Māori and non-Māori may fish in mātaihai reserves.

Taiāpure

Taiāpure may be established under the Fisheries Act 1996. The purpose of taiāpure is to make better provision for the recognition of estuarine or littoral

coastal waters that have customarily been of special significance to iwi or hapū either as a source of food or for spiritual or cultural reasons (section 174).

Anyone may apply to have a taiāpure established. Applications are lodged with the Ministry of Fisheries. The process for processing an application is as follows:

- The Minister of Fisheries considers the application, consults with the Minister of Māori Affairs and then decides whether or not to agree with the proposal in principle. If the Minister does not support the proposal at this stage, it cannot proceed.
- Once the Minister has agreed to the proposal in principle, the Ministry of Fisheries publicly notifies the proposal in the *Gazette* and in metropolitan and local newspapers. At least two months must be given for written submissions, which are lodged with the Māori Land Court.
- Anyone can lodge a submission on the proposal, which must identify the grounds for the submission.
- A public inquiry is then conducted into the submissions by a Tribunal consisting of a Judge of the Māori Land Court and possibly other assessors. The proponent, the Minister, councils affected by the proposal and submitters have a right to be heard at the inquiry.
- The Tribunal then makes recommendations to the Minister and may propose changes to the proposal as a result of the submissions.
- The Minister then decides whether to accept or decline the recommendations. The report and recommendations of the Tribunal and the decision of the Minister must be published in the *Gazette*.
- Appeals against the recommendations of the Tribunal can only be made to the High Court on points of law.

The Minister can then recommend that the taiāpure be created, but only if he or she is satisfied that that the proposal meets the overall purpose of taiāpure and that the proposal is appropriate having regard to (section 176):

- The size of the proposed taiāpure
- The impact on the general welfare of the community in the vicinity of the area
- The impact on persons having a special interest in the area
- The impact on fisheries management.

The taiāpure is formally created by the Governor-General by Order-in-Council (section 175).

A management committee representative of the local Māori community, is appointed by the Minister of Fisheries, and advises the Ministry of Fisheries on regulations to manage the area. These may include restrictions on the species and quantities that may be harvested, size limits, when fish may be taken, the fishing methods that can be used, and the areas from which species may be taken. Unlike in a mātaihai reserve, taiāpure allow commercial fishing, although it may be constrained by its regulations.

Foreshore and Seabed Reserves

These reserves may be established under the Foreshore and Seabed Act 2004 to acknowledge the exercise of kaitiakitanga over a specified foreshore and seabed area and to enable the area to be held for the common use and benefit of New Zealanders (section 40).

A reserve can be created on application to the High Court. In order to create a reserve the applicants need to:

- Establish that they had territorial customary rights to the area (prior to the vesting of the foreshore and seabed in the Crown). This requires that the group has held continuous title to the land next to the proposed reserve amongst other things.
- Identify who will be represented in the management board (called guardians)
- Identify what resources will be required to manage the reserve and how they will be provided
- Propose a charter by which the reserve will be administered.

The area is managed by the guardians who are charged with preparing a management plan for the reserve. This must not be inconsistent with the NZCPS or any national policy statement. The guardians are not able to restrict public access to the area or to charge for people to access the area.

The Foreshore and Seabed Act is currently under review and these provisions may change. At the time of writing no Foreshore and Seabed Reserves had been created under the Act. Some which were close to approval have had final decisions deferred until after the review is completed.

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- 1 Some groups of iwi – such as the 12 Hauraki iwi, or the 11 Te Arawa Waka iwi – have a single entity to represent them identified in the statute.



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Opportunities for Public Involvement

Overview

There are many opportunities for members of the public to become involved in statutory processes in order to improve environmental outcomes for the marine environment. Some of these processes involve establishing policies that govern decision making. Others involve developing rules and regulations that apply to activities in the marine area. Yet other processes involve considering specific proposals to undertake marine activities.

Because there are so many different pieces of legislation applying to the marine area, which are administered by a range of statutory authorities, there is little consistency in the processes that apply to different management jurisdictions. In some areas, such as under the RMA, there are quite extensive rights of public participation. In other areas, such as under the Fisheries Act 1996, consultation rights are focused on directly affected parties, although the Ministry of Fisheries will typically provide the opportunity for written public submissions to be made before delivering final advice to the Minister who then makes a decision.

Where there are no rights of submission or appeal, a statutory decision may still be able to be challenged by initiating judicial review proceedings in the High Court.

Preparing policies and plans

Resource Management Act policies and plans

There is a hierarchy of policies and plans that guide decision making under the RMA, and members of the public have the opportunity to influence the content of these documents. At the national level is the NZCPS. There is also the possibility of other national policy statements being prepared which could affect the marine environment.

Members of the public must be given the opportunity to make submissions on proposed national policy statements. If a board of inquiry process is adopted by the Minister for the Conservation, submitters will also have the opportunity to present their case (including submissions and evidence) at a public hearing convened by an independent board of inquiry. The board of inquiry makes recommendations to the Minister who makes the final decision. There are no

appeal rights against the Minister's decision on national policy statements, and the only redress is through judicial review proceedings in the High Court.

At the regional level, regional policy statements, regional coastal plans and regional plans are prepared. At the district level, district plans are prepared. These documents typically contain objectives, policies, rules, methods and anticipated environmental results which guide decision-making on resource consent applications.

There is a very open process for public participation in council policy and plan-making under the RMA. Councils are required to consult with a range of parties prior to proposed plans and policy statements being formally notified, including Ministers of the Crown, local authorities and iwi (clause 3, Schedule 1). Councils also commonly consult with a range of other stakeholders and this can be a useful opportunity to have some input outside the formal statutory process.

Once the proposed policy statement or plan is formally notified, members of the public can make written submissions. The following persons are then able to lodge further submissions in support or opposition to the submissions already lodged:

- Any person representing a relevant aspect of the public interest. This is likely to include public interest environmental groups.
- Any person that has an interest in the proposed policy statement or plan greater than the interest that the general public has. This is likely to include owners of land and users of resources directly affected by plan provisions. It is also likely to include iwi and hapū where their interests are directly affected.
- The local authority itself

All submitters then have the opportunity to present their case before a council-initiated public hearing which the RMA requires to be held.

When preparing a plan or a plan change councils must take into account relevant iwi planning documents which have been lodged with them. This provides an opportunity for tangata whenua to have proactive input into planning initiatives.

Once the council has made decisions on submissions, any submitter has the right to appeal the decision to the Environment Court. The Court will consider the matter afresh at a public hearing.

Other parties can become party to any appeal (referred to as section 274 parties) only if they can show that they fall within one of the following categories:

- They have lodged a submission at the local authority stage; or
- They have an interest in the proceedings greater than the public generally (excluding trade competitors). This will normally include neighbours, downstream resource users and iwi authorities. It is uncertain whether public interest groups will be able to become parties under this category as well, due to recent changes to the RMA.

Decisions of the Environment Court can be appealed to the High Court on questions of law, but not on the merits.

Fisheries plans

Under fisheries management, there is statutory provision for the Minister of Fisheries to approve fisheries plans. Before approving a plan, the Minister must consult with representatives of stakeholder groups such as fishers and environmental organisations. The Minister must also provide for the ‘input and participation’ of tangata whenua (section 12, Fisheries Act). Although members of the public have no explicit consultation rights in relation to fisheries plans, in practice draft plans are usually circulated for public written comment before approval.

The Ministry of Fisheries is in the process of developing standards, which are intended to guide decision-making on fisheries management measures. The standards are also intended to guide the preparation of fisheries plans. Because there is no statutory provision for these standards, there is no statutory right to participate in their preparation. However, in practice, the Ministry of Fisheries circulates draft standards and invites written comment from members of the public, before finalisation. Advise staff at the Ministry of Fisheries if you have an interest in particular fisheries plans so that you can be placed on the Ministry’s circulation list.

Conservation strategies and plans

Under the Conservation Act 1987, the Minister of Conservation can prepare general policy and conservation management strategies and plans. These guide DoC’s marine protection as well as other work. The public process for preparing each of these documents is similar. Members of the public must be given the opportunity to make written submissions on draft documents. Submitters must then also be given the opportunity to present their submissions to the Director-General of Conservation in person (for submissions on general policy) or to a meeting consisting of representatives of the Director-General and affected conservation boards (for strategies and plans).

Population management plans

Population management plans can be prepared by the Minister of Conservation for marine mammals under the Marine Mammals Protection Act 1978 and for other wildlife such as seabirds under the Wildlife Act 1953. Members of the public must be given the opportunity to make written submissions on the draft plan and also to present their submissions in person to the Director-General or his or her representatives.

Reserve management plans

Reserve management plans are prepared for each reserve protected under the Reserves Act 1977 by the administering body. Members of the public must be given the opportunity to make written submissions on the draft plan as well as to present the submissions in person. Opportunity may also be given to make written submissions at an earlier stage in the plan preparation process if the administering body publicly notifies an intention to prepare a plan.

Marine protected areas

The government has adopted a Marine Protected Areas Policy Statement and Implementation Plan, which although official policy, has no statutory basis. The plan provides for the establishment of regional marine protection planning forums which prepare recommendations for marine protected areas within biogeographic regions. The forums are made up of local stakeholder representatives. The plan and the recommendations of the forums have no legislative basis. There are consequently no statutory rights for the public to participate in forum deliberations. However, in practice, members of the public will be given the opportunity to make written submissions on the proposed recommendations.

Figure 4.1: Opportunities to participate in the preparation of policies and plans

Policy/plans	Application	Opportunity for public involvement
<i>Resource Management Act 1991</i>		
New Zealand Coastal Policy Statement	Activities within catchments and the territorial sea of the country	Written submissions on proposed policy statement
Other national policy statements		Presentation of submissions and evidence to Board of Inquiry (where the board of inquiry process is adopted)
Regional policy statements	Activities within catchments and the territorial sea of the region	Written submissions on proposed policy or plan
Regional coastal plans		Written further submissions by some parties
	Regional plans	Presentation of submissions and evidence to council hearing
District plans		Appeal to Environment Court
		Presentation of submissions and evidence to Environment Court hearing
		Appeal to High Court on points of law
	<i>Fisheries Act 1996</i>	
Fisheries plans	Fisheries or groups of fisheries	<i>No statutory rights to participate.</i> Minister must consult with persons or organisations representative of those classes of persons having an interest in the stock or effects of fishing on the aquatic environment in the area concerned including Māori, environmental, commercial and recreational interests. In practice, a draft plan is usually circulated for public written submissions
Fisheries standards	All fisheries	<i>No statutory rights to participate and the standards are non-statutory.</i> In practice, written submissions are sought on draft standards.
<i>Conservation Act 1987</i>		
General policy	Conservation areas and natural and historic resources	Written submissions on draft policy
Conservation management strategies		Presentation of submissions to the Director-General of Conservation
Conservation management plans	Natural and historic resources	Written submissions on draft strategy
		Presentation of submissions at a meeting of representatives of the Director-General of Conservation and affected conservation boards

*Marine Mammals
Protection Act 1978*

Population management plans	Marine mammals	Written submissions on draft plan Presentation of submissions to the Director-General of Conservation or his or her representatives
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Wildlife Act 1953

Population management plans	Wildlife including seabirds	Written submissions on draft plan Presentation of submissions to the Director-General of Conservation or his or her representatives
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Reserves Act 1977

Reserve management plans	Land-based reserves	Written submissions on proposed plan where intention to prepare plan is publicly notified Written submissions on draft plan Presentation of submissions to reserve administration body or nominee
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No legislative basis

Fisheries standards	All fisheries	<i>No statutory rights to participate and the standards are non-statutory.</i> In practice, written submissions are sought on draft standards.
Recommendations for regional marine protected areas	Marine areas in territorial sea and EEZ	<i>No statutory rights to participate and the process is non-statutory.</i> In practice, stakeholders will be represented on a marine protection planning forum and written submissions will be sought on the forum's recommendations

Making rules and regulations

There are a large number of processes through which rules and regulations are made to manage activities affecting the marine environment.

National environmental standards

Under the RMA, plans contain rules as well as objectives and policies and the opportunity to participate in plan-making processes has been described in the previous section.

There is also the ability under the RMA to make national rules which are called national environmental standards. These are regulations issued under the RMA which prescribe technical standards, methods and other requirements for environmental matters.

Regional councils and territorial authorities must enforce these standards (or they can enforce stricter standards when the standard provides for this). In this way, national environmental standards can be used to ensure that consistent minimum standards are maintained throughout all of New Zealand's regions and districts

There is no prescribed process for the preparation of national standards other than that the public must be given adequate time and opportunity to comment (section 44). Usually written submissions will be sought on a proposed standard before it is adopted, but there will not be a public hearing to consider submissions.

Fisheries sustainability measures

For the setting of sustainability measures under the Fisheries Act 1996 there are no statutory rights of public participation. Sustainability measures include setting the TAC, regulatory measures such as restricting fishing methods or locations, and measures to manage the effects of fishing-related mortality on protected species. The Minister must consult with representatives of groups having an interest in the stock or the effects of fishing on the aquatic environment in the area concerned. This will normally include commercial and recreational fishing interests and environmental organisations. The Minister must also provide for the 'input and participation' of tangata whenua (section 12, Fisheries Act).

When setting or varying the TACC, the Minister must consult with representatives of groups having an interest including Māori, environmental, commercial and recreational interests (section 21(2), Fisheries Act)

In practice, the Ministry of Fisheries makes available an initial advice paper and will accept written submissions from any party, before finalising its advice to the Minister. If you have an interest in particular fisheries sustainability matters, contact staff at the Ministry so they can add you to the relevant circulation lists.

Mātiatai and taiāpure

When mātaiatai reserves are proposed, consultation is undertaken with members of the local community and people with a fishing interest in the area. Public participation provisions for taiāpure proposals are broader. Any person can make written submissions on a proposal and present their submissions to a Māori Land Court judge. There are also rights to appeal the decision of the judge to the High Court on points of law. These processes are described in more detail in Chapter Three.

When marine protection rules are developed under the Maritime Transport Act 1994, when minerals programmes are developed under the Crown Minerals Act 1991 and when marine mammal sanctuaries are established by the Minister of Conservation, there are statutory provisions requiring members of the public to be given the opportunity to make written submissions, but no requirement for public hearings and no appeal rights.

When an application is made to establish a marine reserve, the application is publicly notified and any person can make a written objection to the proposal. The objections are lodged with the Director-General of Conservation.

There are no statutory rights for the public to participate in the preparation of import health standards, although there is an obligation in the Biosecurity Act 1993 for the chief technical officer to consult with representatives of persons having an interest in the standard. In practice, written submissions are sought from the public on draft standards before they are finalised.

Figure 4.2: Opportunities to participate in the development of rules and regulations

Rules and regulation	Application	Opportunity for public involvement
<i>Resource Management Act 1991</i>		
National Environmental Standards	Activities within catchments and the territorial sea	Must be adequate time and opportunity for public comment. Usually written submissions on the proposed standard will be sought
Regional coastal plan	Activities within the territorial sea of the region	Written submissions on proposed plan Written further submissions Presentation of submissions and evidence to council hearing
Regional plan	Activities within the catchments of the region	Appeal to Environment Court Presentation of submissions and evidence to Environment Court hearing
District plan	Activities within the district	Appeal to High Court on points of law
<i>Fisheries Act 1996</i>		
Sustainability measures	Fish stock	<i>No statutory rights to participate.</i> Minister must consult with persons or organisations representative of those classes of persons having an interest in the stock or effects of fishing on the aquatic environment in the area concerned including Māori, environmental, commercial and recreational interests. In practice, written submissions are sought on an initial advice paper

Total allowable catch	Fish stock	<i>No statutory rights to participate.</i> Minister must consult with persons or organisations representative of those classes of persons having an interest in the stock or effects of fishing on the aquatic environment in the area concerned including Māori, environmental, commercial and recreational interests. In practice, written submissions are sought on an initial advice paper
Total allowable commercial catch	Fish stock	<i>No statutory rights to participate.</i> Minister must consult with persons or organisations representative of those classes of persons having an interest in the setting of the TACC including Māori, environmental, commercial and recreational interests. In practice, written submissions are sought on an initial advice paper.
Mātaihai reserves	Spatial marine area	Written submissions from local community and people having a fishing interest in the area of the proposed reserve Local community consultative meeting
Taiāpure	Spatial marine area	Written submissions on proposed taiāpure Presentation of submissions to judge of Māori Land Court Appeal to the High Court on points of law
<i>Maritime Transport Act 1994</i>		
Marine Protection Rules	Vessels and/or offshore installations	Written submissions on the proposed rule
<i>Marine Mammals Protection Act 1978</i>		
Marine mammal sanctuaries	Spatial marine area	Written submissions on sanctuary after it has been notified by the Minister
<i>Marine Reserves Act 1971</i>		
Marine reserves	Spatial marine area	Written objections on the proposal to establish a marine reserve
<i>Crown Minerals Act 1991</i>		
Minerals programme	Individual minerals	Written submissions on draft minerals programme
<i>Biosecurity Act 1993</i>		
Import health standard	International vessels	Consultation with persons who are representative of classes of persons having an interest in the standard. In practice, written submissions are sought from the public on draft standards

Deciding individual proposals

Resource consents

The opportunities to become involved in decisions about individual proposals for activities in the marine area are much more limited than the opportunities provided for in the development of policies and rules.

Under the RMA, there are broad rights of participation in resource consents, when they are publicly notified, and these usually include being heard at a council-initiated hearing, appealing the council's decision to the Environment Court and being heard again, and appealing the Environment Court's decision to the High Court on points of law only.

Matters can be directly referred to the Environment Court for decision on the request of the applicant. In this case there will be no hearing by the council in the first instance. The decision of the Environment Court can be appealed to the High Court on points of law only.

Matters of national importance can be 'called-in' by the Minister of Environment (or Minister for Conservation for applications within the coastal marine area). In this case there will also be no council hearing in the first instance, but the matter will be processed by the Environmental Protection Authority and determined either by the Environment Court or by a board of inquiry appointed by the Minister after hearing the applicant and submitters. Decisions can be appealed to the High Court on points of law only.

There are no appeal rights against a council's decision not to notify a resource consent and the only redress is through initiating judicial review proceedings in the High Court. In cases where the council decides not to publicly notify the application, it may still be given limited notification. This means that parties who are directly affected by the proposal, and who have not given consent, will have the opportunity to lodge a submission and subsequent appeals.

Dumping permits

Under the Maritime Transport Act 1994, a permit must be obtained from the Director of Maritime New Zealand, before dumping waste into the EEZ or the continental shelf area beyond (section 261). The process to apply for a permit is set out in Part 180 of the Marine Protection Rules and requires applications to be publicly notified if they have more than a minor adverse effect on the marine environment or are clearly in breach of the London Convention.

When an application is publicly notified, interested parties can make written submissions and the Director must have regard to the submissions when making

a decision on the application. There is no requirement for a public hearing and no appeal rights against decisions not to notify or to grant applications. The only redress is through initiating judicial review proceedings in the High Court.

Prospecting, exploration and mining permits

There are currently no statutory public participation rights in decisions made on applications for prospecting, exploration or mining permits within the EEZ, although there may be opportunities to participate in RMA process for related on-shore mining facilities and activities.

Figure 4.3 Opportunities to participate in the determination of individual applications

Application		Opportunity for public involvement
Resource consent	Activity within the catchment or territorial sea	<i>Statutory right to participate only where application publicly notified</i> Written submissions on application Presentation of submissions and evidence to council hearing (where no direct referral or call-in) Appeal to Environment Court Presentation of submissions and evidence to Environment Court hearing (or board or inquiry for some call-in)s Appeal to High Court on points of law
Dumping permit	Dumping waste within the EEZ or continental shelf area beyond	<i>Statutory right to participate only where application publicly notified</i> Written submissions on application
Prospecting, exploration and mining permits	Mining activity within the EEZ or continental shelf area beyond	<i>No statutory rights to participate</i>

Publicly initiated processes

In most cases, the initiation of a process to develop new policy or rules or to amend existing instruments applying to the marine environment, is left to the statutory authority concerned. Members of the public will need to persuade the relevant Minister or agency that a change is needed. There are, however, two statutory processes that can be initiated by any person, a private plan change under the RMA and establishment of a *taiāpure* under the Fisheries Act 1996.

Private plan change

Any person can seek a change to a regional or district plan, including a regional coastal plan. Such a request can only be made, however, when the plan is operative. The council can only reject the request if:

- It is frivolous or vexatious
- The local authority or the Environment Court has considered the substance of the request within the last two years
- The request is not in accordance with sound resource management practice
- The request would make the plan inconsistent with the provisions of the RMA
- The policy statement or plan has been operative for less than two years.

If none of these criteria apply the council must process the private plan change, and has the option of adopting the plan change as though it was initiated by the council.

Requesting a private plan change can be an expensive exercise as the application will need to include an assessment of environmental effects of the proposal and an assessment of its benefits and costs. Preparing these assessments is likely to require the assistance of environmental professionals. In addition, the council can charge the applicant the full costs of processing the proposed plan change including the costs of the submission process and hearing.

Taiāpure

The second statutory process that can theoretically be initiated by any member of the public is a proposal to establish a taiāpure under the Fisheries Act 1996. The application is lodged with the Minister of Fisheries in the first instance. Support, however, is required from the Minister after consulting with the Minister of Māori Affairs, before the proposal can proceed. The purpose of a taiāpure is to make ‘better provision for the recognition of rangatiratanga and of the right secured in relation to fisheries by Article II of the Treaty of Waitangi’ (section 174, Fisheries Act). This means that, in practice, a person would need to be acting on behalf of an iwi or hapū that had an Article II right in order to succeed.

Reporting offences

Another way in which members of the public can initiate action to improve the health of the marine environment, is through reporting issues of concern

to the relevant authorities, particularly where there is suspicion of an offence being committed. Before reporting the issue, it is useful to record as much relevant information as possible, including taking photographs or filming a video. Most authorities have hotlines and after hours numbers and these are often prominently displayed on their websites.

- Pollution incidents and suspected illegal structures in the marine area should be reported to the regional council
- Suspected illegal fishing or shell fish harvesting outside marine reserves should be reported to the Ministry of Fisheries
- Suspected illegal fishing or shell fish harvesting inside marine reserves should be reported to the Department of Conservation
- Harm to protected species such as marine mammals and seabirds should be reported to the Department of Conservation
- Suspected new alien marine species should be reported to Biosecurity New Zealand

Judicial review

Judicial review involves challenging the exercise of a statutory discretion. The challenge cannot be based on the merits of the decision, but only on the basis that it was 'unlawful', that it was unreasonable or that the decision making process did not follow the proper procedure. Judicial review has been used to challenge decisions such as the setting of the TACC under the Fisheries Act and the failure to notify resource consents under the RMA.

In order to overturn a decision on the basis that it is unlawful, it will normally be necessary to establish one of the following grounds:¹

- The decision was motivated by an improper purpose, which was not contemplated by the legislation
- The decision-maker took into account irrelevant matters
- The decision-maker failed to take into account relevant matters that were mandatory under the legislation
- The decision was made under an invalid delegation
- The decision-maker made an error of law
- The decision-maker had insufficient information on which to base the decision

It is also possible to challenge a decision on the basis that it was unreasonable. This is a hard test to meet and involves establishing that the decision was 'so unreasonable that no reasonable [decision-maker] could ever come to it' (known as the *Wednesbury* test).

Another ground on which a decision may be set aside is on the basis that the procedure which the decision-maker followed did not accord with the principles of natural justice, including acting in good faith and listening fairly to both sides.

You will need a lawyer to assist with initiating judicial review proceedings.

References and further information

Websites

Department of Conservation – <http://www.doc.govt.nz>

Environmental Protection Authority – www.epa.govt.nz – to see what called-in resource consents are open for submission

Maritime New Zealand – <http://www.maritimenz.govt.nz> – to see what matters are open for consultation

Ministry of Fisheries – <http://www.fish.govt.nz> – to see what processes are open for consultation

Resource Management Act Guide – <http://www.rmaguide.org.nz>

Publications

Kirman C and C Whata, 2005, 'Environmental litigation and dispute resolution, in Nolan D (ed), *Environmental and resource management law*, 3rd edition, LexisNexis, Wellington

Peart R, 2008, *The New Zealanders Guide to the Resource Management Act 1991*, 3rd edition, Craig Potton Publishing, Nelson

1 Kirman C and C Whata, 2005, 1024



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Marine Protected Areas

Overview

An important tool to help preserve marine biodiversity, and to maintain the overall health of the marine environment, is the identification of spatial marine areas which are managed under a protective regime. Such areas commonly fall under the umbrella term 'marine protected areas'. This term is not currently defined in statute in New Zealand. However, the 2005 *Marine protected areas policy statement and implementation plan* prepared jointly by the Department of Conservation and the Ministry of Fisheries, has defined a marine protected area as being:

An area of the marine environment especially dedicated to, or achieving, through adequate protection, the maintenance and/or recovery of biological diversity at the habitat and ecosystem level in a healthy functioning state.¹

Establishing individual marine protected areas can result in significant improvements in the health of the marine environment within the protected area itself. It can also benefit nearby areas as a result of the 'spillover' of marine life, which travels outside the boundaries of the protected area. Much greater benefits are likely to be achieved when there is a 'network' of protected areas, which cover a wide range of habitat types and which encompass the life cycle patterns of important species. Such a network should also be complemented by good management of activities across the entire marine environment.

The objective of establishing a network of marine protected areas has been adopted by the New Zealand government. In 2000 the government released the New Zealand Biodiversity Strategy in order to fulfil, in part, the country's commitments under the Convention on Biological Diversity. As an action to achieve the objective of protecting marine habitats and ecosystems (objective 3.6), the Biodiversity Strategy identifies (para (b)):

Achieve a target of protecting 10 percent of New Zealand's marine environment by 2010 in view of establishing a network of representative protected marine areas.

Marine protected areas may include anything from a fully protected area where all potentially harmful activities are excluded such as fishing, mining, aquaculture and dredging, to areas that are the subject of varying degrees of protective regulation, such as excluding trawling, dredging and dumping.

Under New Zealand's legislative framework there are many types of special marine areas, which can be created under statute and which may constitute marine protected areas. These include marine reserves, marine mammal sanctuaries, areas closed to fisheries, marine parks, areas established under

special legislation, areas of significant conservation value, areas to be avoided by shipping and cable exclusion areas. These mechanisms are summarised in figure 5.1. and described in more detail in the following sections.

The *Marine Protected Areas Policy Statement and Implementation Plan* provides that in order to meet the requirements for a MPA, a management tool must (planning principle 2):

Enable the maintenance or recovery of the site's biological diversity at the habitat and ecosystem level to a healthy functioning state. In particular, the management regime must provide for the maintenance and recovery at the site of:

- Physical features and biogenic structures that support biodiversity
- Ecological systems, natural species composition (including all life-history stages), and trophic linkages
- Potential for the biodiversity to adapt and recover in response to perturbation

Traditionally, Māori used the temporary closures of rāhui to manage fisheries resources. While there is some consistency between the concept of rāhui and that of a marine protected area, the permanent ban on harvest in fully protected marine reserves can conflict with the exercise of traditional customary fishing rights.

In order to provide a more strategic approach to identifying and creating the network of representative protected marine areas identified in the Biodiversity Strategy, the Department of Conservation and Ministry of Fisheries have been working together to progress the *Marine protected areas policy statement and implementation plan*.

Figure 5.1: Overview of mechanisms to spatially protect marine areas

Type of protection	Key features
Marine reserves <i>Marine Reserves Act 1971</i>	Purpose is for ‘preserving them [areas of the sea] in their natural state as the habitat of marine life for scientific study’ Cannot be created outside the territorial sea A range of parties can apply to create a marine reserve but the general public can not Can only be created if the Ministers of Conservation, Fisheries and Transport agree Created by the Governor General by Order-in-Council (meaning it must be approved by cabinet) Provides a high level of protection – all fishing usually excluded Managed by the Department of Conservation
Marine mammal sanctuaries <i>Marine Mammals Protection Act 1978</i>	Purpose is the protection, conservation, and management of marine mammals (whales, dolphins, seals) Can be created within the territorial sea and EEZ Created by the Minister of Conservation by <i>Gazette</i> notice. Requires the consent of other Ministers if they control any Crown-owned land, foreshore, seabed, or waters of the sea May specify activities that may or may not be engaged in within the sanctuary, and may impose restrictions in respect of the sanctuary Managed by the Department of Conservation
Area-based fisheries restrictions <i>Fisheries Act 1996</i>	Purpose is to provide for the utilisation of fisheries resources while ensuring sustainability Can be created within the territorial sea and EEZ Created by the Minister of Fisheries by <i>Gazette</i> notice or Governor-General by Order-in-Council Can restrict or prohibit certain types of fishing activity in order to avoid, remedy or mitigate any adverse effects of fishing on the aquatic environment Managed by the Ministry of Fisheries
Areas of significant conservation value <i>Resource Management Act 1991</i>	Can cover any area within the territorial sea Created by inclusion of the area in a regional coastal plan Managed by the regional council which enforces rules in the regional coastal plan and conditions on resource consents
Areas to be avoided by shipping <i>Maritime Transport Act 1994</i>	Purpose is to protect sensitive marine environments from the risks, principally of marine oil spills, posed by shipping operations Excludes ships in transit from entering defined marine areas Draft rules are initiated by Maritime New Zealand and approved by the Minister of Transport Rules are enforced by Maritime New Zealand
Protected areas <i>Submarine Cables and Pipelines Protection Act 1996</i>	Purpose is to protect submarine cables and pipelines Excludes fishing and anchoring within the protected area Enforced by Maritime New Zealand
Special legislation	Can be tailored to specific issues and conditions Initiated by Member of Parliament (usually the majority party) Approved by Parliament

Mechanisms to protect marine areas

Marine reserves

Highly protected areas, called marine reserves, can be established under the Marine Reserves Act 1971. These can currently only be established within the territorial sea.

The purpose of establishing marine reserves under the Marine Reserves Act focuses on scientific study rather than preserving biodiversity. The long title of the Act states that it is 'An Act to provide for the setting up and management of areas of the sea and foreshore as marine reserves for the purpose of preserving them in their natural state as the habitat of marine life for scientific study.'

Marine reserves are managed by DoC with the aim of maintaining them as far as possible in their natural state. This means that they are usually 'no-take' areas. For example, within the Whangarei harbour marine reserve, the following activities are prohibited:²

- Fishing and the taking or disturbing of any marine life, including shellfish, seaweeds and sea urchin
- Taking any part of the seafloor, including rocks and shells
- Erecting any structure

Although non-commercial fishing can legally be permitted in a marine reserve, it is only currently permitted within a small part of the Kapiti Marine Reserve and is restricted to whitebaiting. Mining activity is not necessarily excluded from marine reserves although it is not currently undertaken in any of them.

An application to establish a marine reserve can only be made by a limited category of persons including a university, a body administering a reserve adjacent to the coast, an organisation involved in the study of marine life and natural history, iwi or hapū with tangata whenua status over the area and the Director-General of Conservation (section 5(a)).

Currently, marine reserves must be approved by the Minister of Conservation, with the concurrence of the Ministers of Fisheries and Transport, before they can be established through Order-in-Council (section 5(9)). A Marine Reserves Protocol has been developed to guide the interactions between DoC and the Ministry of Fisheries when considering marine reserve applications.

A marine reserve may not be established if it would (section 5(8)):

- Unduly interfere with any interest in land in or adjoining the reserve

- Interfere unduly with any existing right of navigation
- Interfere unduly with commercial fishing (within the broader marine area – commercial fishing will almost always be excluded from the marine reserve area)
- Interfere unduly with or adversely affect any existing usage of the area for recreational purposes
- Otherwise be contrary to the public interest.

The process to establish a marine reserve is as follows:

- An application to create a marine reserve is lodged with the Director-General of Conservation
- Notice of intention to apply to create a marine reserve must be publicly notified in a range of newspapers
- Notice in writing of the application must also be given to owners of land adjoining the proposed reserve and other public bodies
- Members of the public have two months in which to lodge a written objection to the proposal with the Director-General of Conservation
- The applicant is given three months to answer the objections in writing
- The Director-General forwards the objections and answers received to the Minister of Conservation
- The Minister of Conservation decides whether any objection should be upheld. If the objection is upheld the marine reserve cannot be declared.
- The Minister of Conservation then decides if the reserve would be in the best interests of scientific study, if it would be for the benefit of the public and if it would be expedient. There are no appeal rights against the Minister's decision although it can be judicially reviewed.
- If the Minister decides to create the marine reserve, concurrence is sought from the Ministers of Transport and Fisheries
- If concurrence is granted the Minister of Conservation recommends that the Governor-General makes an Order-in-Council creating the reserve

A list of current and proposed marine reserves, and information about them, can be found at <http://www.doc.govt.nz/conservation/marine-and-coastal/marine-protected-areas/marine-reserve-information/>

Marine mammal sanctuaries

Marine mammal sanctuaries can be established under the Marine Mammals Protection Act 1978. The purpose of the Act as set out in the long title is the 'protection, conservation, and management of marine mammals within

New Zealand and within New Zealand fisheries waters'. The sanctuaries are managed by DoC.

Marine mammal sanctuaries are created by the Minister of Conservation declaring the sanctuary by notice in the *Gazette*. The consent of other Ministers must also be obtained if they control any Crown-owned land, foreshore, seabed, or seawater. Within 28 days after publication of the notice members of the public may lodge written submissions with the Minister who may vary, redefine or abolish the sanctuary after considering the submissions (section 22). The Minister may specify restrictions on activities that may take place in the sanctuary at the time the declaration is made.

Area-based restrictions on fishing activity

Areas can be closed to specified types of fishing activity by the Minister of Fisheries, under the sustainability provisions of the Fisheries Act 1996, after taking into account the effects of fishing on any stock and the aquatic environment. Before making a decision to restrict or prohibit fishing in an area the Minister must consult with stakeholders (section 12) and take into account any conservation or fisheries services and any relevant fisheries plan (section 11(2A)).

The process to impose a restriction on fishing activity will normally include the following steps:

- The Ministry of Fisheries will prepare an Initial Position Paper
- The paper is publicly released and written submissions called for. Any person is able to make a written submission
- Once the submission period has closed the Ministry summarises and considers the written submissions and prepares a Final Advice Paper
- The Final Advice Paper is forwarded to the Minister who makes a final decision. The decision is recorded in a short letter which sets out the decision and the reasons for it.
- The decision is then given effect to, either through a notice in the *Gazette* or by regulations promulgated under section 298 of the Fisheries Act

There are no appeal rights, but the decision can be judicially reviewed in the High Court.

Numerous area-based restrictions have historically been placed on fishing activities, particularly in harbours and coastal areas. Many of these were not originally intended to provide marine biodiversity benefits, but were put in place to deal with issues such as conflicts between recreational and commercial fishers. They may, however, in practice result in improvements in biodiversity.

More recently, restrictions under the Fisheries Act have closed 17 areas around and over seamounts and 17 large benthic protection areas within the EEZ to dredging and trawling activity as a result of biodiversity concerns.

Marine parks

Two marine parks were established under the now repealed Harbours Act 1950 and regulations under the Fisheries Act 1996. These are the Tawharanui Marine Park in the northern part of the Auckland region and the Mimiwhangata Marine Park in the southern part of the Northland region. The fisheries regulations, which apply to the parks, prohibit the taking of all marine life in the Tawharanui Marine Park and exclude all commercial and some recreational fishing within the Mimiwhangata Marine Park. The Auckland Regional Council has lodged an application with the Director-General of Conservation seeking to establish the Tawharanui Marine Park as a marine reserve with minor boundary changes.

Areas protected by special legislation

Three areas are protected by special legislation, which is tailored to each case and does not have more general application. The Sugar Loaf Islands Marine Protected Area Act 1991 establishes the Sugar Loaf Islands Marine Protected Area covering an area of eight square kilometres. The protected area is managed by DoC. Mining is prohibited in the protected area and there are also restrictions on fishing activity under the Fisheries Act.

The Hauraki Gulf Marine Park Act 2000 (HGMPA), amongst other things, establishes a Hauraki Gulf Marine Park which primarily consists of reserve land, Ramsar wetlands, foreshore and seabed owned by the Crown, marine reserves and seawater (but not marine life). Provisions in the legislation provide for council-owned and privately-owned reserve land to be added to the marine park with the consent of the owner and some private land has been added. Taiāpure and mātaihai reserves may also be added to the park with the consent of the Minister of Fisheries and the committee of management or tangata whenua.

Anyone administering an area within the park must recognise and give effect to the purpose of the park. The purposes of the marine park are (section 32, HGMPA):

- (a) to recognise and protect in perpetuity the international and national significance of the land and the natural and historic resources within the Park
- (b) to protect in perpetuity and for the benefit, use, and enjoyment of the people and communities of the Gulf and New Zealand, the natural and historic resources of the Park including scenery, ecological systems, or

natural features that are so beautiful, unique, or scientifically important to be of national significance, for their intrinsic worth

- (c) to recognise and have particular regard to the historic, traditional, cultural, and spiritual relationship of tangata whenua with the Hauraki Gulf, its islands and coastal areas, and the natural and historic resources of the Park
- (d) to sustain the life-supporting capacity of the soil, air, water, and ecosystems of the Gulf in the Park

The HGMPA also establishes the Hauraki Gulf Forum and provides a set of management objectives which apply to the Gulf's coastal marine area, islands and catchments. These are described more fully in Chapter Two.

The Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 establishes a Fiordland (Te Moana o Atawhenua) Marine Area and establishes the Fiordland Marine Guardians which has an advisory and facilitative role, amongst other things. Members of the Guardians are appointed by the Minister for the Environment with a requirement that one member be nominated by Te Rūnanga o Ngāi Tahu, that at least five members are ordinarily resident in the Otago or Southland regions and that the membership reflects a balanced mix of knowledge and experience in relation to the Marine Area (section 15). Unlike the situation with the Hauraki Gulf, the Fiordland legislation does not establish any common management objectives for the marine area.

Areas of significant conservation value

Areas of significant conservation value are sometimes spatially identified in regional coastal plans under the Resource Management Act 1991 and measures such as protective rules to ensure that these areas are not adversely affected, are specified. Usually activities such as aquaculture, marinas and moorings, reclamations and building structures are to be avoided.

Fishing methods, which have an impact on the marine environment, can potentially be restricted in regional coastal plans in order to promote sustainable management. Regional coastal plans cannot, however, 'control the harvesting or enhancement of aquatic organisms for the purpose of conserving, using, enhancing, or developing any fisheries resources controlled under the Fisheries Act 1996' (section 30(3)).

Such provisions are included in plans after a public process, which involves the public notification of the proposed plan provisions, opportunity for written public submissions, opportunity for further submissions and then a public hearing. Decisions of the council on submissions can be appealed to the Environment Court.

It is also possible for any person to lodge a private plan change to insert provisions into regional coastal plans identifying significant conservation areas and providing for their protection. Private plan changes go through a similar process, although they can be rejected by the council on a number of grounds.

Areas to be avoided by shipping

Areas to be avoided by shipping can be established under the Maritime Transport Act 1994 through marine protection rules within the umbrella of the International Maritime Organisation. There are three categories of such areas: recommended areas to be avoided; mandatory areas to be avoided; and particularly sensitive areas.

New Zealand has mandatory areas to be avoided around the Three Kings Islands and Poor Knights Islands, which means that ships in transit are prohibited from entering the areas. In the case of the Poor Knights Islands the restriction applies to every ship of more than 45 metres in length except for fishing vessels and barges under tow (so long as their cargo does not include oil or other harmful liquid substances). The restrictions around the Three Kings Islands apply to all ships of 500 gross tonnes or more.

The following process is followed to develop a new marine protection rule, or to amend an existing rule:

- Maritime New Zealand prepares a draft rule and undertakes informal consultation
- An invitation to provide written comment on the rule is advertised widely to the maritime community and general public
- Maritime New Zealand considers the written comments and prepares a revised rule which may be circulated to key stakeholders for comment before finalisation
- The Minister of Transport approves the rule and it is notified in the *Gazette* (section 448).

Cable protected areas

Protected areas can be declared under section 12 of the Submarine Cables and Pipelines Protection Act 1996. All fishing operations and anchoring are prohibited in such areas. This exclusion is not for environmental purposes, but is intended to protect cables and pipelines from damage. However, it may have the indirect effect of restricting activities which could also impact on biodiversity within the area.

Marine protected areas policy

In 2005, the Department of Conservation and Ministry of Fisheries formally released a policy statement and implementation plan for marine protected areas. The purpose of the policy is to 'protect marine biodiversity by establishing a network of MPAs [marine protected areas] that is comprehensive and representative of New Zealand's marine habitats and ecosystems'. The Ministry of Fisheries and the Department of Conservation are jointly responsible for implementing the policy.

A coastal and marine classification system, which identifies different types of marine habitat, is being used to help ensure that the network of marine protected areas is comprehensive and representative. Fourteen biogeographic regions have been identified, which collectively cover the entire New Zealand marine area out to the 200 metre depth contour. They are intended to reflect major different coastal biological patterns. These regions have been further classified, through the application of layering criteria, into 44 different types of habitat. The layering used to identify these areas included:

- Biogeographic region (14 regions)
- Environment (estuarine or marine)
- Depth (intertidal, shallow subtidal, deep subtidal)
- Exposure (low, medium, high)
- Substrate type (mud, sand, gravel, cobble, boulder, rocky, biogenic reef)

The policy takes a regional approach to the planning and establishment of a network of protected areas around New Zealand, although it is not aligned with regional council boundaries.

As part of the agreement with the fishing industry to establish the 17 benthic protection areas in the EEZ, the government will not implement the marine protected areas policy within the EEZ until 2013. It is anticipated that planning in this area will be undertaken by an expert panel. Current implementation efforts are therefore concentrated on the territorial sea.

Each of the biogeographic regions is to be the focus of a planning process. Planning will be undertaken by community-based Marine Protection Planning Fora supported by staff from DoC and the Ministry of Fisheries. The fora will have up to 14 members. Expressions of interest in becoming a member of the fora will be sought through advertising and directly approaching relevant stakeholders and marine user groups. The fora will include representatives from:

- Tangata whenua
- Commercial fishers
- Recreational users including fishers, charter fishers and divers
- Conservation groups
- Tourism
- Aquaculture industry
- Marine science
- Minerals industry

The fora will be tasked with identifying potential sites and tools for area-based protection of biodiversity. They will be expected to seek consensus on proposed areas and to make recommendations to the Ministers of Conservation and Fisheries. Fora will be expected to constructively involve and engage stakeholders in each region and will be required to seek written submissions on any recommendations being made. It is expected that each forum will produce a set of recommendations within 18 months of being appointed. The process is shown in Figure 5.3

The *Marine protected areas: classification, protection standard and implementation guidelines*, prepared by the Ministry of Fisheries and the Department of Conservation provide design guidelines to assist the fora in identifying and selecting potential protected areas. These include guidance on identifying potential protected areas, on selecting candidate areas from among potential sites and for selecting appropriate tools for protecting the areas. In summary they include the following:

Identifying potential MPAs

- Protect whole habitats and ecosystems
- Be of sufficient size to provide for the maintenance of populations
Fewer, larger areas are to be preferred than numerous smaller areas
- Maximise and enhance linkages among individual and groups of protected areas
- Include differences in habitats and ecosystems that cover both latitudinal and longitudinal or cross-shelf ranges (i.e. from the intertidal zone to deep offshore waters)
- Consider adjacent uses which may affect the protected area
- Use simple boundaries and minimise boundary to area ratio

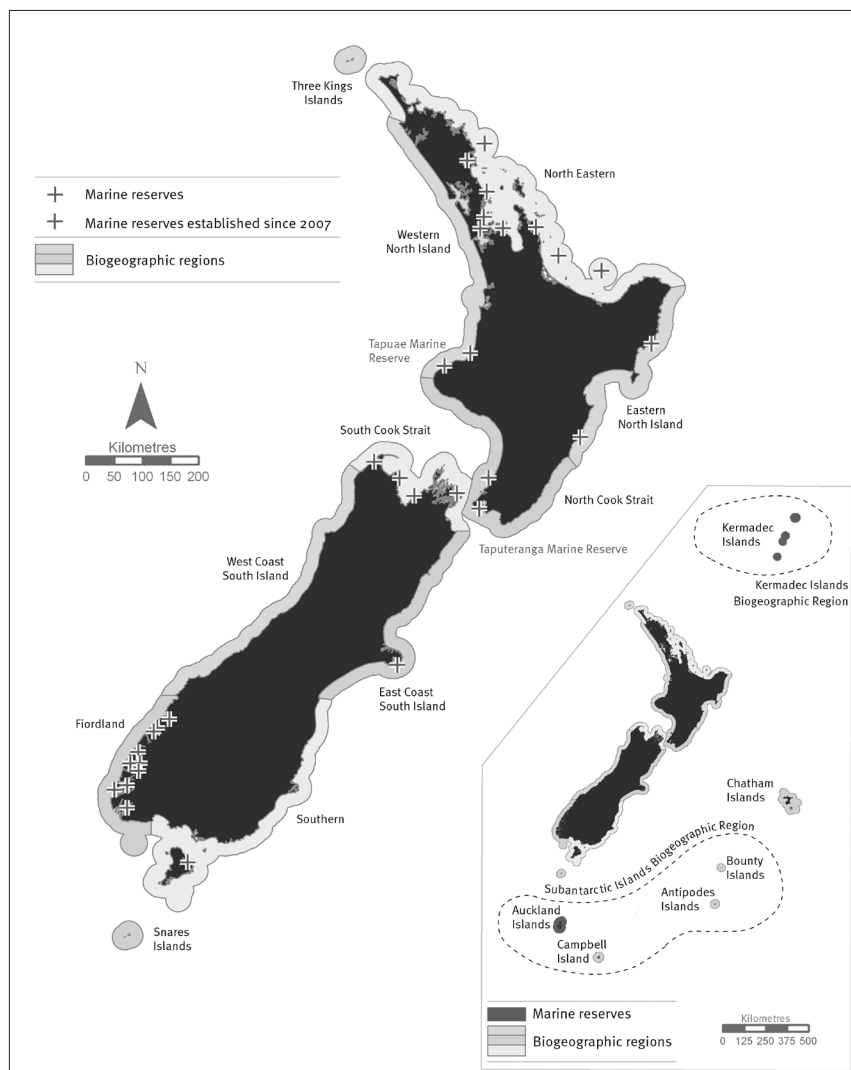


Figure 5.2: New Zealand's biogeographic regions

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Figure 5.3 Process to prepare marine protection recommendations

Phase	Tasks	Timeframe
One	Preparatory work by DoC and MFish Appointment of chair and members of the Forum	Six months
	Forum convened Consultation with community	Six months
Two	Review information Develop recommendations	Six months
Three	Public consultation on recommendations Finalise report to Ministers	Six months

Selecting candidate MPAs

Primary considerations

- Protect the full range of marine habitats and ecosystems
- Consider information on cultural use and values and Treaty settlement obligations
- Consider information on social and economic interests and users
- Minimise adverse impacts on existing users and Treaty settlement obligations

Secondary considerations

- Protect multiple habitats within each protected area
- Have fewer, larger rather than numerous smaller protected areas
- Consider activities such coastal structures, dredging or dumping sites which may negatively impact on the site
- Align, where possible with other protected areas
- Provide, where possible, replication of habitats and ecosystems

At the time of writing two fora had been convened for the West Coast South Island and Subantarctic biogeographic regions. The timetable for progressing the development of marine protection plans in other areas was uncertain.

Elements of best practice

The IUCN World Commission on Protected Areas has recently produced a document that represents a synthesis of the expertise, knowledge and views of world-leading experts in marine protected area network design and implementation. The following elements of best practice have been drawn from this document.

The Commission estimates that, in order to meet all fishery and conservation goals, networks of fully protected reserves should cover 20 per cent or more of all biogeographic regions and habitats. To maximise its effectiveness, a network needs to include a group of marine protected areas that are carefully designed to function collectively and synergistically as an ecological network. This includes the following design considerations:³

- Does the MPA network protect some of all the habitat types and habitats linked by species life cycle patterns found in a biogeographic region?
- Does the MPA network include both transition zones between biogeographic areas and core zones within each biogeographic region?
- Does the MPA network include representation of ecosystem processes as well as habitat types?
- Does each MPA include a variety of depths and transition zones and areas which cover the life-stage movements of species?
- Are the MPAs large enough to be effective, self-sustaining ecological units? Do they protect and maintain ecological processes such as nutrient flows and food-web interactions? Are they large enough for populations to be self-seeding?
- Does the MPA network include replicates of each representative habitat? It is recommended that at a minimum three replicates of each habitat type should be included.
- Does the MPA network include ecologically significant areas such as rocky reefs, oyster reefs, mussel beds, bryozoan reefs, salt marshes, seagrass beds and mangroves?
- Does the MPA network include unique habitats?
- Does the MPA network protect spawning aggregations of fish, egg sources and nursery areas?
- Does the MPA network protect roosting sites, foraging grounds, socialising areas and migratory routes for marine mammals and seabirds?
- Does the MPA protect source populations including larger, older individuals who have a greater reproductive capacity?

- Are the MPAs spaced so that larvae from as many species as possible can reach other MPAs with appropriate habitat?
- Are the MPAs in the network located so that they are connected by the exchange of eggs, larvae recruits, other propagules, juveniles and adults?
- Are individual MPAs large enough to cover a range of movement patterns of adults, juvenile fish and invertebrates and to include larval dispersal distances?
- Are MPAs located in a wide variety of places in relation to prevailing currents to facilitate larval dispersal?

References and further information

Websites

The Department of Conservation marine protected area webpages at <http://www.doc.govt.nz/conservation/marine-and-coastal/marine-protected-areas>

The Ministry of Fisheries webpages on taiāpure and mātaītai reserves at <http://www.fish.govt.nz/en-nz/Maori/default.htm>

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- 1 Department of Conservation and Ministry of Fisheries, 2005, 8
 - 2 <http://www.doc.govt.nz/conservation/marine-and-coastal/marine-protected-areas/marine-reserves-a-z/whangarei-harbour/facts/marine-reserve-rules/>
 - 3 These design considerations have been synthesised from IUCN World Commission on Protected Areas (2008),

PART TWO

ACTIVITIES IMPACTING ON THE MARINE ENVIRONMENT



six

Fishing

Overview

The commercial fishing industry exports around \$1.3 billion of seafood a year and annually takes around 440,000 tonnes of fish from New Zealand's oceans. There are just under 1300 commercial fishing vessels operating within New Zealand waters, with the industry providing over 7000 full-time equivalent jobs. Commercial fishers target 130 different species, which include sharks, finfish, squid, shellfish, lobster, crabs, sea cucumbers and seaweeds.¹

The most valuable wild caught species is rock lobster, with \$180 million worth exported in 2008. This is followed by hoki (\$151 million) and squid (\$71 million). In terms of tonnage exported, squid is the leader with over 46,000 tonnes exported in 2007, followed by hoki (39,000 tonnes) and jack mackerel (33,000 tonnes).²

There are three main types of commercial fisheries within New Zealand's oceans. *Inshore fisheries* are those that take place over the continental shelf in relatively shallow waters up to 200 metres deep. They include target species such as paua, rock lobster, snapper, tarakihi and seaweed.

Deep-water fisheries take place within the deeper waters of the EEZ and include many of the most commercially valuable species such as squid, hoki, orange roughy, ling, hake and jack mackerel. Fishers also target *migratory species* that travel into and beyond New Zealand's waters including tuna and swordfish.³

Much of the deepwater commercial fishing effort is concentrated off the South Island. Around 60 per cent of the fish harvest comes from the Chatham Rise and Subantarctic areas, including the main hoki, hake, ling, silver warehou, squid, orange roughy and oreo fisheries. A further 30 per cent is caught off the country's west coast, mainly when the fish spawn there in winter and spring, including hake, hoki, ling and silver warehou.⁴

Just under 20 per cent of adult New Zealanders engage in non-commercial fishing each year, around 600,000 people.⁵ Although the total amount of seafood harvested by non-commercial fishers is unknown, it is significant, particularly for popular species such as snapper, blue cod and shellfish.

Methods

A wide range of methods are used to harvest seafood in New Zealand. Each method is described in the sections below.

Collecting



Diving



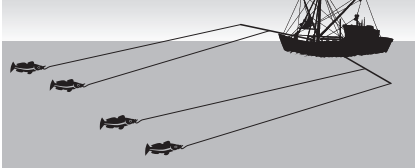
Squid jigging



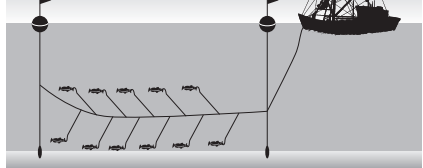
Trapping and potting



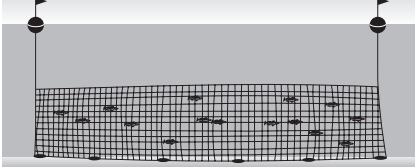
Trolling



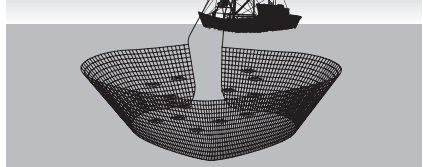
Long lining



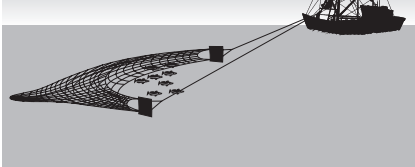
Gill netting



Purse Seining



Trawling



Dredging

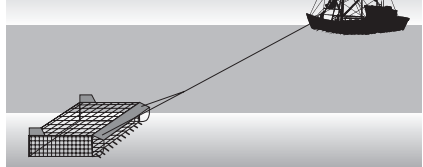


Figure 6.1 Sketches of fishing methods

Collecting and Diving

Divers harvest kina, paua and scallops by hand. Cockles are harvested by hand and by mechanical digging and raking.

Squid Jigging

Jigging for squid involves using powerful lights, which illuminate seawater at night, and attract squid to the area under the boat. Lures are attached at short intervals along lengths of line that are dropped over the side of the fishing boat. The lures are 'jigged' up and down by mechanically operated pulleys. These entice the squid which are caught on the hooks and hauled onboard.

Trapping and potting

Pots and traps are used to catch rock lobster, blue cod, octopus and crabs. They are usually constructed from steel mesh. A baited pot is lowered to the sea floor and its location marked by a line and float. The targeted species are attracted to the bait and enter the pot or trap. Once inside they are unable to escape. The pots are lifted by winch back onto the boat and the catch removed.

Trolling

Trolling is mainly used to catch species such as tuna and yellowtail kingfish. A boat slowly drags a set of hooked lures through the water. Weights can be attached to the lines so that they sink in order to target fish at lower depths.

Long lining

Long lining is used to catch high value inshore species such as snapper and groper; deepwater species such as ling and bluenose; and migratory species such as marlin and several species of tuna and shark. Long lining involves setting a long main line, which has numerous branch lines connected to it, each containing a baited hook.

Long lining can be used to target fish on the surface, in mid-water or on the bottom. Surface long lines are used to target tuna and other large schooling and migratory species. They can be many kilometres long and each line can have up to 3000 hooks. Surface long lines are held up by a series of floats. The line is gradually let out as the fishing boat slowly moves ahead. The entire process of setting and hauling in a surface long line can take up to 12 hours.

Bottom long lines are used to target flatfish, sharks, skates, ling and snapper. They are shorter than the surface lines and are attached to the sea floor with an anchor at one end and a weight at the other. The lines are left on the bottom for between six and 12 hours and are then hauled in.

Gill netting

Gill nets are used to catch a variety of inshore fish including flounder, rig shark, elephant fish, butterfish and yellow-eyed mullet. The nets are normally long and narrow with the bottom weighted down and the top held up with floats. Driftnets are left to drift in water currents and are usually set at or near the water surface. Setnets are anchored to the seafloor and are usually set near the bottom. The net effectively forms a wall in the water which fish swim into and get caught, often by their gills. Using a net with a large mesh size can help avoid small fish being caught.

Seining

Seining is a method where a net is used to encircle fish and capture them. Purse seining is used to catch fish that feed on the surface, such as tuna, trevally, kahawai or mackerel. When a school of fish has been sighted from the main fishing vessel, a small boat is launched which tows one end of the net around the school to encircle the fish. Ropes at the bottom of the net are drawn in to completely enclose the fish. The net is then pulled close to the larger boat and the fish are scooped or pumped into the hold.

Other variants on this method are beach seining and Danish seining. Beach seining, also called drag netting, is used to target mullet, flatfish, snapper, trevally and crabs. The beach seining net has long ropes attached to each end. One of the ropes is left on the beach and the net is then extended out into the sea and brought around in a semi-circle until the other rope reaches further down the beach. The net is then hauled in towards the shore. Danish seining is similar, but is undertaken in deep water with the net weighted down and dragged along the seafloor. It is used to target snapper and John Dory.

Trawling

Trawling is used to catch the greatest quantity of fish in New Zealand. This method is particularly important in deep sea fisheries. It involves one or two boats towing a net through the sea. Strong steel cables attach the net to the specially designed trawling boat. The mouth of the net is held open by two large doors. The fish enter the net through the mouth and are forced down to the smaller cod end. The mesh size helps to avoid catching small fish, although as the net fills with fish, the meshes can get blocked by fish already in the net.

Trawling can be undertaken in mid-water or along the sea floor. Bottom trawling is used to catch species such as orange roughy, ling, hake, squid and scampi. The net is often protected along the leading edge by chains or rollers,

which are dragged along the sea floor. Mid-water trawling is also used to catch a range of species including hoki, jack mackerel, squid and hake.

Dredging

Dredging is used to harvest oysters and scallops. It involves dropping a steel-toothed dredge to the sea floor and then dragging it across the sea bed. The width of separation between the teeth of the dredge can be adjusted to help avoid harvesting under-sized shell-fish.

Potential environmental effects

Fishing activity can impact on the marine environment in many ways. Some of these effects can be mitigated through careful fisheries management. The effects can be classified into four broad categories, which are described in more detail below:

- Impacts on target fish populations
- Impacts on non-target species
- Damage to seabed habitats
- Disruption of marine ecosystems

Impacts on target fish populations

Fishing activity can significantly reduce the population size of a targeted fish species, sometimes to the level where the stock has difficulty recovering. This is a particular problem with long-lived species, which reproduce slowly, such as orange roughy and sharks. Fishing also preferentially removes larger and older fish, changing the size and age structure of exploited populations and reducing their genetic diversity.

Impacts on non-target species

Some fishing methods result in the harvest of species that are not targeted by the fisher. Non-targeted fish species that are in the quota management system cannot generally be discarded at sea and must be landed. The application of deemed values, as described in chapter 2, is designed to discourage bycatch. In general, this is a more serious problem for fishing nets (including gillnets and trawling) than for hook and line methods and fish traps.

Of particular environmental concern is the capture of protected species such as marine mammals, seabirds and turtles in fishing equipment. Dolphins,

fur seals and sea lions can become entrapped in fishing nets, with some being released successfully and others drowning. Seabirds and turtles are attracted to baited hooks in long line fisheries and can get hooked or tangled in the line and drown. Others can become entangled in trawling equipment and gill nets. A high proportion of marine mammal and seabird species in New Zealand are vulnerable or endangered.

Damage to seabed habitats

Bottom trawling and dredging can destroy important benthic habitats. Heavy chains or rollers dragged along the seafloor crush many benthic organisms including sponges, bryozoans, seaweeds and soft corals. Trawling and dredging can flatten outcrops which provide refuge for juvenile species of fish. Such activities can also suspend sediment in the water column which can disrupt filter feeders.

Dredging is more destructive to the seabed in the localised areas where it occurs. Trawling is cumulatively more destructive within New Zealand's marine area as it is undertaken over much larger areas. Other methods which involve the use of equipment on the seabed can also damage the seabed including seining, long lining and potting, although the effects are less.

Disruption of marine ecosystems

The reduction of targeted fish populations may have significant ecological flow-on effects. These impacts are not well understood. Large scale purse seining of inshore species such as kahawai, mackerel and trevally may have reduced food for seabirds such as white-fronted terns and Huttons and fluttering shearwaters which typically feed in association with schooling fish. These larger schooling fish drive the smaller fish to the surface where they can be captured by these seabirds.

Squid is an important part of the diet of many bird species, toothed whales, dolphins, seals and sea lions. Harvesting large quantities of squid may have an impact on these species, but the effects are poorly understood.

Reductions in targeted fish populations can significantly impact on the entire food web. When top predators are removed, their prey species may increase in number, in turn increasing predation further down the food web. For example, because predators such as snapper and crayfish have been extensively fished in coastal areas around New Zealand, numbers of their prey kina (sea-egg) have increased dramatically. Kina graze on kelp, and their increase in numbers has resulted in some rocky reefs being stripped bare of their kelp forests –referred

to as 'kina barrens'. Damage to seabed habitats can also affect the wider marine ecosystem.

Potential effects of different fishing methods

Each fishing method can have very different impacts on the marine environment and these are described below. Where more than one fishing method is used to target the same species, use of the method with the least environmental impacts should be encouraged.

Hand collection has minimal impacts other than on the targeted species. This harvesting method enables only target species to be harvested and undersized specimens to be left undisturbed. Mechanical methods of collection are more indiscriminate and can impact on habitats and other organisms living on the sea floor.

Jigging is a relatively low impact fishing method that only catches the targeted species. If lines or jigs are lost at sea, they can become a hazard for other marine species, which may become entangled in the fishing equipment.

Potting is also a relatively low-impact fishing method, although fragile species located on the seafloor such as soft corals, sponges and bryozoans can be damaged if heavy pots are dropped on top of them. Undersized and non-target specimens can be returned to the sea relatively unharmed. Some seals and shags can drown as a result of being caught in rock lobster pots. Marine mammals can become entangled in pot lines.

Trolling is a low-impact and selective method of fishing. Few non-target species are caught and it has no impact on benthic habitats. Occasionally fur seals and seabirds can become hooked on the lures but this is thought to be a rare occurrence.

Long lining results in little damage to the fish caught, which can then be sold for higher prices. Long lines generally have little impact on sea floor habitats, although if lines and weights are dragged across the seafloor they can damage sensitive benthic organisms. The size and shape of the hook and type of bait can be tailored to attract specific species, thereby reducing bycatch. Because this method of fishing uses a large amount of bait, it can place pressure on stocks of bait fish if these are not managed sustainably.

Long lining can result in the death of seabirds, turtles and marine mammals which drown as a result of becoming caught on the hooks or entangled in the lines. Seabird deaths can be reduced if the lines are set at night when many species of birds are less likely to be actively feeding.

A device, which can help address this problem is a tori line, which consists of hanging coloured bands over the long lines when they are being set, so that the lines are more visible to the birds. Another approach is to run the line

through a plastic tube which extends from the back of the boat to the surface of the sea to prevent birds coming in contact with the hooks. Hook size and shape can also be varied to reduce impacts on seabirds and turtles. Other mitigation methods being developed include weighting lines so that they sink quickly, not disposing of offal when the lines are being set, and using sonic guns to scare away the birds.

The following restrictions currently apply to the use of long lines:

- All commercial fishers using surface long lines within the EEZ are required to carry and use streamer lines. In addition they are not permitted to set surface long lines from half an hour before dawn to half an hour after dusk unless they are weighted.
- Bottom long liners on boats seven metres long or greater are required to use streamer lines and all bottom long liners are also required to weight their line if it is set during daylight hours. In addition, there are restrictions around discharging offal or fish when setting bottom long lines.

Gill netting has little impact on benthic habitats. However it results in substantial by-catch of non-target species including dolphins, penguins, shags, shearwaters, seals, sharks and reef fish which get caught in the nets. Such nets are a particular problem for the endangered Hector's and critically endangered Maui's dolphins which live in the shallow coastal waters where nets are commonly set.

The most effective measure to reduce dolphin mortality is to avoid using set nets in coastal areas where the dolphins live. Experiments with acoustic alarms on nets overseas showed promising results initially, but have not proved to be practically applicable.

Set net prohibitions have been put in place along areas of the coast to protect the Hector's and Maui's dolphins, but do not extend over the dolphin's full range. The prohibitions around Banks Peninsula were put in place under the Marine Mammals Protection Act 1978 as part of the marine mammal sanctuary established in 1988. Further prohibitions have been put in place under the Fisheries Act 1996. At the time of writing, some of the more stringent set net prohibitions put in place during 2008 were under legal challenge by the Federation of Commercial Fishermen and others.

Seining has different impacts on the marine environment, depending on the type of seining adopted. Because purse seining targets schooling fish, it usually results in little bycatch, although dolphins are occasionally caught. It has little impact on benthic habitats as the net does not normally come in contact with the seafloor. Beach and Danish seining can be less selective than purse seining,

depending on the extent to which schools of fish are targeted. Danish seining can damage sensitive benthic habitats as a result of the weighted net and ropes being dragged across the sea floor.

Trawling is a relatively indiscriminate method of fishing and can result in a large amount of bycatch. Fur seals, sea lions, dolphins and seabirds can become entangled in the trawling equipment and drown. Fur seals and sea lions often deliberately swim into trawl nets to feed on the fish which have been caught.

In the squid trawl fishery, a Sea Lion Exclusion Device (SLED) has been used in all trawl nets in recent years. This consists of a grid that is placed across the net in front of the cod-end where the fish accumulate. The gaps in the grid are large enough for squid and other fish to swim through, but are too small for larger creatures such as dolphins, seals, sea lions, or turtles. The SLED is designed to enable marine mammals to escape through an open hatch, but they may sustain injuries or drown in the process. Since 1992 the Minister of Fisheries has set a fishery-related mortality limit for sea lions, and when it is estimated that the limit is reached, the squid fishery is closed for the year.

Other fishing practices which may reduce the capture of marine mammals in trawl fisheries include avoiding dumping offal when the net is being shot (launched) or hauled, leaving the area when marine mammals are sighted, shooting and hauling the net as quickly as possible, and recovering any accidentally lost fishing gear.

Since 2006, regulations have been in force requiring fishing boats over 28 metres in length to carry and deploy a seabird scaring device if using a trawl net, but currently no regulations apply to smaller inshore trawlers.

Bottom trawling can be particularly damaging to benthic habitats as a result of heavy nets and equipment being dragged along the sea floor. Delicate organisms such as sponges and corals can be crushed and the profile of the seafloor flattened. Bottom trawling can also suspend sediment in the water column, affecting filter feeders. These impacts can be reduced by ensuring that trawl equipment does not come into contact with the sea floor, by avoiding trawling in areas which have not already been impacted, and by excluding trawling activity from areas which have ecologically important benthic habitats such as seamounts. Within the EEZ, 17 areas around and over seamounts and 17 benthic protection areas have been closed to trawling to protect benthic habitats.

Dredging can also be harmful to benthic habitats. The heavy dredge scrapes the seafloor, crushing benthic organisms and flattening the seabed profile. It also suspends sediment in the water column, which reduces light levels, smothers remaining benthic organisms, and clogs up the apparatus of filter feeders.

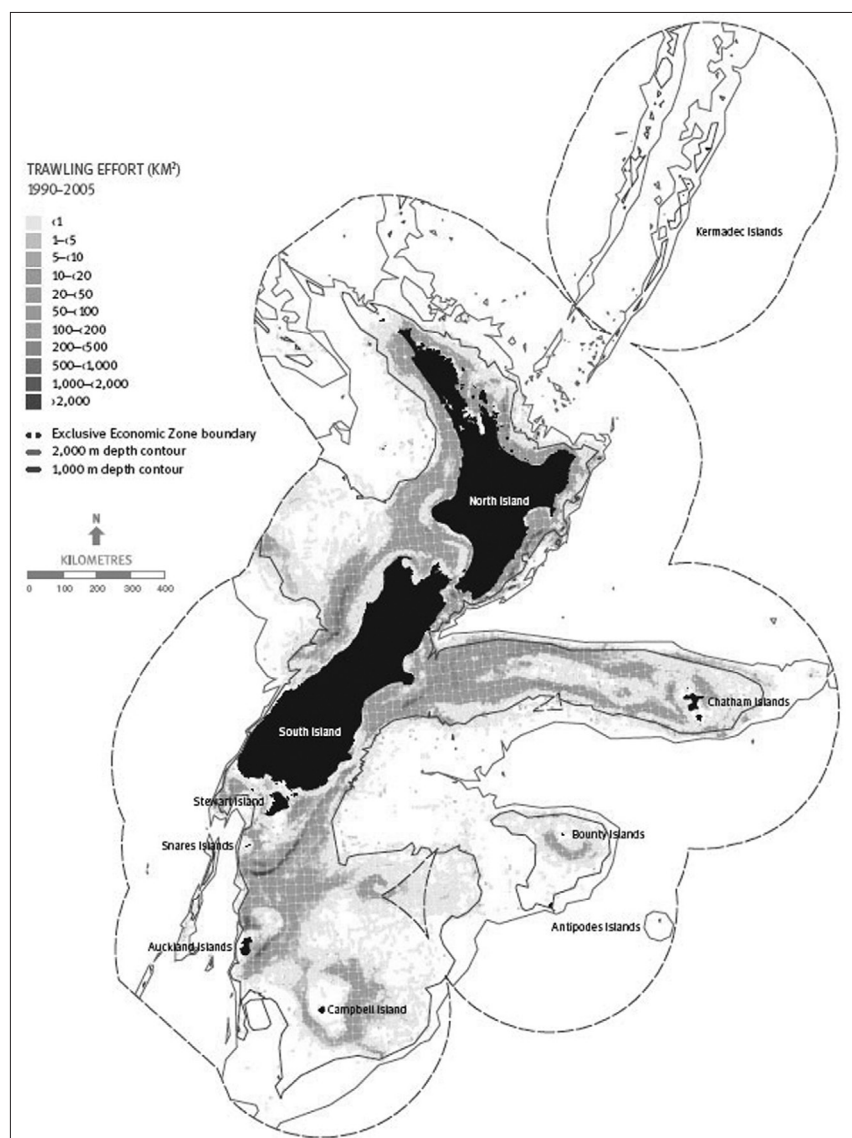


Figure 6.2 Trawling activity in New Zealand's marine area

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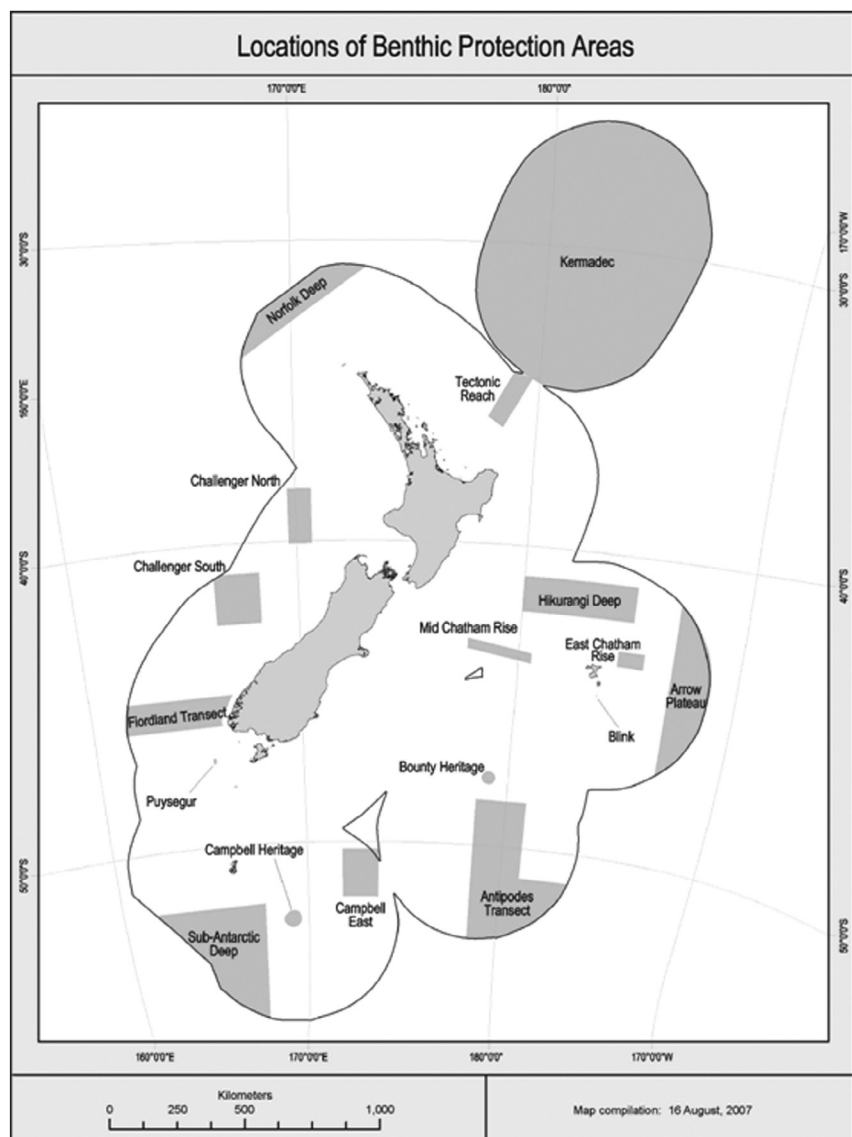


Figure 6.3 Locations of Benthic Protection Areas (MFish)

Figure 6.4: Summary of potential environmental impacts of commercial fishing methods

(Adapted from Weeber, 2007, 14-15)

Fishing method	Main features	Potential environmental effects		
		Bycatch	Protected species	Benthic habitats
Collecting and diving	Harvesting by hand or with a mechanical digger or rake	Low	Low	Generally low although trampling and disturbance of intertidal area may occur
Squid jigging	Lighting up the sea to attract squid and jigging hooked lures up and down	Low	Low	Low
Trapping and potting	Baited pots and traps are placed on the sea floor	Low	Low – seals and shags can be drowned and marine mammals can become entangled in pot lines	Low – damage can occur to sensitive species located under pots
Trolling	A set of hooked lures are slowly dragged through the water	Low	Low – occasionally fur seals and seabirds caught	Low
Long lining	Numerous baited hooks are placed in the sea on small branch lines attached to long main lines	Medium – sharks and billfish can be caught	High – seabirds, marine mammals and turtles can become hooked or entangled	Low – although lines and weights dragged across the seafloor can damage sensitive benthic habitats
Gill netting	Nets are placed in the sea and fish swim into them	High - catches a wide range of species which swim into the net	High – Hector's and Maui's dolphins, seabirds and seals can become entangled	Low
Seining	Nets are used to encircle fish	Purse seining – Low as targets fish schools Beach and Danish seining – Low to high depending on the extent to which fish schools are targeted	Low	Purse seining – Low Beach and Danish seining – Low but may impact on sensitive benthic habitats
Trawling	A large net is dragged through the water or along the seabed	High – catches wide range of species	High – seals, sea lions and dolphins can be caught	Mid water trawling – Low if net does not touch seabed Bottom trawling - High
Dredging	A steel-framed, toothed structure is dragged along the sea floor	High	Low	High

Decision-making framework

As already described in Chapter Two, fishing activity is managed under the Fisheries Act 1996. Key decisions under the legislation include those determining the TAC and the TACC and the application of other sustainability measures such as restricting fishing methods or locations. These decisions are informed by a cycle of work undertaken by the Ministry of Fisheries which has three main components:

- The *research planning and contracting process* - which influences the availability and type of scientific knowledge that informs fisheries management decisions. It can take two to three years for the planned research to influence management advice given by the Ministry to the Minister of Fisheries.
- The *stock assessment process* - which uses scientific information to estimate the current state of fish stocks in relation to the level of the stock that can produce the maximum sustainable yield (B_{MSY})
- The *management advice process* which involves the preparation of position and advice papers for the Minister of Fisheries on sustainability measures and fisheries management rules including the setting of the TAC and TACC. These are informed by the above research and stock assessment processes.

More recently, fisheries plans are being developed to help identify longer term objectives to help guide shorter term management decisions. The process is illustrated in Figure 6.5.

Research planning and contracting process

The Ministry of Fisheries contracts out around \$17 million of research each year. Much of this cost is recovered from the fishing industry. The Chief Scientist within the Ministry is responsible for overseeing this process and ensuring the quality of the scientific advice that informs management decisions.

Research planning groups, which include stakeholder representatives, meet during the year and make recommendations to the Ministry on what research should be carried out in future years.

Research needs are identified from a range of sources including from the work of science working groups (described below), Ministry operations staff, medium term research plans developed by the Ministry, science providers and stakeholders.

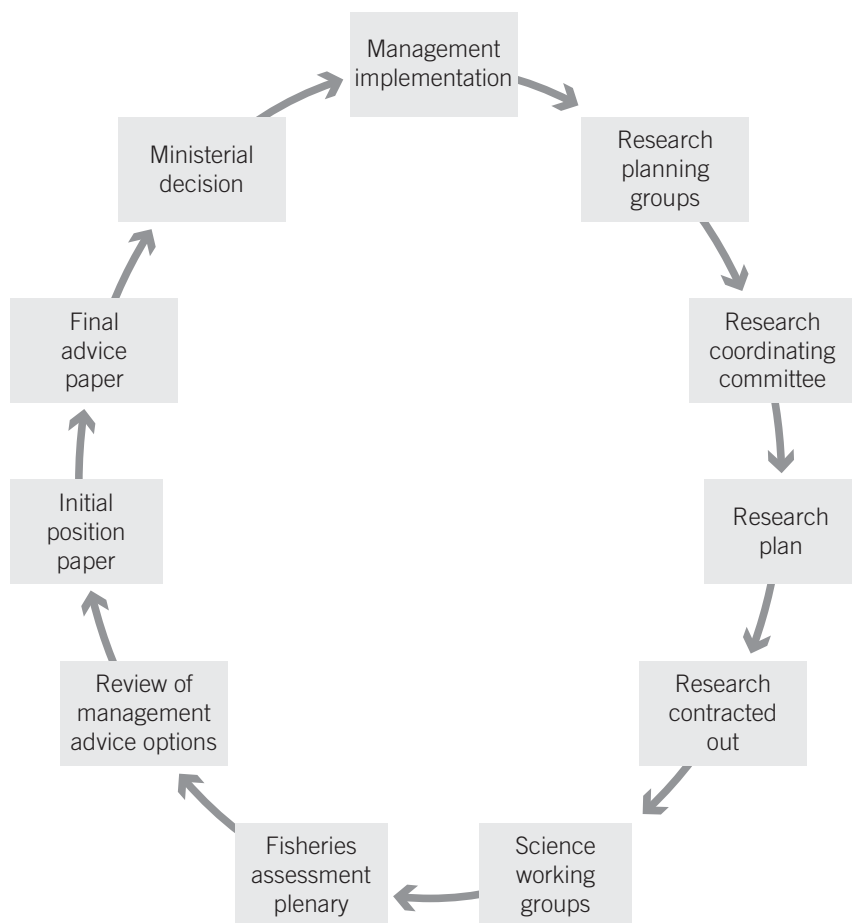


Figure 6.5: The fisheries management cycle

A Research Coordinating Committee meets annually to evaluate research proposals and determine overall research priorities. Meetings are held in Wellington. These priorities, combined with the level of budget available for each year, form the basis of a research plan which is prepared by the Ministry. This plan identifies what research projects will be funded for the year. Membership of the Research Coordinating Committee is open and participation by Ministry staff, stakeholder representatives and science providers is encouraged.

The projects identified in the research plan are contracted out to scientists. The draft results of each research contract are often considered by the relevant science working group (see below) before finalisation. Usually the final reports are published as either a Fishery Assessment Report or an Aquatic Environment and Biodiversity Report. These reports can be downloaded from <http://fpcs.fish.govt.nz/science/ResearchDocuments.aspx>.

Stock assessment process

Science working groups meet regularly to evaluate research and to contribute to the assessment of size and health of fish stocks and the development of sustainability measures. The role of the working groups is advisory only. The groups prepare reports and provide advice to the Ministry of Fisheries Chief Scientist. They endeavour to reach consensus on the findings presented in their reports, and any dissent is recorded.

Membership of the working groups is open to all interested parties and usually includes a range of representatives of fishing interests and environmental groups. If you are interested in attending a working group, contact staff at the Ministry of Fisheries and ask to be put on the contact list for meetings.

Meetings are chaired by a Ministry of Fisheries' scientist. Other scientists and representatives of relevant fisheries operations teams also attend meetings. Members must commit to minimum standards of participation which include maintaining confidentiality of group discussions and adopting a constructive approach.

Some working groups focus on the assessment of fish stocks and have the purpose 'to assess, based on scientific information, the status of fisheries and fish stocks relative to MSY-compatible reference points and other relevant indicators of stock status; to conduct projections of stock size under alternative management scenarios; and to review results of relevant research projects'.¹²

Other working groups have a more cross-cutting focus. For example, the aquatic environment working group focuses on the environmental effects of fishing activity. This includes:¹³

- Bycatch and unobserved mortality of protected species, fish and other marine life, and consequent impacts on populations
- Effects of bottom fisheries on benthic biodiversity, species and habitat
- Effects on biodiversity, including genetic diversity
- Changes to ecosystem structure and function, including trophic effects
- Effects of aquaculture and fishery enhancement on the environment and on fishing

Each fisheries assessment working group prepares a report, which includes information on fishing interests in the stock, sustainability issues and estimates of stock size. These reports are incorporated into a fisheries assessment plenary report. Two plenary reports are produced each year – one in May which covers most fish stocks and one in November for highly migratory species, rock lobster and Antarctic toothfish. The reports summarise the biological, fishery, stock assessment and stock status information for commercially fished species and inform the fisheries management advice process. The most recent plenary reports can be found on the Ministry's website at <http://fpcs.fish.govt.nz/Science/Plenary.aspx>.

Fisheries management advice process

Every year the Minister reviews a small number of fish stocks and makes a decision on whether to change the management measures in place. This can include the amount of fish that can be harvested (TAC and TACC), and any restrictions on by-catch and the manner and locations in which fishing is undertaken. The processes to undertake these reviews are called 'sustainability rounds'.

Currently there are two sustainability rounds a year, resulting in new management measures being in place by April or October – called the April or October round. Of the stocks that are reviewed each year, most are reviewed in the October round, but rock lobster, southern blue whiting and some shellfish species are reviewed in the April round. Ministry officials undertake a priority setting process that determines which stocks will be included in the sustainability round.

Each round takes five to six months and consists of the following steps:

- An *Initial Position Paper* is prepared for each stock that is being reviewed. This is normally prepared by the Ministry of Fisheries, but in some cases is prepared by a fisheries management group such as the National Rock Lobster Management Group. This will set out the issues to be addressed, identify and assess options to address them and make recommendations as to what management measures should be adopted. The paper will also often include a summary of the relevant science related to the fish stock in question, usually drawing on the information contained in the latest plenary report.
- The Initial Position Paper is publicly released and *written submissions* called for. Usually six weeks is allowed for submissions but this time may be extended for more contentious issues. Any person is able to make a written submission. The paper is placed on the consultation page of

the Ministry of Fisheries website. In addition, the Ministry emails out a notice of the paper to a list of stakeholders. For those interested in a particular issue, it pays to contact the Ministry and ask to be placed on the relevant stakeholder database. If a proposal has local implications it may also be advertised in local newspapers.¹⁴

- Once the submission period has closed, the Ministry summarises and considers the written submissions. The Ministry then prepares a Final Advice Paper, which is forwarded to the Minister of Fisheries. This provides the information on which the Minister will base his or her decision. This paper will normally have attached to it the Initial Position Paper and a summary of written submissions received.
- The Minister of Fisheries then makes a decision and this is recorded in a short letter, which sets out the decision and the reasons for it. Approval is then sought from cabinet for the decision. Once cabinet approval has been obtained, a copy of the final decision is sent to submitters and placed on the Ministry of Fisheries website.
- A notice is then placed in the *Gazette* to give legal effect to the decision.

There is no legal right to appeal the Minister's decision on its merits. It can only be legally challenged through initiating judicial review proceedings in the High Court.

In addition to the sustainability rounds, the Ministry operates regulatory rounds, which operate in a similar manner but apply to changes that require regulations to be adopted by the Governor-General on the recommendation of cabinet, rather than becoming operative through a notice in the *Gazette* after the decision of the Minister. The regulatory round is more appropriate for decisions that will have longer term implications.

Fisheries plans

The Ministry has embarked on a process of preparing fisheries plans for key fish stocks. The plans are intended to provide a more strategic approach to fisheries management through guiding decisions made through the sustainability and regulatory rounds. Four plans will be prepared covering the following species:

- Inshore
- Deepwater
- Highly migratory and pelagic
- Fresh water

Draft plans will be prepared by Ministry of Fisheries' staff with stakeholder feedback being sought on draft documents. The development and implementation of each plan will be overseen by the relevant fishery manager. The inshore and deepwater fisheries managers are located in Wellington and the highly migratory and pelagic and fresh water fisheries managers are located in Auckland

The Fisheries Act 1996 provides a broad statutory framework for the preparation of fisheries plans (section 11A):

- Plans may relate to fish stocks, fishing years, and/or areas
- Plans may include fisheries management objectives, strategies to achieve the objectives, and/or performance criteria to measure the achievement of the objectives amongst other things
- Plans are prepared within the overriding purpose of the Act which is 'to provide for the utilisation of fisheries resources while ensuring sustainability' (section 8(1)).
- Plans are approved by the Minister

There is no statutory submission or appeal process for the preparation of fisheries plans although there is a requirement in the Act (section 12) that before approving such a plan the Minister must consult with 'such persons or organisations as the Minister considers are representative of those classes of persons having an interest in the stock or the effects of fishing on the aquatic environment in the area concerned' and provide for the input and participation of tangata whenua.

It is possible for other parties to prepare fisheries plans, although they must be approved by the Minister before they have legal effect.

Information about current and future proposed fisheries plans is available on the Ministry's website at <http://fpcs.fish.govt.nz/>

Environmental strategy and standards

In order to help address the environmental impacts of fishing activity, the Ministry of Fisheries has prepared a *Strategy for Managing the Environmental Effects of Fishing*. The purpose of the strategy is to provide policies through which the Ministry and Minister can meet its environmental obligations under the Fisheries Act in an efficient and consistent manner. The core approach adopted in the strategy is the development of environmental standards, which will specify the limits of acceptable effects of fishing on the aquatic environment. Such standards have no statutory force, but are used as a guide to decision-making under the Fisheries Act.

An opportunity will usually be provided for written submissions to be lodged on a draft standard before it is finalised.

The *Harvest Strategy Standard*, finalised in October 2008, provides guidance for the setting of the target sizes for fish stocks. It contains three core elements:

- A specified target about which a fishery or stock should fluctuate
- A soft limit that triggers a requirement for a formal, time-constrained rebuilding plan. The default soft limit is the higher of one half of B_{MSY} or 20 per cent of the estimated virgin biomass of the stock.
- A hard limit below which fisheries should be considered for closure. The default hard limit is one quarter of B_{MSY} or 10 per cent of the estimated virgin biomass of the stock.

At the time of writing, a seabird standard was also under development.

Iwi and recreational fora

In order to develop a closer working relationship with some key groups involved in fisheries management the Ministry has facilitated the establishment of a number of iwi and recreational fora. These fora meet regularly (every 3 months) and provide the opportunity for the participants to provide feedback to Ministry staff and also for the Ministry to update participants on current and upcoming issues and processes. There is also an annual national meeting of all regional recreational fora.

Elements of best practice

There has been a growing awareness that fishing activity needs to be managed by an approach which takes cognisance of the ecosystems within which the fisheries are located. An ecosystems-based management approach takes into account the interactions of fishing activity with the marine ecosystem. Its central focus is on incorporating ecosystem considerations into fisheries management decisions. The aim is to allow the utilisation of fisheries while maintaining the natural structure, balance and functioning of marine ecosystems and important species. It also focuses on protecting habitats of importance to fisheries.

Such an approach requires an understanding of the ecosystems within which fishing is taking place, and the interactions between fishing and the affected ecosystems – knowledge which is gradually being developed. An ecosystems-

based approach to fisheries management could be expected to take into account the following issues:¹⁵

Targeted fish stock

- Is the size of the exploited population such that it retains its productivity, provides margins of safety for error and uncertainty, and retains its capacity for yields over the long term?
- If the fish stock is depleted, is it able to recover, and if so are management measures in place to enable its recovery within a reasonable time frame?
- Is fishing conducted in a manner to ensure that the age or genetic structure, or sex composition, of the stock is not changed to a degree that impairs reproductive capability?

By-catch species

- Are any threatened or endangered species affected by the fishery such as sea lions, Hector's and Maui's dolphins and seabirds. If so, are effective measures in place to ensure that any negative effects on populations are avoided?
- Are any protected species affected by the fishery such as seals, dolphins, whales or seabirds? If so, are effective measures in place to ensure that the effects are minimised?
- Does management ensure recovery of depleted populations?
- Are fisheries impacts on bycatch species reversible over a reasonable time period (20-30 years)?

Associated or dependent species

- Is the fishery depleting food sources important to other species such as seabirds or marine mammals and is this having a detrimental effect on these populations?
- Does management ensure recovery of depleted populations?
- Are fisheries impacts on associated and dependent species reversible over a reasonable time period (20-30 years)?

Biodiversity

- Is the fishing activity threatening marine biological diversity at the genetic, species or population levels?
- Is the fishing activity enabling the natural functional relationships among species to be retained? Is it leading to trophic cascades or ecosystem state changes? Trophic cascades can occur when large quantities of

top predators are removed, releasing the next lower trophic level from predation and allowing numbers to increase, which result in increasing predation on the trophic level lower down the food chain, and so on.

- Is the natural structure and function of the marine ecosystem being maintained, including its biodiversity and productivity?
- Are there sufficient no-take areas to ensure the maintenance of biological diversity?

Habitats

- Are habitats of particular significance for fisheries management well-managed or protected?
- Are changes to the marine habitats caused by fishing activity reversible over a reasonable time period (20 -30 years)?
- Are sufficient representative and unique benthic habitats protected from trawling and dredging activity?

Information and uncertainty

- Has the decision been based on the best available information and has the adequacy and reliability of the information been assessed?
- Has a recent assessment of the biological status of the fish stock and impacts of the fishery been conducted?
- Is sufficient research being undertaken into the environmental impacts of the fisheries to enable informed management decisions to be made?
- Where information is inadequate, has a precautionary approach been adopted?
- Has uncertainty in the available information not prevented action being taken when the sustainability of the fisheries resource or the health of the aquatic environment is at risk?

Monitoring and enforcement

- Is adequate monitoring being undertaken?
- Is there adequate enforcement of fisheries regulations?

Management

- Have clear management goals been established?
- Have management plans been developed to ensure progress towards those goals?
- Do these management plans include clear targets and timelines for progress?

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seven

Aquaculture

Overview

The aquaculture industry is worth around \$360 million to New Zealand today. The industry has grown significantly in the last 10 years, and the sector now employs nearly 30 per cent of the total seafood industry workforce. In 2006 members of the industry, working with the Ministry of Economic Development, released a strategy aimed at making aquaculture a billion dollar industry in New Zealand by 2025.¹ The consumer demand for fish is rising internationally, at a time when global wild stocks are under pressure, creating growing markets for farmed seafood products.

The most commonly farmed species in New Zealand is the greenlipped mussel, which is not found anywhere else in the world. In 2007, 99,500 tons of farmed greenlipped mussel were harvested. The next biggest harvests were of king salmon (9,400 tons) and pacific oysters (3,000 tons).² Mussels are predominantly farmed in the Marlborough Sounds, as well as at Golden Bay, the Coromandel Peninsula, Canterbury and Stewart Island. King Salmon are farmed in the Marlborough Sounds, Canterbury and Stewart Island. Pacific oysters are mainly farmed in the Far North.

New Zealand also has growing industries in paua and seaweed and the potential for commercial farming of other species including kingfish, eels, geoduck clams, butterfish and Hapuka.³

Methods

Mussel farms

A mussel farm consists of a series of buoys held together by long lines attached to either side of the buoy. The line is anchored to the seafloor at either end. A continuous rope, on which the mussels are grown, is strung from the long lines.⁴ A three hectare farm would typically have nine long lines of 110 metres each. Each long line is supported by 50 to 70 large plastic floats. Each float may support one ton of mussels. One 110-metre long line will generally support 3,500 to 4,000 metres of crop rope.

Most crop lines are seeded with mussel spat (baby mussels), which are gathered on Ninety Mile Beach in the Far North. Here, at irregular times throughout the year, large quantities of spat attached to seaweed are washed up on the beaches. This spat is collected and transported by air or truck to farmers

in other parts of the country. Local spat catching also occurs on a smaller scale at some mussel farm sites in Golden and Tasman Bays, in Jackson Bay and in some bays in Marlborough. Some farmers also seed their lines with spat from hatcheries which has been selectively bred.

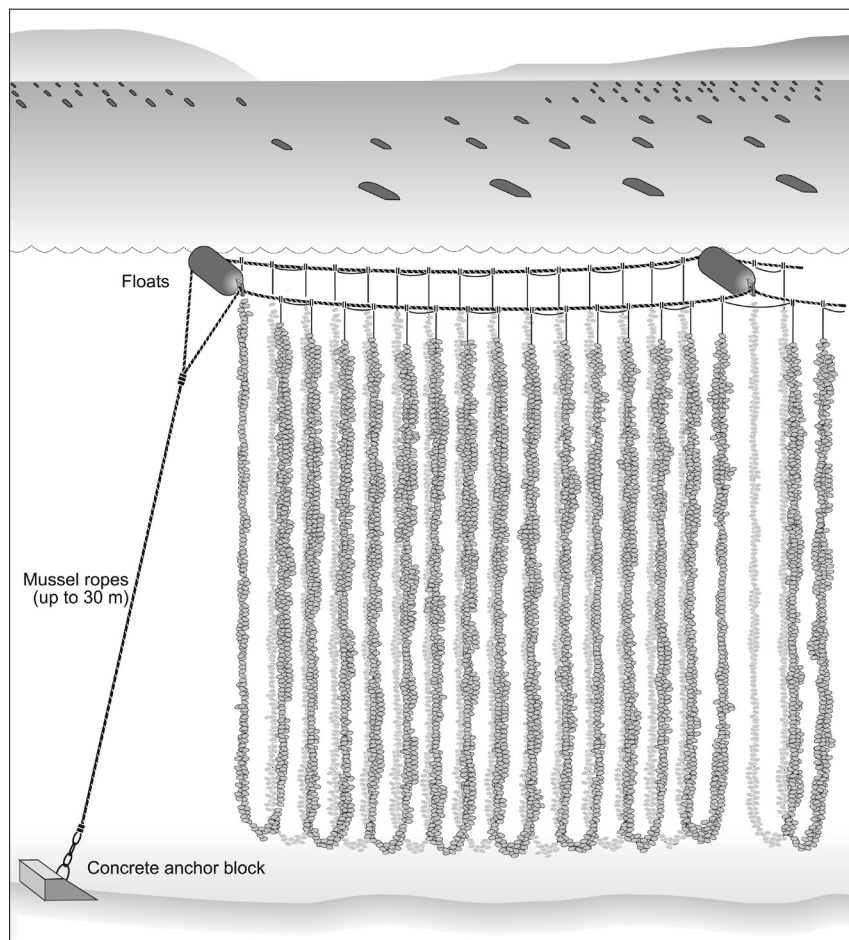


Figure 7.1 Typical mussel farm (Lloyd, 2003, 5)

Approximately 1000 to 5000 spat per metre of rope are wrapped onto the lines, using cotton stocking mesh. This mesh gradually disintegrates, by which time the spat have attached themselves to the rope. The baby mussels are left to grow for three to six months, and then these nursery lines are lifted. The young

spat are stripped from the ropes and reseeded on a final production rope at approximately 150 to 200 per metre, and returned to the sea to grow. Mussels take between 12 and 18 months to grow to a harvestable shell size of 90 to 100mm.⁵ The mussels are harvested by pulling the ropes up onto barges and removing the shellfish from the lines.

Offshore marine farms are now being developed, located several miles offshore, although still within the coastal marine area. A proposed large farm offshore from Napier is still at the experimental stage. The mussels will be on lines 20 metres below the water surface. Lines will be spaced to enable the movement of shipping between them and all but large vessels across lines. The sunken lines should also offer no obstruction to marine mammals.

Pacific Oysters

Oysters have been farmed in New Zealand since the 1960s. Pacific oysters were accidentally introduced to New Zealand from Asia in the 1970s. They started displacing the New Zealand native oyster and so farmers adopted them as their principal commercial species.

Today Pacific oysters are the main farm-raised oyster in New Zealand. Oysters are grown to marketable size within 12 to 18 months.⁶ Pacific oysters are grown on wooden racks in sheltered and shallow intertidal bays around the northern North Island. The seed to stock most farms is caught from the wild, predominantly from the Kaipara harbour. Bundles of wooden sticks catch the larvae drifting in the water, and the sticks are then sent to the marine farm where they are placed on the racks for on-growing the baby oysters. Sometimes, the oysters are transferred to mesh bags, hanging baskets or wire racks for growing. A growing proportion of oyster spat is obtained from hatcheries where it has been selectively bred.

Using baskets suspended on wires can be an effective method of farming and it allows oyster farms to be located in more exposed locations. In rough weather the baskets swing and thereby resist the damage that would be caused to the traditional fixed racks. Baskets are also beneficial because they prevent shell from dropping to the seabed. Oyster spat that has come from hatcheries is farmed in baskets because they are not attached to sticks and need to be contained.

Finfish

Sea cage farming of fish has become widespread across the world, driven by a growing demand for fish and pressure on wild stocks globally. In New Zealand there is the potential for farming Hapuka and other species. However, it is

salmon farming that currently makes up the majority of the finfish farming industry in New Zealand.

New Zealand has no native salmon, and only king salmon (chinook), which is native to the cool waters of the northern Pacific rim, have been successfully farmed here. After their introduction, king salmon became established on the east coast of the South Island and to a lesser extent also on the South Island's west coast. Commercial farming of salmon started in the 1970s, both in inland ponds and offshore. Today most commercially sold salmon are grown in cages in the sea.⁷

No imports of live salmon have been allowed into New Zealand for the last 50 years – instead eggs and milt are collected from captive broodstock. Fertilised eggs are incubated in a freshwater hatchery, where newly hatched fry are reared for another six to 12 months. The fish are then transferred to the sea cages where they remain for the rest of their lives, about two to three years.⁸

Potential Environmental Effects

General effects of marine farming

Landscape effects

For successful aquaculture, unpolluted and sheltered waters are required, away from centres of high population density, to ensure that runoff from land-based activities does not affect the marine environment where the farm is located. Shellfish filter food from the water column and can concentrate toxins from the water in their flesh, which can pose a danger to consumers.⁹ There are, therefore, relatively few places where aquaculture can occur, and these tend to be places which are also valued for other activities and for their unspoilt landscape values and natural character.

Marine farms are spread out over fairly large areas of coastal water and tend to have elements that are visible from the shore. Mussel farms have lines of buoys holding up the mussel lines, which are visible as horizontal structures on the water plane. Some lines are deliberately suspended below the surface to avoid swells. Barges working the farms, although not permanent fixtures, may be perceived as intrusive vertical structures when stationary. In addition, navigational lighting marking the farms is typically visible from two nautical miles away at night time.

Finfish farms are usually visible as vertical structures on the surface of the water. They can be particularly noticeable as they use high nets to keep predators out. Similar to mussel farms, service structures and craft working on the farm will also be highly visible. However, finfish farms tend to have a smaller visual footprint than mollusc farms. Oyster farms are generally constructed from rows of wooden racks located in the intertidal zone

In terms of location, several key variables affect the ability of the marine environment to visually absorb marine farming activities. These are the visual scale of the affected coastal environment, the degree of visual interest in the view, and the elevation and distance from which the farm is likely to be viewed. Structures are also generally more visible in calmer, lighter coloured water.

When deciding where to site marine farms, consent authorities must consider the effects of the proposal on the natural character of the area. Consent will be more difficult to obtain where the coastline is undeveloped and has a high degree of natural character, and where the addition of a marine farm would introduce a 'working landscape' into the area where this is not already present.

Effects on marine mammals

The effects of marine farming on marine mammals are not fully understood. There is concern that marine farms could adversely affect species such as dolphins if they block access to food sources, limit the space available for hunting, or interfere with normal behaviour. Marine farms may also increase the risk of vessel disturbance, where marine mammals are injured or killed through hitting boats or are disrupted by increased underwater noise.

Currently the limited extent of marine farming in New Zealand means that few such effects have been observed here, but this may become an issue as the industry grows. In other countries where marine farming is much more widespread, such effects have been reported.¹⁰

There is potential for marine mammals to become trapped in marine farm equipment. To date there have been no reported cases of dolphins or seals becoming entangled in mussel farm lines. Since 1996, two Bryde's whales have been reported as casualties of entanglement in mussel spat catching lines.¹¹ Entanglement for whales may become more of a problem if large offshore marine farms are established in areas which overlap with the migratory paths of humpback and southern right whales.¹²

New Zealand fur seals are attracted to salmon farms. Most farms use predator exclusion nets to reduce the risk of entrapment. In approximately 25 years of sea cage salmon farming in New Zealand, there have been two reported cases involving seals and three involving dusky and Hector's dolphins becoming trapped in farm nets.¹³ Advances in net design may reduce the likelihood of

future marine mammal entanglement. Refined net changing and other practices can also reduce these risks.

Attraction of pest species

The fixed structures found in marine farms can provide ideal habitats for some pest species, particularly biofouling species – microorganisms, plants, algae, and animals such as barnacles, tubeworms and seaweed that grow on submerged structures and on the shellfish themselves. These can multiply on a marine farm, and then spread to the surrounding environment, potentially altering its ecology irreversibly.

Such biofouling can include invasive species such as the Asian kelp (*Undaria pinnatifida*) which has been strongly associated with mussel farms in the Hauraki Gulf. The nitrogen discharges produced by finfish farming may increase the growth of such unwanted species.¹⁴

Biofouling is a serious concern to marine farmers, as it can lead to lower yields through poor growth of the shellfish. It can also result in higher maintenance costs, as a result of equipment becoming damaged or unuseable because of the growths. When biofouling organisms grow on finfish nets, they can prevent water flow through the nets and increase their weight so that they can be pulled from their moorings. Nets must therefore be changed or cleaned regularly.

Removal of biofouling organisms – which is usually done by hand – can be a very expensive and labour intensive job for marine farmers. However it is necessary to ensure the health of the farm and the surrounding environment.

Accumulation of heavy metals in seabed sediments

Marine farmers, particularly finfish farmers, often use antifouling paint to prevent the build up of biofouling organisms on some structures. Many of these paints contain copper, which is toxic in high concentrations. These paints gradually wash off the structures and can accumulate on the seabed under the farms. The extent to which such concentrations of copper may enter the food chain and impact on marine life is not fully understood. The greatest impact is likely to be on the organisms living within the contaminated sediment.¹⁵

Although manual defouling is possible for farm nets, there are few alternatives to putting antifouling paint on items such as predator nets which must be kept clear of debris to allow water to flow through. The use of antifouling paints, however, should be avoided wherever possible.

Spread of pests and diseases

The transportation of spat, gear, stock and vessels around the country has the potential to spread pests and diseases. It is therefore important to ensure that all stock and spat comes from healthy sources and that all equipment is cleaned properly before transportation. Most marine farmers follow a code of practice that contains good management practices and environmental objectives for managing risks around spread of pests and diseases.

Obstructing navigation

In most cases, members of the public can fish in the vicinity of marine farms and small vessels can navigate through the lines of mussel farms. Marine farms should be adequately lit at night to ensure safety of navigation and should not be located within shipping routes or where navigation is obstructed by their presence.

Debris

Where marine farms are not correctly maintained, debris such as lines, baskets, nets and poles can come adrift and pose a risk to other water uses and marine life. All non-biodegradable waste material should be removed from the coastal marine area, and structures should be adequately maintained to ensure their integrity. In particular, lines should be kept under tension to reduce the likelihood of nuisance for other users and marine mammal entanglement.

Shellfish – ecological effects

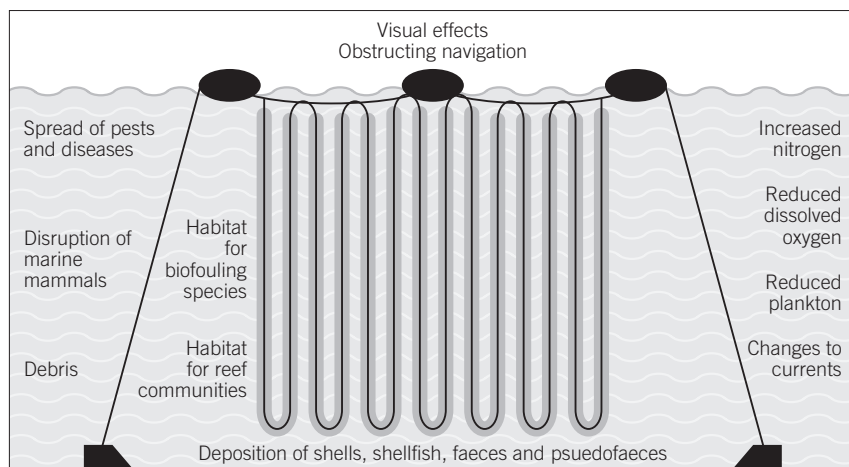


Figure 7.2 Potential environmental effects of mussel farming

Changes to seabed

The principle ecological effect of mussel and oyster farming is localised changes to the seabed caused by shells, shellfish, faeces and pseudofaeces (particles which are not digested bound in mucous) dropping from the mussel lines and oyster racks. These items can accumulate on the seabed and alter its physical, chemical and biological nature. The waste products are nutrient-rich and thus increase microbial activity, which can lead to oxygen depletion within the sediment.

These effects are particularly pronounced where the seabed is sand or pebble, rather than mud. In these areas benthic species are less able to cope with smothering by fine material from the farm. The extent of these effects will largely depend on the type of substrate under the farm and the flushing capacity of the site.

Changes to water column

Mussel farming can increase levels of nitrogen and reduce levels of dissolved oxygen in the water column, but these effects are localised and are only likely to cause problems in poorly flushing sites.

The structures required for mussel farming create obstructions in the water column that can alter the direction and speed of currents in the area, which in turn can alter the rate at which plankton and nutrients are replenished and waste dispersed. The severity of these effects will depend on the particular characteristics of the area, the scale and density of the farm and the orientation of the lines relative to the predominant direction of the currents. Larger, denser and poorly orientated farms will have greater effects.

Effects on wild species

Material falling from mussel farms can attract small fish and encourage the development of reef-type communities including sponges, crabs, starfish and tube worms. Juvenile fish commonly use artificial structures placed in the sea for shelter, which can lead to the creation of new communities and increased biodiversity. The effect of marine farms on fish abundance is not well understood but the effects may be positive. Research on farms in the Marlborough Sounds indicates that there is more abundance and diversity of fish within the farms than in surrounding areas. Anecdotal evidence from recreational fishers indicates that species such as snapper and kingfish can be found in high numbers around marine farms, attracted to the food supply provided by the mussels and juvenile fish, especially during harvesting.

Marine farms can also change the diversity of species within their vicinity because of the way they change the seabed substrate, increasing the amount of

shell and nutrients. This results in a shift towards species favouring the new conditions. For example, studies of farms in the Marlborough Sounds found that heart urchins and brittle stars declined in the locality of mussel farms and were replaced by other species.¹⁶

Consumption of naturally occurring plankton

Mussels and oysters feed by filtering naturally-occurring plankton from the water column. This means that no artificial feeding is required. A shellfish farm can potentially deplete the amount of plankton available as food to wild species, but these impacts have generally been found to be only localised.¹⁷ While mussels consume plant and animal plankton, they also excrete nutrients into the water which stimulate the growth of plant plankton.

Abandonment of farm structures

Abandonment of marine farms can result in degrading structures being left in the marine area. The structures required for finfish and mussel farming are generally of a high enough value to motivate farmers to remove them, whereas oyster racks are not.

If oysters are left behind on the racks they can outgrow the structures and cause the farms to collapse into the mud and build up on the sea floor. The abandoned structures can be very visible for many years. This problem can be addressed by conditions of resource consents requiring structures to be removed on the cessation of farming.

Siltation and alteration of natural currents

Oyster farms can cause siltation on the seabed by disturbing the natural flow of water in the area and changing the height of the seabed beneath and around the farm through scouring or silt buildup. These effects can be minimised by ensuring that oyster racks are oriented parallel with, rather than at right angles to, currents and wave action. The growing of oysters in baskets, which is becoming more common, minimises the trapping of sediment.

Genetic effects on wild populations

Although farming marine species can potentially affect the genetic distinctness of local wild populations, the adverse genetic effects of mussel farming are thought to be low. The farmed greenlipped mussels are native to New Zealand and are found throughout the rocky coastlines of the country.

Spat for mussel farms is predominantly sourced from Ninety Mile Beach in Northland, but there is currently no evidence that wild greenlipped mussels

are declining in population fitness, although mixing with farmed mussels has probably already occurred. This could be due to the long-lived dispersive larval stage of greenlipped mussels, which facilitates wide genetic mixing.

Pacific oyster farming may result in the establishment of populations of a non-native oyster in the wild. However, Pacific oysters are already widespread around the coastline and are present in areas where there are no marine farms. Farming of Pacific oysters is generally not permitted in areas where there are not already wild populations.

Finfish / king salmon – ecological effects

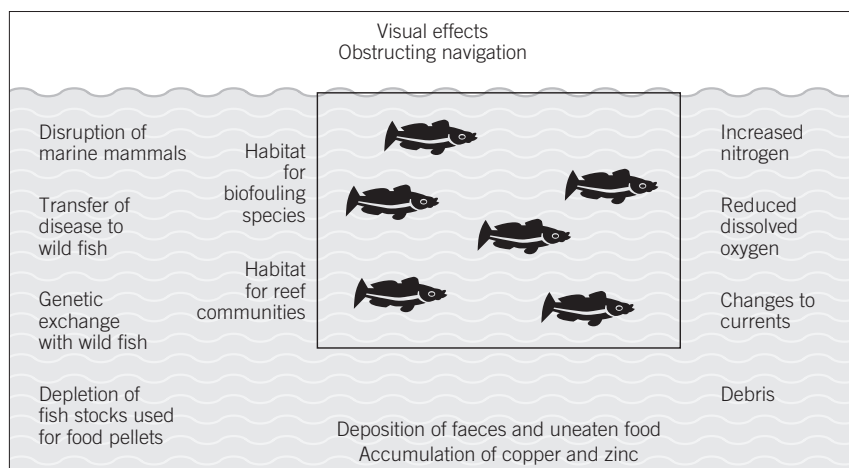


Figure 7.3 Potential environmental effects of finfish farming

Depletion of fish stocks

Whereas shellfish feed on naturally occurring plankton, farmed fish such as kingfish and salmon are top-end carnivores. They are fed an artificial diet of food pellets. These pellets are made of waste from fish processing plants, fishing bycatch and small fish such as sandeel, sprats, capelin, anchovies, herring, mackerel, blue whiting and krill. Fish pellets are not currently manufactured in New Zealand because of the small size of the finfish industry here.

Recent estimates of the amount of wild fish required to produce one tonne of farmed fish are between 3.8 and 5.8 tonnes. About a quarter of the global harvest of wild fish is used to produce fish meal.¹⁸ It is therefore important that the fisheries of wild species used in the manufacture of fish feed are managed

sustainably to avoid depletion and that the fish is not an important food source for local communities.

Experimentation is being undertaken with the use of soy and vegetable products in fishmeal, but this is not currently cost-effective. Some finfish species which may be farmed in the future, such as butterfish, are vegetarian and could therefore be directly fed on vegetable products.

Changes to the seabed and water column

Deposition of faeces and uneaten feed causes changes to the seabed beneath finfish farms. These particles are organic-rich and can alter the physical, chemical and biological nature of the seabed, increasing microbial activity. This in turn can decrease oxygen levels in the sediment and lead to changes to the biodiversity of the area.

These effects tend to be greater in finfish farming than in shellfish farming. However, they can be minimised by siting farms in deep, well-flushing locations and by good farm management such as using higher quality feed to gain an improvement in feed conversion rates (meaning that less feed is needed to grow a fish), and using automated feeders with shut-off signals linked to underwater cameras that detect waste feed.¹⁹ Farmed areas can also be rotated so that areas of the seabed are regularly fallowed to allow recovery.

Such approaches are also important for reducing the effect that uneaten food and fish waste have on the water column. Too many nutrients in the water can cause algal growths, which can in turn, cause reduced water quality and possibly an increased likelihood of toxic algal blooms.

Changes to local biodiversity

Finfish farms can cause changes to local biodiversity, in particular a decrease in those benthic species that are intolerant of the enriched conditions and a corresponding increase in species that are adapted to them. Wild fish and other species can be attracted to finfish farms because they feed on the dropped fish feed and use the structures for shelter.

Impacts on wild populations

There is the potential for escaped farmed fish to spread disease to wild populations, to breed with wild fish, altering the genetic structure in the wild stock and to outcompete wild stock. Although a major issue in salmon fisheries overseas, this is not currently a problem in New Zealand. It is something that will need to be addressed, however, if indigenous species are farmed in the future.

Toxic chemicals

Therapeutants (vaccines, anti-fouling paints and feed) are all potential sources of chemical pollutants to the marine environment. Overseas, farmed finfish are very susceptible to diseases, due to the densities at which they are stocked. Large numbers of chemical treatments are used to combat these diseases, and also for colouring, to obtain the red/orange colour found in wild salmon, which intensively farmed salmon do not achieve naturally.

Currently, in New Zealand, feed for farmed salmon does not contain antibiotics, vaccines, steroids or colourants. However their feed contains zinc, necessary to prevent cataract formation and other health problems, which can accumulate in sediments beneath salmon farms at levels probably toxic to benthic organisms. The effects are thought, however, to be limited to the footprint of the farm.²⁰

Therapeutants for treating parasites and diseases have not been required in New Zealand salmon farms, due to the relatively low density and small scale at which they are farmed and absence of diseases. However, the expansion of finfish farming into new areas and other types of fish may require new methods of disease prevention. Kingfish trials undertaken in the Marlborough Sounds indicate that there will be a need for disease treatments because kingfish are unusually susceptible to some parasites. It is proposed that these parasites will be managed by low concentration in-water baths of hydrogen peroxide.

Decision-making framework

The aquaculture reforms of 2004 placed all aquaculture planning and consent processing under the RMA, whereas previously marine farmers had to obtain separate consents from the regional council and the Ministry of Fisheries.

Under the RMA, there is a three-step process to obtaining consent to establish a new marine farm. First, the area for the proposed farm must be identified as an Aquaculture Management Area (AMA) in the operative regional coastal plan. Secondly, an authorisation must be obtained for space within an AMA. The granting of authorisations is the mechanism that is used to allocate space within an AMA - authorisations give the holder the right to apply for a coastal permit for a marine farm. Thirdly, a coastal permit must be obtained, which addresses the environmental impacts of individual marine farm proposals.

Since January 2005, regional councils have been able to identify AMAs in their regional coastal plans. AMAs are the only areas where new marine farms can be established. Transitional provisions provide that areas where marine

farms are already established become AMAs²¹. AMAs will usually be drawn on maps attached to regional coastal plans and be accompanied by a set of objectives and rules which apply to the area. At the time of writing no new AMAs had been created through plan changes under the new provisions, and the regime was under review.

Where there is good information about the likely effects of marine farming within the AMA, aquaculture activities may be given the status of controlled activity (consent cannot be refused for a controlled activity). Where effects are less certain and require consideration on a case-by-case basis, marine farming may be designated as a restricted discretionary, discretionary or non-complying activity.

The AMA can specify the character, intensity and scale of activities, and the size of the space that may be the subject of a coastal permit. AMAs can also specify the types of aquaculture that can occur in a particular area, and what other activities involving occupation of the area may be undertaken, although the AMA must principally be used for aquaculture. Lastly, the AMA can specify that no applications for a coastal permit to occupy the space may be made before a date specified in a public notice.

Regional councils are not required to take action to identify AMAs. However, if they do decide to do so, there are two ways in which this can be achieved – through a council-initiated plan change or through an invited private plan change procedure. A third option available is for a private party to initiate a private plan change to create a new AMA. These three options are summarised in Figure 7.4 and are described more fully in the sections below.

Councils identify AMAs through initiating plan change

First, local authorities can decide to proactively identify AMAs in any part of the coastal marine area that is vested in the Crown or in the regional council (s165A, RMA). When identifying potential AMAs, the council will consider the effects of the proposed AMA on other users of the marine area, marine life and the marine environment more generally. The council may use constraints mapping to identify unsuitable areas for marine farming. This involves spatially plotting, usually on GIS, different layers of ecological, cultural, social and economic information which indicate marine areas that may not be suitable for marine farming.

Once the proposed plan change identifying potential AMAs has been prepared, it must be submitted to the Chief Executive of the Ministry of Fisheries, who must make an ‘aquaculture decision’ under the Fisheries Act (Part 9A subpart 1). This decision considers whether there are any undue adverse

Figure 7.4: Processes to create Aquaculture Management Areas

Option 1 – Regional council initiates plan change	Option 2 – Regional council invites private plan changes	Option 3 – Private party initiates private plan change
Council decides to prepare plan change to identify AMAs in any area vested in the Crown or regional council	Council may decide to identify excluded areas	
Council consults with Ministers of the Crown, affected local authorities, tangata whenua and other interested parties over location of AMAs	Council consults with Ministers of the Crown, affected local authorities, tangata whenua and other interested parties over location of excluded areas	
	Council issues public notice identifying excluded areas	
Council prepares proposed plan change	Council calls for private individuals to initiate a private plan change (outside any excluded areas)	Private party seeks private plan change
	Council decides to accept, adopt or reject the request for a private plan change	
Council submits proposed plan change to the Chief Executive of the Ministry of Fisheries for consideration under the Fisheries Act 1996		
Ministry of Fisheries consults with commercial, customary and recreational fishers		
If an area in a proposed AMA has an 'undue adverse effect' on customary, recreational or non-quota commercial fishing it will be deleted from the proposed AMA. If it has an undue adverse effect on commercial quota fishing this will be noted in the proposed plan change – an agreement with the affected quota holders will be needed for the space to be used for aquaculture		
The decision of the Chief Executive of the Ministry of Fisheries is publicly notified and can be appealed to the High Court within 3 months of decision		
The council publicly notifies the proposed plan change, receives written submissions, holds a public hearing and makes a decision on submissions		
The council's decision can be appealed to the Environment Court by submitters		
20% of each AMA is allocated to the Māori Trustee – this decision is publicly notified		
Minister of Conservation approves the plan change which then becomes operative		
The AMA takes effect		
The council issues authorisations for the remaining space within each AMA – this will be through public tender unless alternative methods are provided for in the regional coastal plan	The council issues authorisations for the remaining space within each AMA to the party who initiated the private plan change	The council issues authorisations for the remaining space within each AMA – this will be through public tender unless alternative methods are provided for in the regional coastal plan
Holders of authorisations can apply for a coastal permit to establish a marine farm		

effects on commercial, customary or recreational fishing activities. The Ministry of Fisheries primarily looks at the extent and type of fishing at the site, and whether there are any alternative areas available for fishing if the site is used for aquaculture. 'Fishing' in this context means the catching, taking or harvesting of fish, aquatic life or seaweed.

The Ministry must use the best available information to make its decision, including fishing records and reports, research documents, institutional knowledge, regulations and management plans. The Ministry will consult with commercial, recreational and customary fishers in order to make its decision. The decision of the Chief Executive of the Ministry of Fisheries must be publicly notified and can be appealed to the High Court within three months of decision.

Proposed AMAs (or parts of a proposed AMA), that have an undue adverse effect on customary and recreational fishing will be removed from the plan. Proposed AMAs, that have an undue adverse effect on commercial quota fishing, will be identified in the plan as a 'reservation'. Any prospective marine farmer wishing to farm an area subject to a reservation must first reach an agreement (known as an 'aquaculture agreement') with the affected quota holders before he or she can obtain an authorisation to apply for a resource consent.

After the aquaculture decision has been made by the Ministry of Fisheries, the plan change for the proposed AMA is publicly notified under the RMA. There is then an opportunity for the public to make submissions and any decision of the regional council on the submissions can be appealed to the Environment Court. Unlike normal plan changes, the proposed AMA does not take effect upon notification, but only once it is part of an operative change to the regional coastal plan. This requires the final approval of the Minister for Conservation.

Before a plan change creating new AMAs has become operative, the council must allocate twenty per cent of all new space to Te Ohu Kai Moana Trustee Limited which will then further allocate it to iwi. This space must be representative of all new space being created within the region and/or of economic size (sections 9 and 10, Māori Commercial Aquaculture Claims Settlement Act 2004). The Crown can also instruct councils through an Order-in-Council to allocate a further 20 per cent (bringing the total to 40 per cent) as a mechanism to meet the Crown's commitment to provide iwi with the equivalent of 20 per cent of existing aquaculture space.

Under the procedure of council identified AMAs, the council pays the cost of identifying potential AMAs, and leads the process with input from industry, the public and other interested parties. Councils have encountered difficulties with this process, because of the large amount of information required and

the sometimes high levels of conflict. It is also expensive and ratepayers can be reluctant to fund the cost of a process they see of prime benefit to commercial interests.

Councils can remove or modify AMAs via a plan change, which would undergo a similar procedure. A resource consent for a marine farm has a fixed term, so even if the relevant AMA is removed, the consent can continue until the end of its term. The AMA status of the area is only significant when the farmer wants to renew or modify the consent.

Once a plan change creating new AMAs has become operative, the council may allocate the remaining 80 (or 60) per cent of space within each AMA by way of authorisations (which provide the holder with the right to apply for a resource consent). Allocation must be through a public tender process unless the plan provides for an alternative allocation mechanism.

Council invites private plan change

The second option is for the regional council to invite private interests to initiate the process of changing the plan to create AMAs. Currently the first stage is for the council to call for interested parties to submit proposals. Under proposed amendments in the Aquaculture Legislation Amendment Bill (No 2) 2008, there will be an 'expressions of interest' phase prior to AMA applications to avoid competing applications for the same space. Anyone can propose the creation of an AMA by way of a private plan change. The applicant for the plan change will pay the costs of the identification of the AMA, which may be considerable, but will be entitled to priority for space in the newly created AMA. Twenty per cent of the AMA will still be allocated to Māori. The proponent of the private plan change will be authorised to apply for consents for the remaining 80 per cent.

The council may, by public notice, indicate areas in which it will not seek or accept requests for private plan changes, called excluded areas. Such areas are to be identified following consultation with the Minister for the Environment, Ministers of Conservation, Fisheries and Transport, any other Ministers affected by the proposal, affected local authorities, and tangata whenua. The council may consult with anyone else it feels necessary. The consultation process follows the same rules as for a proposed plan. Whether or not excluded areas are identified, councils must undertake this consultation process before they can invite applications for private plan changes.

Councils can accept or adopt a request for a private plan change. The process then follows the same path as a council initiated plan change, except that if the council adopts the request, the council may combine it with any other request or part of a request made by another private party.

The council can reject a request for a private plan change if:

- It has already accepted a request from another person to establish an AMA in the same area of the region
- The council is preparing a change to establish an AMA in the same area
- The request is frivolous or vexatious
- The substance of the request has been considered by the council in the last two years
- The request or part of the request is not in accordance with sound resource management, or
- The request or part of the request would make the plan inconsistent with Part 5 of the RMA which sets out the purpose and content of plans.

If, following the process of notification and submissions, an AMA resulting from an invited plan change is established, the local authority must identify 20 per cent of the newly created space for allocation to Te Ohu Kai Moana Trustee Limited. As soon as practicable after doing this, the local authority must allocate authorisations for the remaining space to the person who requested the plan change, and publicly notify this allocation. This authorisation enables the holder to apply for a coastal permit for the establishment of a marine farm in the area covered by the authorisation.

Normal private plan change

The last option is for private individuals to request private plan changes in the normal manner in order to create AMAs. Under this option, the applicant would not be able to claim priority for obtaining an authorisation to occupy space in the new AMA (unless the plan provides for this), but would still have to pay for the costs of the process. However, it is possible for a proponent to apply for a second plan change which would amend the allocation rules in the plan to provide that private plan change proponents receive 80 per cent of the space created. Although it has less certainty, this is currently the only option available to a proponent where the council fails to take action in relation to AMAs.

Resource Consents

Once an AMA is established, prospective marine farmers will still have to apply for a resource consent, but the intention is that the process will be less time consuming because many of the planning matters will have already been addressed in the identification of the AMA.

Marine farms existing before AMAs created

All leases and licences issued under the Marine Farming Act 1971, and all marine farming permits issued under the Fisheries Act 1983, are deemed to be coastal permits (resource consents) under the RMA. Any conditions stay the same, although these may be reviewed by the council within a specified time frame. Marine farms, which hold a resource consent or Marine Farming Act authorisation and are sited in an area where marine farming is not prohibited under a proposed or existing plan, are deemed to be AMAs.

Marine farming licences and leases existing at the time the new legislation came into force on 1 January 2005 automatically received a term of 20 years from that point, and after this their rights take priority over other consent applications for the space, provided that they meet certain conditions (section 124B). These conditions are that the new application seeks to continue the existing aquaculture operation, and that the proposed area is still within an AMA. The consent authority will also consider the consent holder's track record in compliance and whether they use industry best practice.

The newer marine farming permits (issued under the old regime, after the Marine Farming Act was closed off) expire at the end of the term that they were issued under the old regime, and the holder then has to apply under the new rules.

Elements of best practice

Effects – all marine farms	Best practice response
Degradation of landscape values and natural character of the area	Marine farms should be sited away from areas with high natural character and where the visual effects of the farm could be significant
Alteration of the behaviour of marine mammals as a result of marine farms obstructing access to feeding grounds and other areas of their habitat	Marine farms should be carefully located in order to ensure that access by marine mammals to habitual feeding and other areas is not obstructed
Potential for marine mammals to become caught in long line or finfish nets used on the marine farm	Marine farms should use the best available technology and best techniques to minimise the risk that marine mammals will become trapped in nets or lines
Fixed structures used by the marine farm can attract biofouling species which can spread to the surrounding environment	Fixed structures should be regularly cleaned to prevent the build up of biofouling species
Copper from antifouling paints can build up on the seabed to toxic levels	Marine farms should either manually defoul equipment or use paints that contain minimal amounts of copper
Transporting spat and machinery used on marine farms around the country can spread diseases and pests	Good management, including careful washing and drying of equipment before transportation, should be adopted to prevent spread of disease and pests
Marine farms can pose a hazard to watercraft and navigation	Marine farms should be adequately lit at night to ensure safety of navigation and should not be located in areas which obstruct navigation
Debris from poorly maintained farms can be a hazard to other water users and to marine fauna	All non biodegradable waste material should be removed from the coastal marine area, and structures should be adequately maintained to ensure their integrity. Lines should be kept under tension (this reduces likelihood of nuisance for other users and of marine mammal entanglement)
Shellfish	
Changes to the seabed through deposition of shell, shellfish and faeces	Marine farms should be sited in areas where benthic habitats are less sensitive to impacts such as areas with a muddy substrate rather than rocky or sandy areas
Effects on water quality through excretion of faeces and alteration of naturally occurring currents	Marine farms should be designed to minimise effects on currents, should be sited in areas with adequate flushing, and should not be so intensive as to allow build up of faeces in the water column
Effects on wild species as a result of changes to their environment	Marine farms should be located away from areas where they can have negative impacts on species or habitats with special conservation value or of ecological importance
Effects on light-sensitive marine flora as a result of shading by the farm structure	Marine farms should be located away from areas where they can have negative impacts on seagrass and other light sensitive species

Depletion of naturally occurring plankton leaving less available for wild species	The effects of shellfish farming on the amount of plankton available should be carefully considered, including the cumulative effects of having more than one shellfish farm in the same area
Abandonment of marine farm sites and structures when farming ceases	If marine farming activity ceases, all structures associated with its use should be removed and the site rehabilitated. The consent authority can require this as condition of consent or require a bond or payment to cover the cost of removal by the local authority
Changes to patterns in the current and natural flow of water as a result of farms, particularly oyster racks, causing sand build up under the farm	Proposals for marine farms should be accompanied by an assessment of the hydrodynamic patterns at the subject site, with current speeds and directions, flushing times and water dispersion patterns and modelling of how this will be affected by the proposed farm. Marine farms should not be sited in poorly flushing areas
Finfish	
Depletion of fish stocks when farmed fish are fed on pellets of fish from capture fisheries	Only sustainable sources of food should be used including waste fish material from processing plants, sustainable capture fisheries and fish from stocks which are not an important food source for local communities
Deposition of faeces and uneaten food causing changes to the physical, chemical and biological nature of the seabed and water column	Extremely intensive farming should be avoided, high quality feed should be used to reduce the amount required, and automated feeding systems should be used which stop dispensing when they detect uneaten food in the water. Farms should be located in areas which flush well
Effects on local biodiversity as a result of changes to their environment	Marine farms should be located away from areas where they can have negative impacts on species or habitats with special conservation value or ecological importance
Increased likelihood of disease from intensive fish farming. Escape of farmed fish and spread of diseases to wild fish	Fish farms should be designed to minimise escapees. Fish should not be farmed at high intensity to minimise the likelihood of disease
Breeding of escaped fish with wild fish that adversely affects the genetic robustness of the wild population	All reasonable efforts should be made to prevent the escape of farmed fish, including utilising new technology where available
Harm to the marine environment and to humans consuming the products from chemicals used to treat the fish	Finfish should be farmed at a density which limits the likelihood of disease and therefore does not require the use of antibiotics and other therapeutants, as has been the case up to this point in New Zealand

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1 http://www.aquaculture.govt.nz/government_response.php

2 Marine Farming Association <http://www.nzmfa.co.nz/industryinfo.asp>

3 Seafood Industry Council – Aquaculture <http://www.seafood.co.nz/aquaculture>

4 <http://www.seafood.co.nz/greenshell>

5 Marine Farming Association - <http://www.nzmfa.co.nz/faq.asp>

6 <http://www.niwa.cri.nz/ncfa/aquaspecies/pacific>

7 Some salmon are also farmed in freshwater locations. Ocean ranching was another method tried in New Zealand in the 1980s. This involved releasing large numbers of smolts into the sea to fend for themselves until adulthood and then relying on their homing ability to guide them back to their point of release to be harvested. This method proved inefficient as marine survival rates were too low.

8 New Zealand Salmon Farmers Association <http://www.salmon.org.nz/aboutsalmon.shtml#production>

- 9 New Zealand Greenshell Mussels – safety <http://www.greenshell.com/safe.html>
- 10 See information on finfish farming effects on marine mammals below
- 11 Lloyd B D, 2003
- 12 Du Fresne, 2008, 13
- 13 Forrest B et al, 2007, 28-29; Du Fresne, 2008, 14
- 14 Du Fresne, 2008, 2
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- 17 Keeley N, 2009, 25-27
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Sand mining

Overview

Sand is mined from the seabed in New Zealand for a number of purposes, depending on the quality and content of the sand. It is commonly used as an aggregate in the production of concrete, as a component of asphalt for roading and as an ingredient for other cement based products.

The sand required for such products must be very clean, and in particular must be free from high levels of mud and silt. It must also have a high quartz content. Such sand is quite rare, and growing increasingly so as intensifying land use causes more mud and silt to run off the land and into the sea.

There are many other uses of sand. Sand is mined for restoration of beaches, such as those in Auckland's inner city, which do not currently retain sand but are popular destinations for recreation.

It is also used in the manufacture of glass, which requires extremely clean silica sand. Kokota Sandspit, at the head of the Parengarenga Harbour in the Far North, is the only coastal source of silica sand with less than two per cent impurities within New Zealand.

The titanomagnetite contained in the black ironsand off the west coast of New Zealand is used for the manufacture of iron and steel. As yet, no iron sand mining operations have taken place in the coastal marine area. However, there is growing commercial interest in mining such sand from the seabed and some areas with significant commercial potential have been found.

Methods

Sand is usually mined from the seabed using a barge and tug operation that pumps sand slurry through a pipe up into the barge. The sand is pushed through wire mesh gratings which screen off coarse material. As the barge fills, the sand settles and the seawater drains back over the side. The barge is connected to a tug boat which tows the barge with the sand to its coastal destination.

Potential Environmental effects

Coastal erosion

One of the most noticeable environmental impacts which can be caused by sand mining is accelerated coastal erosion. This can occur when the mining activity reduces the amount of sediment available to be deposited on beaches from the coastal sediment system.

The profile of New Zealand's shoreline is constantly changing as a result of natural processes. When a storm occurs, high-energy waves rise up over the shore and sculpt out large amounts of sand, which are carried out to sea leaving eroded beaches and sand dunes. Often the sand which is removed by waves accumulates in underwater sand bars situated some distance offshore from the beach.

During calmer conditions, longer-period waves rushing up the beach drag with them sand from the seabed and underwater sand bars. The returning waves take sand back to the sea, but being less powerful, they leave some sand behind. When the tide recedes, the newly deposited sand dries on the shore and the wind blows it into the dunes. This natural process can gradually repair dunes eroded during storms, particularly where sand binding grasses trap the sand and retain it on the dunes.

In what are known as 'closed' systems, there is little or no new sand entering the sediment cycle. A fixed amount of sand is constantly recycled between the shore and the seabed during alternating stormy and calm periods. If sand is removed from these systems, there is less available to be deposited on the beach, and erosion of the coastline may result as seabed sand reserves are depleted.

In 'open' systems new sand is constantly entering the cycle. This can include sand that is being transported along the coast from other areas by marine currents, sand being carried down rivers into the sea, and sand being produced by the erosion of nearby coastal cliffs and headlands. In some areas, such as along the west coast of New Zealand, sand may also be transported into coastal areas from further out on the continental shelf. If the amount of sand removed from open systems by mining is less than the quantity of new sand entering the system from outside sources, coastal erosion is much less likely to occur.

Ultimately, to avoid erosion, it is desirable to maintain a positive sediment budget. That is to say, the amount of sand removed from the system, either by natural or human means, should be less than that available to the coast.

A key issue for any seabed sand mining proposal is therefore whether or not the affected sand system is open or closed, and if it is open, whether the

amount of new sand entering the system is greater than that proposed to be extracted. Unfortunately, marine sediment systems are very dynamic and often the available scientific information is not sufficient to provide conclusive answers to these questions.

Ecological effects

Sea bed mining can have a range of ecological effects which include:

- Depletion or destruction of shell fish beds lying within the sand being extracted
- Suspension of sediment in the water column, which reduces light levels, impacting on light-sensitive species and disrupting and smothering filter feeders
- Where mining results in coastal erosion, there can be negative impacts on shorebirds, on indigenous dune vegetation, and on the intertidal zone habitat. These effects will be exacerbated if hard protective barriers are built to protect buildings and infrastructure from coastal hazards
- Disruption to marine mammals such as Hector's and Maui's dolphins, through the movement of equipment, increased underwater noise levels, and ecological changes to the marine environment
- Disruption to fish populations, particularly if prey is affected or if sand mining is undertaken in fish spawning areas

Māori values

Mining may negatively impact on areas and resources of importance to iwi and hapū. These could include kaimoana areas and wāhi tapu or urupa sites which lie on or under the seabed. This may be through the direct impacts of the mining activity on the seabed or through accelerated coastal erosion caused by the removal of sand from the system.

Amenity effects

Seabed mining can potentially have the following amenity effects:

- Changes to the profile of the shoreline if erosion occurs – once healthy dunes may become eroded
- Noise from the operation of barges, motors and equipment, can affect residents and visitors to the areas, particularly if operations are close to the shoreline

- Barges and tugs passing regularly through the marine area can create an unwelcome visual impact, particularly in marine areas which are normally little used by vessels.

Cultural and recreational resources

Mining may impact on marine archaeological sites including sites of ship wrecks and structures and historic anchorages. Sand mining can also interfere with recreational uses of an area. For example, changing the underwater profile of the seabed may in turn have an impact on the location and shape of waves, which could impact on swimming activities and the quality of waves for surfing. Mining activities may also disrupt recreational fishing and dolphin-watching activities.

Decision-making framework

Sand mining within the coastal marine area is managed under the RMA. In order to undertake seabed mining within the coastal marine area, a proponent will normally need to apply to the regional council for a resource consent. Such a consent is required unless the activity is expressly permitted by a rule in a regional coastal plan and any applicable proposed regional coastal plan. Given the intrusive nature of sand mining activities, this would be unusual.¹

Extensive sand mining proposals will be listed as restricted coastal activities in the regional coastal plan. Under the New Zealand Coastal Policy Statement, the ‘disturbance of the seabed including removal of sand, shingle, or shell’² is a restricted coastal activity unless it:

- relates to an area of less than four hectares, extending less than 1000 metres over the foreshore and seabed, and amounts to extraction of less than 50,000 cubic metres per year; or
- relates to an area of less than 10 hectares, extending less than 10,000 metres over the foreshore and seabed and amounts to extraction of less than 300,000 cubic metres per year, and sand mining is listed as a discretionary activity in the proposed or operative regional coastal plan, with rules about where and when it can be carried out, and what conditions must be put in place to control adverse effects.

There is a slightly different application procedure for consents to undertake restricted coastal activities than for other resource consents. All resource consent

applications for restricted coastal activities must be publicly notified. The regional council must delegate its role of hearing and deciding the application to a body consisting of one or more council employees and/or hearings commissioners and this must include one person nominated by the Minister of Conservation. Applicants and submitters cannot request that an independent commissioner be appointed to consider a restricted coastal activity application, as they can with other types of resource consent applications (section 117, RMA).

When considering resource consent applications, the consent authority must have regard to the aims and purpose of the RMA. The purpose of the RMA is stated in section 5 as the sustainable management of natural and physical resources. Although in respect of most activities under the RMA, this includes the need to sustain the resource to meet the needs of future generations, minerals are explicitly excluded because section 5(a) states:

‘Sustaining the potential of natural and physical resources (excluding minerals) to meet the reasonably foreseeable needs of future generations’

This means that finite resources of sand can be exploited without concern about retaining sand for future use. However, the consent authority is still required to assess whether the sand mining activity proposal would promote the sustainable management of natural and physical resources in all other respects.

When considering an application for consent, the consent authority must also have regard to the relevant provisions of the applicable statutory instruments made under the Act, for example the NZCPS, regional policy statements and the regional coastal plan.³ The New Zealand Coastal Policy Statement contains a number of policies that are relevant to applications for sand mining activities:

- It is a national priority for the preservation of the natural character of the coastal environment to protect significant habitats for indigenous fauna (policy 1.1.2) which would include areas of importance to indigenous migratory species
- It is a national priority to protect characteristics of the natural character of the coastal environment which are of special spiritual, historical and cultural significance to Māori (policy 1.1.3(b))
- The natural character of the coastal environment must be protected, including the natural movement of sediments and natural substrate composition (policy 1.1.4)
- A precautionary approach is to be adopted (policy 3.3.1) – which is particularly relevant when accurate sediment budgets are difficult to determine

- Alternatives to the proposal must be considered when coastal permits are sought for the removal of sand, shingle, shell or other natural materials from lands of the Crown in the coastal marine area (policy 4.1.6)

Often the scientific evidence supporting an application contains considerable uncertainties, because the marine environment is a dynamic system subject to constant change, and obtaining scientific data can be expensive and difficult. Demonstrating a cause and effect linkage can be problematic. For example, is coastal erosion a result of sand mining or simply a natural process that is occurring as a result of other factors such as climate change and sea level rise? Lack of certain scientific information means that there is often disagreement about how coastal sand systems work, and how frequently they are replenished.

Experience of sand mining cases⁴ indicates that, for those opposing a mining application, it may not be enough to merely show that the other party's scientific evidence is incorrect. It may be necessary to provide a scientifically valid alternative interpretation of the data to persuade the hearings committee that the negative environmental impacts of the activity would outweigh the social and economic benefits to be gained by allowing more mining of sand.

Because there is little certain information about the effects that sand mining will have on the surrounding environment, and because there are relatively few sites where sand mining can take place, it can be difficult for consent authorities to give effect to the precautionary approach required by the NZCPS, when faced with the high demand for sand. Increasingly, consent authorities deal with this problem by granting consent accompanied by strict conditions providing that a programme of adaptive management and monitoring must be followed. Conditions can provide that reviews of the consent must be undertaken once particular extraction quantities have been reached.

The right to prospect, explore or mine minerals that are owned by the Crown is governed by the Crown Minerals Act 1991. Under that Act, the Crown establishes a minerals programme, which sets out policies, procedures and provisions to be applied in respect of Crown-owned minerals. In particular, the programmes address the efficient allocation of permits and the obtaining of a fair financial return for the minerals.⁵ They do not address environmental impacts. A minerals programme must be created as soon as practicable for minerals that are, or are likely to be, the subject of a mining application.⁶

The current minerals programme provides that mining for sand in the coastal marine area does not require a permit under the Crown Minerals Act, but mining for minerals that are a component of sand, such as titanomagnetite, does require a permit. Hence ironsand mining is governed by a different regime to the mining of sand for aggregate. The minerals programme also applies to

all other minerals in the coastal marine area, although permits for uranium and thorium minerals will not be granted.⁷

Elements of best practice

Because sand mining can have significant impacts on the marine environment, it is important that it is only undertaken in appropriate areas. Sand mining activities should avoid, where possible:

- Closed sand systems
- Sediment with a high silt content
- Benthic habitats of high ecological value
- Significant shell fish beds
- Fish spawning areas and areas of high fish density
- Areas frequented by marine mammals including whales, dolphins and sea lions
- Areas of high cultural or archaeological significance
- Areas of recreational significance including areas near high quality surf breaks

Where sand mining is authorised, strict conditions should be included as part of the consent, including a requirement for a robust monitoring regime to identify any negative environmental effects. Where there is considerable scientific uncertainty about environmental impacts a smaller operation should be undertaken, incorporating rigorous monitoring, to identify any adverse effects before the operation is allowed to expand to its full size.

As a matter of best practice, information from monitoring should be made publicly available in a timely fashion. In some cases, monitoring programmes have been overseen by tangata whenua and stakeholders.

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- 1 This is because they include the following activities: disturbance of the seabed by extracting sand (s12(1)(B) and s 12(3), removal of sand and other natural material such as shells (s12(2)) extraction of sand (s12(3) RMA, discharge of excess seawater, shells, sand and oversize material into water (section 15(1) RMA).
 - 2 New Zealand Coastal Policy Statement, Appendix 1 S1.6
 - 3 S 104(b) RMA
 - 4 *Oruawharo Marae Trust v Auckland Regional Council* A/083 2006
 - 5 Crown Minerals Act 1991 s12
 - 6 Crown Minerals Act 1991 s13
 - 7 Ministry of Economic Development, 2008, para 1.8
 - 8 *Oruawharo Marae Trust v Auckland Regional Council* A/083 2006



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Deep Sea Exploration and Mining

Overview

New Zealand's EEZ contains numerous seamounts, trenches, methane seeps and hydrothermal vents. These areas are often extremely rich in minerals that have a high commercial value. Until recently, technology has not allowed mining machinery to reach the depths at which the minerals are found, but new technology is opening up the economic potential of the seabed. Although there is currently no mineral mining activity within New Zealand's EEZ, several companies have made moves towards setting up commercial mining operations in the area.

Phosphorite is found in nodules on the Chatham Rise east of Christchurch. Rock phosphate can be extracted from phosphorite. Phosphate is an essential ingredient in fertilisers, and is therefore commercially valuable. The existence of the nodules was discovered in 1952, and studies have shown that in some places on the crest of the Chatham Rise, there are up to 80 kilograms per square metre of phosphorite, containing about 20 per cent phosphate.¹

Seafloor massive sulphides (SMS) are found on the Kermadec Ridge off the north-east coast of the North Island. Seafloor massive sulphides are undersea deposits of high grade polymetallic sulphides, rich in copper, zinc and lead, and with a high gold and silver content. They are produced in underwater volcanic regions, usually on seamounts, by hydrothermal plumes or black smokers.

Black smokers are chimney-like structures, formed when seawater seeps into the porous seafloor, and is heated. The hot water re-emerges through fractures, at a temperature of between 300 and 500 degrees Centigrade, and at high pressure. The water carries dissolved minerals, and when the hot pressurised water hits the cold sea floor, the minerals precipitate out as sulphide particles creating chimneys over time. Fine particles from the chimneys fall to the ocean floor and build up as deposits and mounds. Eventually the vent becomes inactive. Both inactive, and active vents with a low level of hydrothermal activity, have the potential to be mined.²

Manganese nodules are also found within New Zealand's EEZ. These are rocky lumps found on the seabed in areas where sedimentation rates are very slow. They are created when manganese dissolved in sea water reacts with oxygen to create insoluble manganese oxide. A dense bed of manganese nodules was discovered in the 1980s at the base of Bollons seamount, an enormous underwater mountain 600 metres south of the Chatham Islands. Manganese nodules often contain metals such as cobalt, nickel and copper, which have commercial potential. However the nodules found within New Zealand's EEZ have been found to be relatively low in valuable metals.³

Another resource of potential economic value is methane hydrates, a large quantity which are located on the Hikurangi Margin off the east coast of the North Island. Deposits have also been identified off Fiordland. The methane is formed when organic material in the form of mud is dragged deep underground with the Pacific Plate as it slides under the Australasian Plate. As the organic matter decays, it breaks down into simpler organic substances, including methane. The methane then travels up through fractures in the crust and soft sediments. Much methane becomes trapped in a frozen layer just below the seabed. But some of the gas escapes through the surface of the seabed in areas called 'cold seeps'. Commercially feasible technology to mine this resource is currently unavailable, but may be available as soon as 2020.⁴

At the time of writing, no mineral mining was being undertaken on a commercial basis within New Zealand's EEZ. However, a company has been prospecting for SMS deposits on the Kermadec Arc and Colville Ridge off the north-east coast of the North Island. Particular seamounts in these areas have been identified as promising areas for mining operations. The company has applied for mining licences to commence mining in specified areas within the near future.

Prospecting has also taken place in respect of phosphorites on the Chatham Rise. The first phosphate prospecting occurred in the 1980s, but technological limitations and resistance from the commercial fishing industry meant that it did not develop further.⁵ Presently, the New Zealand agriculture industry imports phosphates from Morocco (having been reliant on imports from Nauru until its supplies were exhausted) and so a phosphate mining operation within New Zealand could help meet strong demand.

In August 2007 a company applied for a prospecting licence over 3,048 square kilometres of seabed on the Chatham Rise to locate economic deposits of phosphorite, along with fluorine, rare earth minerals and other valuable by-products. Another company has also applied for a prospecting licence for phosphorites in the same area.⁶

Methods

Technology that can be used to mine the seabed efficiently is only just being developed and work is still being undertaken to ascertain the most effective method of mining. Prospecting has been undertaken using geophysical and sonar mapping as well as a remotely operated vehicle (ROV) that can grab samples from the seabed and return them to the surface.

Generally it is proposed that equipment used in the mining of marine diamonds, in the offshore oil and gas industries, and for undersea trenching and cabling will be adapted to the mining of seabed mineral deposits.

The company which proposes to mine an area on the Rumble II West seamount on the Kermadec Ridge for SMS deposits, has investigated the use of a cutting and suction system to pump material from the seabed to the surface, and onto a surface support vessel. The chimneys formed by black smokers, along with material from the seabed, could potentially be dug up by a marine excavator. Hardened material, such as chimneys, would then be put into a crushing device on the seabed. The material would then be combined with seawater in a vacuum system and pumped up as a slurry to the surface. Barges would then take the material to an onshore plant to be processed.

Potential environmental effects

Seabed mining has a range of potential environmental effects, which primarily arise from the direct impacts of excavating rock from the seabed and from the indirect impacts of suspending material in the water column and mixing oceanic waters. As methodologies for mining are still being developed, it is unclear what their specific environmental effects will be.

Direct benthic impacts

Seabed mining will result in significant physical changes to the direct area that is being mined. The removal of rock from the seabed will not only kill organisms located on the sea floor, but it will result in the permanent modification of the profile of the seabed. Recovery of the benthic communities could take considerable time.

The ecological significance of these effects largely depends on the distribution of the habitats and species that are impacted upon. If they are widely distributed in areas that are not subject to mining, or other activities which have an impact on the seabed such as trawling, the ecological impact may not be great. If however, the affected species are endemic to the area, mining activity could result in a significant reduction in the range and number of individuals of a species. It could even result in extinctions.

In particular, deep sea vents, which are the source of SMSs, are associated with highly productive and endemic biological communities that rely on sulphur compounds as an energy source. These communities were only

discovered in 1976 and remain poorly understood. Some scientists argue that such vent systems are where life first evolved on Earth.

As only a very small proportion of the EEZ has been thoroughly sampled scientifically, it is important that adequate research is undertaken to determine the range of species present in target areas and their frequency elsewhere, before mining activity takes place.

Suspended sediment

Mining activity may result in the suspension of particles in the water column as rock is extracted, and possibly crushed on the seabed, before being transported to the surface. If this happens, the heavier particles will sink rapidly to the sea floor and are likely to smother benthic organisms. Lighter particles may become suspended in a plume, which could travel some distance with ocean currents before settling out. Such sediment can be a considerable problem for suspension feeding organisms, such as sponges and corals, as the fine particles can clog their filtering apparatus. It may also impact on important vent communities and deep-sea fish living within the water column.

Where the sediment remains in the water it can reduce light levels in upper levels of the sea. Bacteria can grow on the suspended particles resulting in oxygen depletion within the plume. As the sediment resettles on the sea floor it could smother and even entomb slow moving and sessile organisms living within or on top of seabed sediments.

Mixing oceanic waters

Mining activity could change the nutrient values of seawater, by raising water containing concentrated nutrients from the deep sea area to relatively nutrient poor areas near the surface. This could result in increased planktonic growth in the surface water. If water is pumped back down to the deep ocean area, it will be warmer than the surrounding deep sea water and could have localised impacts.

Other impacts

There may be other ecological impacts resulting from increased light and noise in the deep ocean environment. These are poorly understood but could impact on marine mammals that are dependent on sound for locating prey, navigating and communicating.⁷ There is also the risk of accidents leading to spillages and loss of equipment within the marine environment.

Decision-making framework

Deep sea exploration and mining of minerals which takes place outside the territorial sea are managed under the Continental Shelf Act 1964. This legislation vests the right to explore and exploit minerals from the seabed of the EEZ and continental shelf beyond in the Crown (section 3).

Any party wishing to undertake prospecting or mining activities within this area must obtain a licence issued by the Minister of Energy. Applications for prospecting and mining licences are processed by Crown Minerals located within the Ministry of Economic Development.

Prospecting permits are generally granted to allow the applicant to undertake investigations of the area to establish whether a mining operation would be feasible. They are often granted over very large areas of seabed and will normally involve seabed sampling, geophysical mapping of the area, and other fairly non-invasive activities. Mining permits allow commercial mining enterprises to be undertaken on the site, for a stated period of time.

The Minister will ordinarily grant a permit if he or she is satisfied that the applicant has an approved work programme, that the subject area is appropriate and adequate to enable activities authorised by the permit to be carried out, and the proposed exploration or mining will result in increased knowledge or development of New Zealand's mineral resources. When allocating such permits there is no legal requirement that the Ministry of Economic Development notify the public of applications for permits, or consider environmental matters.

The provisions of the Resource Management Act 1991 only apply out to the limits of the territorial sea. Within the EEZ, the Continental Shelf Act 1964 applies, and there is no equivalent of the RMA to require public consultation and environmental assessment to help determine what activities can take place there. In respect of seabed mining this has not been a problem until recently, because mining has not been technically feasible. However, this situation is changing, and there is growing interest in exploiting the deep seabed.

In June 2008 the previous government agreed to the drafting of a Bill regulating environmental effects in the EEZ. Work has progressed on the Bill, which is part of the government's legislative programme, but at this time no formal decisions have been taken on the final policy or on timing.

Elements of best practice

The following best practice considerations can be applied to deep sea exploration and mining activities:

- Require a thorough assessment of the environmental effects of any exploration or mining proposal
- Consult with the wider public before making a decision on exploration and mining applications
- Ensure that sufficient research has been undertaken to identify what species are likely to be present in the area to be mined and their distribution
- Avoid areas of high ecological value including areas of high endemism such as active vent sites
- Consider the cumulative impacts of mining activity and other activities such as deep sea trawling on benthic habitats
- Minimise the area which is directly affected by mining activity
- Minimise the mixing of deep and surface waters
- Minimise the amount of sediment which is released into the water column
- Avoid disposing of mining tailings at sea
- Monitor and assess the impacts of a pilot operation before expanding to full production
- Undertake sufficient monitoring so that any significant adverse effects on the marine environment can be identified and addressed in a timely manner
- Support the efficient use and recycling of minerals to minimise the need for deep sea mining activity

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Major Marine Developments

Overview

A range of marine developments are commonly undertaken near the coastal edge to provide for commercial, residential and recreational activities associated with the sea.

Wharves, jetties and terminals provide for commercial shipping. There are currently 13 major commercial ports around New Zealand's coastline as well as many smaller wharves and ferry terminals dotted around the coast.

Marinas, moorings, floating pontoons and boat ramps provide for recreational boating activity. New Zealand has one of the highest boat ownership rates in the world, and boats need places to be stored, both on and off the water. Pressure for marinas has been greatest on the east coast of the northern part of the North Island where the greater proportion of New Zealanders live, and where the weather and coastline is particularly favourable for boating. In areas such as the Coromandel Peninsula, there have been high levels of conflict over marina proposals.

Canal developments create new coastal edge for waterfront houses and associated boat moorings. This kind of development is relatively new in New Zealand, with the first canal development being built at Pauanui on the Coromandel Peninsula during the early 1990s. It has become increasingly popular, particularly as accessible waterfront land has become scarce and very expensive, near main population centres. There are now several canal developments which are in the process of being constructed.

Other structures, which are commonly placed in the coastal area, include bridges, causeways, pipelines, groynes and seawalls.

Despite the disparate uses for which these developments are intended, such projects often involve similar types of work. These include the placement of structures within the marine area, the dredging of the seabed, and the reclamation and declamation of land. The effects of these activities are discussed in this chapter.

Methods

Large fixed structures

A wide variety of structures are placed in the marine area. Some may have relatively little impact on the marine environment, such as moorings and

navigational aids, whose construction does not alter the bathymetry of the seabed, shade areas of water or prevent natural water flows. But there are others which have greater potential to alter the coastal marine environment and the natural processes that govern it.

Large fixed structures that are attached to the seabed, such as wharves, breakwaters, seawalls, causeways and groynes, can have a significant impact on the coastal marine area. Some developments, such as ports and marinas, require the construction of a large number of structures within a confined marine area, together with extensive development above mean high water springs.

A marina development is likely to include numerous piles and floating pontoons to create marina berths and fueling facilities, breakwaters to protect the boats from wave damage, slipways, haul out facilities and washdown areas for boat maintenance, boat ramps, and onshore facilities such as hard stand areas, administration and toilet blocks, shops, cafes and car parks.

Canal developments typically include seawalls around the edges of the canals, piles and floating pontoons to provide berths adjacent to the canal-side homes and may include bridges or causeways to provide pedestrian and vehicle access across the canals. Onshore structures typically include houses, apartments and commercial developments. A marina may be included as an integral part of a canal development.

Dredging

Dredging of the seabed is commonly undertaken to increase, maintain or restore navigable water depths in the coastal marine area, particularly in association with ports, marinas and canal developments.

Marinas are typically built in shallow coastal waters adjacent to land. Initial dredging is usually required to create a basin deep enough for boats to moor and navigate within. Such dredging is also usually required to create a navigable channel to connect the marina basin with open water.

Canal developments in New Zealand have, to date, been constructed on low-lying land adjacent to estuaries, inlets and harbours. Some areas, historically, were coastal wetlands. The canals are dug out of the soil, lined with seawalls, and then flooded when the area is opened up to the sea. This approach has avoided the need for canals to be dug within the existing marine environment. Seabed dredging is usually required, however, to create a navigation channel between the canal development and the open sea.

To accommodate the increasing size of international ships, capital dredging has also been undertaken at commercial ports to increase the navigable depths within major shipping channels and near wharves.

Maintenance dredging is often required to maintain access to ports, wharves, marinas and canal developments. This is particularly the case in enclosed harbours and sheltered bays, where there is a large quantity of sediment running off the land into waterways, and little energy from waves or currents to disperse the sediment before it settles onto the seabed.

Reclamation and declamation

Reclamation and declamation of coastal land is used to alter the geography of the coastline to provide for development, by adding or removing material from the coastal marine area. Reclamation has been undertaken extensively in many parts of the country to provide for extensions of the urban environment. Indeed, in cities such as Auckland, Wellington and Dunedin, reclaimed land forms a significant part of the urban area.

Reclamations and declamations are also often integral components of ports, marinas and canal developments. Reclamation and declamation projects tend to occur in estuaries, wetlands and other parts of the coastal marine area where the water is shallow and the new structures will be protected from the worst of the coastal weather.

Potential Environmental effects

Degradation of natural coastal landscapes

The placement of large structures into the marine environment, and changes to the natural form of the coastal edge through the construction of reclamations and seawalls, can have a major impact on natural coastal landscapes. Such human-made elements can create significant visual intrusion, replace natural rounded shapes and forms with straight lines and hard edges, and disrupt natural coastal patterns and processes. Many of the impacts are irreversible in practical terms.

Large single structures such as wharves and boat ramps may be intrusive when seen from both the land and from the sea. They are also likely to bring with them increased activity in the form of cars, boats and people. But it is the cumulative effect of many structures along the coastline that can have the most detrimental impacts on natural coastal landscapes.

Marinas, in particular, often require reclamations and numerous structures to be placed within the coastal marine area. These are occupied by a high density of boats and supported by a wide array of facilities. They are commonly accompanied by residential development. This can have the effect

of 'domesticating' the natural coastal landscape to the extent that few natural features near the coastal edge, apart from the seawater itself and natural tidal rhythms, can be evident.

Canal developments also have a major impact on natural coastal landscapes, through the creation of canals, high density housing, numerous boats and floating pontoons and commercial facilities. The marine area and coastal edge within these developments is often entirely artificially created.

Because of the major impact that these kinds of developments can have on natural coastal landscapes, and the irreversibility of these impacts, it is important that they are only built when really necessary, and that those developments which do occur are carefully located in areas where natural coastal landscapes have already been compromised. A related benefit of such intensive development can be the reduction of development pressure elsewhere.

Public access

Structures built near the coastal edge can obstruct public access both to the marine area and to areas within it. Port developments typically exclude public access to the coastal edge for safety and security reasons. Marina and canal developments vary in the extent to which they provide for public access.

Public access to marina berths is often excluded, particularly at night, for security reasons. The marina berths themselves are occupied by licencees, and are not freely available to other boat users, although some may be available to visiting boats for short-term periods. Some marinas incorporate public walkways, boat ramps, parking, picnic areas and cafes as part of the complex, and this enables members of the public to access and enjoy the coastal environment.

Public access is typically not available around the coastal edge of canal developments as privately-owned residential properties extend right down to the water's edge. The waterways themselves are usually in public ownership and open to all users. Some canal developments also provide small public parks alongside sections of the canals. At the Whitianga waterways development there are proposals to provide public floating boardwalks adjacent to restaurants and shop complexes, where visiting vessels can tie up, and where members of the public can enjoy the coastal edge.

There has been considerable controversy over the extent to which members of the public should have access to private wharves and jetties. The general legal principle is that members of the public do have the right to use private jetties to gain access to the adjacent coastal marine area, so long as they use them in a reasonable manner, and they do not unreasonably impede the owner's access to and use of the jetty.¹ But the situation in any specific case will be affected by the conditions of the coastal permit authorising the construction of the jetty.

Damage to marine ecosystems

Reclamation causes the direct loss of coastal habitat in the marine area that is reclaimed. It can also disrupt the sensitive ecosystems of transitional zones between the land and the sea which, in their undisturbed state, contain a diverse range of organisms and are very ecologically productive. These areas can provide important nursery areas for marine life and habitat for seabirds. Natural systems in these transitional zones can also be important for trapping sediment runoff and controlling erosion. For these reasons, reclamations and major structures should be located away from ecologically important inter-tidal and marine habitats such as salt marshes, shell banks, sea grasses, shellfish beds, mangroves and major fish habitats. They should also be located away from bird roosting, feeding and nesting sites.

Causeways result in the direct loss of marine habitat through burial, and can reduce wave action and the amount of flushing in the marine area on the inland side of the structure, resulting in the build up of sediment and sometimes eutrophication. This can create new coastal wetland habitat, but if the trapped sediments are contaminated, the resulting habitat can be of poor quality. The amount of flushing will be affected by the size of the culverts built under the causeway.²

Human-made structures placed in the coastal marine area may provide new habitat for marine life by creating shelter and surfaces on which vegetation and benthic communities can establish themselves. But they will also impact on the natural substrate on which they are placed. Large fixed structures such as wharves and breakwaters may shade areas of water. The impacts of shading will depend on the habitat, but most plants such as mangroves, sea grasses and sea weeds will deteriorate or die if shaded for significant periods.

Dredging necessarily removes flora and fauna from the seabed and changes its bathymetry. With the exception of deep burrowing animals, dredging may completely remove all life from the sea floor. The recovery rate of the benthic communities will depend on the particular nature of the ecosystem: some ecosystems dominated by opportunistic species may recover in only a few months, whereas those populated by slow growing, sensitive species may take years, or may never recover.

Dredging can also discharge sediment into the water column, during the dredging process itself and when sediment slops off hoppers and barges carrying the dredged material. Where sediment is released into the water column, it can affect the clarity and turbidity of the water, potentially reducing light penetration of the area.

Material dispersed by dredging may settle over areas around the dredged site, blanketing the benthic animals and plants that live there. The effects of this will

vary depending on the resilience of the communities and the conditions that they are used to but may cause stress, reduced rates of growth, or fatalities. Where sediment is released as a result of maintenance dredging, the effects are likely to be minimal as the benthic communities in the area have already been disturbed.

The most serious effects of smothering by dredged sediment are likely to occur where currents or tides cause the dredged material to settle some distance away from the dredge site, over habitats that have not been smothered before. Some animals are more sensitive to siltation than others. Shellfish, which have delicate feeding and breathing apparatus in particular, dislike increases in siltation. Similarly, sediment can become trapped in the gills of young fish causing increased fatalities, and smothering of spawning or nursery areas for fish can result in the death of eggs and larvae. In addition, smothering of intertidal areas may result in decreased availability of food for the birds and fish that feed there.

The release of organic rich sediments in the water column can make more nutrients available to marine organisms. This has the potential to cause algal blooms or result in localised eutrophication of the marine environment. However, the effects of this will generally be minimal, as tides and currents will disperse the sediment before it can have serious adverse effects.

In areas where maintenance dredging is undertaken, such as at the entrances to ports and marinas, the sediment on the seabed often has elevated levels of contaminants. For example, metals, chemicals, dioxins, organochlorines and PCBs are all commonly found in the inner parts of harbours. Dredging disturbs the sediment containing these toxins, and can spread them further around the area, or release them into the water column.³ The effects of these toxins will vary depending on their type and quantity, but there are a large number of common contaminants that are highly dangerous to marine life – for example copper and zinc can build up to toxic quantities in shellfish, and polycyclic aromatic hydrocarbons have been linked with tumours in fish.

Changes to coastal processes

Structures such as jetties, breakwaters, seawalls, reclamations and groynes may disrupt wave energy, currents and tidal flows. This, in turn, has the potential to alter the transport of sediments along the coast, resulting in coastal erosion in some areas and higher levels of sedimentation and accretion in others. Reclamation inside harbours and inlets also reduces water space and consequently the volume of water that moves in and out of the area on each tide, thereby affecting tidal flows and flushing. This volume of water is sometimes referred to as the tidal compartment.

The size, shape and orientation of structures relative to the shoreline and natural water flows will determine its hydrological and sedimentation effects.

A notable example of how a structure can change sedimentation flows is the breakwater that was constructed as part of the Timaru port in the late 1870s. This resulted in major erosion at Washdyke but also in the creation of a new beach at Caroline Bay.

Dredging can also alter the processes that govern the movement of sediments in the coastal marine area. Generally dredging for capital projects, in comparison with maintenance dredging, is likely to have the greatest effect on coastal processes. Maintenance dredging usually involves frequent removal of small amounts of material, involving small changes in water depth that the system can adapt to, adopting the dredging routine as part of its equilibrium.

In contrast, dredging for capital works can cause a significant change to the equilibrium of the system. It can alter the sediment transport regime of an area. It may cause sea water to travel further up the deepened channel than usual, as well as increasing shoreline wave action and tidal range.⁴ Altering the bathymetry of the seabed, and the flow of sea water, can also have an impact on wave patterns which consequential effects on areas such as surf breaks. This was an issue raised in connection with a marina proposal at Whangamata.

Degradation of water quality

During the construction phase of a marine development, there may be a risk that sediment or pollutant-laden water will seep out from the construction site, causing sedimentation and contamination of the wider marine area. It is therefore important that preventative measures, such as the use of silt curtains and settlement ponds, are adopted during marine construction projects. Nowadays, such discharges will normally be controlled by strict resource consent conditions.

After construction has been completed, the potential for degradation of water quality surrounding the structure will depend on what it is used for. Wharves, jetties and marinas typically attract large numbers of people and boats that, in turn, may result in significant discharges into the marine area – including fuel spillage, sewage from boat toilets, rubbish and leaching of antifouling paints which contain toxic chemicals. The parking and boat washing areas found at marinas may also be a source of contaminated runoff into the sea.

If a marina or canal development site is located in a poorly flushing area, or if the design of canals and structures prevent adequate flushing, the area can become a trap for sediment and pollutants causing the water quality of the area to decline significantly. Poor water quality has been an ongoing issue with some of the early canal developments in Queensland. The problem can be minimised through the use of wide canals which have good linkages to allow a circular flow

of water throughout the entire development. The Westpark marina in the upper Waitemata harbour has had ongoing problems with high levels of sedimentation.

These effects can be lessened through good location and design, to ensure that the development is well-flushed by tidal action and that any contaminated run-off is collected and treated. Marinas also require ongoing vigilant management to ensure that spillages into the marine area are minimised through the adoption of good practice.

Disrupting the relationship of tangata whenua with the marine environment

The balance between marine development and impacts on the marine environment are important to coastal Māori. Gathering of kaimoana, both for sustenance and to provide for visitors, is an integral component of life. When those resources are compromised, such as when structures are placed over shellfish beds or in a place that obstructs access to them, a way of life is also compromised. Activities which contaminate water quality can result in kaimoana being unsafe to eat and in a reduction in the size of populations.

Coastal works such as reclamations and the construction of seawalls can result in the loss of intertidal areas, which reduces the areas where kaimoana can be harvested. Construction activities can also result in the destruction of culturally important sites such as wāhi tapu and traditional waka linkages and mooring places.

Degradation of marine heritage

Marine developments can negatively impact on marine heritage if reclamations or structures are built in the vicinity of archaeological sites, wāhi tapu and other sites of historical significance. As the location of marine heritage sites is poorly recorded, research will often need to be undertaken in respect of individual marine development proposals.

Decision-making framework

Under section 12 of the Resource Management Act, a range of activities in the coastal marine area are prohibited unless they are granted a coastal consent or the work is allowed by the regional coastal plan, including:

- Reclamation or drainage of the foreshore or seabed
- Construction, alteration or demolition of a structure in the coastal marine area

- Disturbance of the foreshore or seabed (eg by excavation or drilling) in a manner that is likely to have an adverse effect (except for lawfully harvesting a plant or animal)
- Deposition of substances that are likely to have an adverse effect on the foreshore or seabed
- Introduction of any exotic or introduced plant in, on, or under the foreshore or seabed
- Generally destroying, disturbing or damaging any part of the foreshore or seabed which is likely to have an adverse effect on historic heritage

The 'coastal marine area' is defined in the RMA as being the coastal waters within the territorial sea, the seabed, and the foreshore up to mean high water springs. At river mouths the coastal marine area extends up the river to a distance of one kilometre, or five times the width of the river, whichever is the lesser.

The NZCPS identifies a range of activities that have the potential to cause particularly significant or irreversible adverse effects on the marine environment as restricted coastal activities. These include:

- Reclamations of the foreshore or seabed exceeding 1 hectare or extending more than 100m in any direction
- The erection of a structure containing 4 hectares or more of the coastal marine area (except where the structure is less than 8 hectares and the plan specifies it as a discretionary activity, providing criteria for determining location, materials and use of the structure, and likely adverse effects)
- The erection of solid structures parallel to mean high water springs more than 300m long (except where the structure is less than 1000m long and the plan specifies it as a discretionary activity, providing criteria for determining location, materials and use of the structure, and likely adverse effects);
- The erection of solid structures sited obliquely or perpendicular to mean high water springs of more than 100m in horizontal length (except where the structure is less than 1000m long and the plan specifies it as a discretionary activity, providing criteria for determining location, materials and use of the structure, and likely adverse effects);
- The erection of structures used for storing petroleum and chemicals in quantities greater than 50,000 litres (except where the structure is used for storing less than 100,000 litres and the plan specifies it as a discretionary activity, providing criteria for determining location and likely adverse effects)

- Disturbance of foreshore and seabed amounting to the removal of more than 50,000 cubic metres of sand per year extracted from areas equal to or greater than 4 hectares of seabed, extending 1000m or more over the seabed (except where the activity amounts to maintenance dredging, or amounts to the removal of less than 30,000 cubic metres of sand and is specified in the plan as a discretionary activity, providing criteria for when and where the activity can be carried out, and conditions to control adverse effects)
- Depositing more than 50,000 cubic metres of material on the sea floor (unless specified in the plan as a discretionary activity, stating the time during which the activity can be carried out and the location, together with measures for identifying and controlling adverse effects)
- Exclusive occupation where public access would be excluded from areas of the coastal marine area of more than 10 hectares, or where public access would be excluded from more than 316 metres of coastline, or where exclusive occupation of more than 50 hectares would restrict public access to or through that area.

Those who wish to undertake activities in the coastal marine area that are not permitted activities under the regional coastal plan must apply for a coastal permit. Unless the activity is a restricted coastal activity, the application will follow the normal resource consent application process under the RMA.

Slightly different rules apply to an activity that is a restricted coastal activity. All resource consent applications for restricted coastal activities must be publicly notified. The regional council must delegate its role of hearing and deciding the application to a body consisting of one or more council employees and/or hearings commissioners and this must include one person nominated by the Minister of Conservation. Applicants and submitters cannot request that an independent commissioner be appointed to consider a restricted coastal activity application, although such a request can be made for other resource consent applications (section 117).

When the consent authority considers an application to undertake major construction works in the coastal marine area, it will evaluate the effects of the activity in accordance with the requirements of the RMA. The social, economic, cultural and recreational benefits of the activity will be weighed against the effects that the activity will have on the environmental values of the locality. However, as required by the RMA and by the NZCPS, consent authorities will need to consider carefully the finite and sensitive characteristics of the coastal marine area and ensure that they do not permit the construction of works in the

sea that are not functionally necessary. An applicant should therefore show that alternatives outside the coastal marine area have been considered adequately.⁵

When considering applications to undertake development of structures in the coastal marine area, the consent authority will need to consider the cumulative effects of allowing development in the coastal marine area, in recognition of the fact that it may well be the impact of several small structures in the coastal marine area taken together that has the most effect on the natural character of the area.

The NZCPS requires that sprawling or sporadic development in the coastal environment be avoided and that development in the coastal marine area should occur in localities where the natural character of the area has already been significantly compromised (policy 1.1.1). In practice this means that development should not occur where there are existing structures that adequately perform the function for which the structure is proposed, and structures in the marine environment should not be larger than that required to fulfil the function for which they are designed.

The consent authority will consider whether the structure will have adverse effects on water quality, and that where relevant, tidal flushing rates are adequate to ensure that pre-existing water quality is maintained or enhanced in the vicinity of the development. The consent authority will also consider any adverse effects on indigenous vegetation and fauna and other conservation values.

In addition, the authority will consider the extent to which the activity will affect and be affected by coastal processes, including natural hazards, erosion and inundation, and the effects that climate change and sea level rise are likely to have on the development and its surroundings.

Elements of best practice

Those managing the development of the coastal marine area must walk a fine line between promoting and facilitating the close bond that New Zealanders have with the sea and coastal activities, and preventing overdevelopment that ruins the values for which it is enjoyed. The coastal environment is a finite and fragile resource, so the development of structures in it should be carefully planned and limited to those which are strictly necessary.

A first step in achieving best practice coastal management for large marine developments is to carefully plan for their location in a strategic manner. This can include delineating port management areas and marina areas in regional coastal plans, with provisions clearly indicating that developments proposed outside the delineated areas, are very unlikely to obtain consent.

Potential environmental effects	Best practice response
The natural character and landscape values of the coastal environment are degraded by built structures being located in the coastal marine area	Locate reclamations and structures within the coastal marine area only if they are absolutely necessary and there is no practical alternative Locate reclamations and structures where natural character has already been compromised Design reclamations and structures to visually 'fit in' with the natural coastal environment – by minimising their size and where possible adopting natural shapes
Inadequate provision is made for public access to and along the coastal marine area	Make provision for public access around the coastal edge and to the marine area wherever possible Provide public access to wharves and jetties unless exclusion can be justified for health or safety reasons
Marine ecosystems and species are damaged by the effects of marine development	Locate marine developments away from ecologically important marine habitats such as salt marshes, shell banks, sea grasses, shellfish beds, mangroves, fish spawning areas and bird roosting, feeding and nesting sites Locate marine developments away from marine mammal habitats Design marine developments to minimise impacts on natural coastal processes including sediment transport, tidal flows and local currents Minimise shading of marine habitat in areas where mangroves, sea grasses or seaweeds are present Locate and design marine developments to minimise the amount of capital and maintenance dredging required Time dredging operations to minimise disturbance to sensitive marine animals
Marine developments disrupt natural coastal processes	Avoid reclamations where possible, and where they are constructed, minimise their size Carefully design reclamations and coastal structures in terms of size, shape and orientation to minimise impacts on natural seawater flows Locate and design marine developments to minimise the amount of capital and maintenance dredging required
Marine developments result in degraded water quality	Carefully design marinas and canal developments to allow the free flow of tidal water and only site them in well flushing areas Where possible, avoid dredging in areas where there are contaminants concentrated in seabed sediment Use the best available technology to reduce the amount of sediment released into the water column Adopt environmental management programmes to avoid discharges from marinas into the marine areas from ongoing operations
Marine development disrupts the relationship of Māori with the marine area	Ensure that the development does not adversely impact on sites of significance to Māori Ensure that the development does not obstruct access or adversely impact on kaimoana areas of importance to Māori Ensure that development does not degrade water quality
Marine developments result in the loss of historic heritage	Undertake sufficient research to identify any marine heritage sites in the vicinity of the proposed development area. Locate marine developments, where possible, away from any sites of heritage significance

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1 See *Hume v Auckland Regional Council* [2002] 3 New Zealand Law Reports, 363

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Vessels and Offshore Installations

Overview

Over 99 per cent of New Zealand's exports and imports are transported by sea. There are 13 major commercial ports located around New Zealand's coastline and the country receives around 5,800 port visits from international ships each year. New Zealand has around 3,800 registered small commercial vessels and a very large recreational fleet of some 450,000 vessels.¹

In 2002/03, 13 million tonnes of oil and associated products were transferred to and from tankers in New Zealand waters in around 600 cargo transfers. In addition, 590,000 tonnes of bunker fuels and lube oils were loaded in over 3,000 bunkering operations. A variety of toxic chemicals and noxious liquid substances are also shipped in New Zealand waters.

More than 140 introduced marine species have been recorded in New Zealand waters. Many of these are relatively benign, but some have become invasive pests, outcompeting indigenous species. Many of the new species are thought to have arrived in New Zealand on ships.

Potential environmental effects

Vessels and offshore installations such as oil rigs can result in the pollution of the marine environment. Other potential environmental impacts include biosecurity risks and impacts on marine mammals.

Oil and chemical spills

Diesel, petrol, oil and other toxic chemicals can be spilled into the sea from vessels and offshore installations. Every year many small spills occur, with over 80 reported in 2006 alone. Major oil spills can occur when vessels collide or run ashore and hole their tanks. The risk of oil spills from tankers is reduced if vessels have double hulls. Spillages can also occur when oil cargoes are transferred to and from tankers and during vessel refuelling. The last big oil spill in New Zealand occurred in 2007 when 23 tonnes of oil was discovered along a 10 kilometre stretch of Taranaki coastline and was traced to the floating production, storage and offtake facility *Umuroa*.

Recreational vessels also frequently release oil into the marine environment when oily water is pumped from bilges and when fuel tanks overflow during refuelling.

Oil discharged into the marine area can have serious impacts on wildlife and habitats, depending on where the oil ends up. In open sea the oil is broken down by micro-organisms such as bacteria and fungi into substances that are not hazardous. Sunlight, wind and wave action also help to break down oil. Fish in open waters tend to swim away from oil spills. Oil spills at sea can therefore break down naturally and may cause little ecological damage.

Major ecological impacts can occur, however, if an oil spill moves onto the shoreline. The seriousness of the impact depends on the type of species and habitat affected.

Birds are particularly susceptible to oil spills. Oil damages the waterproofing on their feathers. As a result the birds can get cold and are prevented from swimming, floating or flying. The toxic components of oil can cause serious or fatal illnesses in birds. Oil can also contaminate the habitat and feeding grounds of coastal seabirds.

Oil spills can also have a major impact on sheltered environments such as estuaries where the oil can become trapped. Mangroves, worms, molluscs and crustaceans can be killed as a result. Inshore shellfish beds can become contaminated.

Sewage discharge

Ships, fishing boats and recreational boats can discharge raw sewage into the sea from on-board toilets. This is only likely to constitute an environmental hazard in sheltered areas close to the shore where there is poor flushing and/or recreational users of the marine area, or within the vicinity of marine farms or marine protected areas. It is, however, of significant concern to tangata whenua for spiritual and cultural reasons.

Garbage

Garbage can be thrown overboard, including food scraps from ships' galleys and other waste. Plastic waste is a particular hazard to the marine environment. Plastic can be eaten by seabirds, marine mammals and fish which die as a result. Beaked whales, for example, can mistake plastic bags for jellyfish. Marine mammals and seabirds can become entangled in plastic waste. Plastic waste can also settle on the seafloor smothering marine organisms. Much of the plastic waste in the ocean is sourced from land, but some comes from vessels.

Antifouling

Antifouling paints are applied to the underwater sections of the hulls of commercial and recreational vessels to prevent or slow down the growth of

fouling organisms including barnacles. The toxic substances in antifouling paints can leech into the surrounding seawater especially near marinas and can accumulate in seabed sediments.

Biosecurity risks

Vessels arriving in New Zealand from overseas ports can bring with them new marine organisms, which are transported from other countries on the ship's hull or in ballast water.

Ballast water is carried in empty ships to provide stability. It is pumped into the ship before the voyage begins and can be discharged when the ship arrives at its destination port. The seawater which is pumped into the ship at one port can contain marine organisms that are then expelled at another port if the seawater is discharged.

Offshore installations such as oil rigs can also create significant biosecurity risks. As these large structures are stationary in coastal waters for lengthy periods of time, they tend to accumulate extensive biofouling. Once a rig has finished its job in one location it is typically slowly towed through the ocean to its next destination, and carries with it communities of marine organisms attached to the structure.

Once they have arrived in New Zealand waters from overseas, invasive organisms can be transported around the coast by both commercial and recreational boats which have fouled bottoms.

Dumping waste at sea

The dumping of material dredged from the sea floor is the most common form of waste disposal at sea in New Zealand. The material is often sourced from the dredging of ports, harbours, marinas, canal developments and other coastal waterways to provide navigable channels. Some of the dredged material can be contaminated as a result of pollutants entering the marine area and accumulating in seabed sediments. Much of this sediment is dumped within the territorial sea, but some is disposed within the EEZ. Dredged material is also, sometimes, used in reclamations and to renourish beaches. In recent years, disposal of dredged material on land has become more common.

Other items which are disposed of at sea from time to time include unwanted vessels. These can result in the physical disturbance of the seabed, the discharge of contaminants into the marine environment and may also pose a navigation hazard for ships. Carefully planned and implemented sinkings can however create new dive sites.

Dumping waste at sea can result in the smothering of benthic organisms within the disposal site, and also further afield if the waste material disperses. Long-term impacts can also occur on water quality if sediment is re-suspended into the water column and if contaminants leach out of the dumped material. These contaminants can accumulate within the seabed sediment and in animal and plant life.

Ship strike

Ships moving through the water can hit marine mammals and injure or kill them. Ship strike happens because dolphins and especially whales are moving at a much slower pace than vessels. They also spend considerable time on the surface, which can put them directly in the path of boats. The greatest number of marine mammal deaths from ship strike are caused by large ships travelling at more than 12 knots. This is a particular problem in the Hauraki Gulf because it houses the busiest port in the country and it is also important habitat for dolphins and the nationally critically threatened Bryde's whale. Between 1997 and 2007, 25 whale deaths were recorded in the Gulf, of which 14 were found to have died from injuries consistent with collision with a large vessel.²

Vessel wake

Vessels moving at fast speeds through coastal passages can create a large wake which can result in accelerated coastal erosion. This potential problem is particularly associated with fast ferries which can travel at up to 40 knots, much faster than conventional vessels. Because of the speed at which fast ferries travel, they produce a longer-period wake that disturbs the seabed at greater depths than conventional shipping.

In confined areas such as the Marlborough Sounds this can lead to a reduction in marine species in the inter-tidal and shallow sub-tidal areas. Tangata whenua are also concerned that the ferries can impact on kaimoana resources. These effects result from the potential of the wake of the ferries to move both sediment and marine organisms in greater depths, thereby changing the marine habitat and community composition and also affecting the shoreline.³

Decision-making framework

Marine pollution

Management of marine pollution in New Zealand is undertaken under two international conventions:

- The *International Convention for the Prevention of Pollution from Ships* which was adopted in 1973 and modified in 1978 by a protocol (known as MARPOL 73/78). This seeks to prevent pollution from ships such as accidental oil spills from tankers, but also pollution caused by the discharge of chemicals, harmful substances, sewage and garbage. The convention includes standards which apply to the design and construction of ships, their equipment and operating procedures. It contains, for example, a requirement that all new tankers be built with double hulls and a schedule for the phasing out of single-hulled tankers. It also contains a complete ban on disposal into the sea of all forms of plastic.
- The *Convention on the Prevention of Marine Pollution by Dumping of Wastes or Other Matter*, adopted in 1972 (known as the London Convention) and its 1996 Protocol. The convention prohibits the dumping of waste at sea except for a restricted group of materials. Even then, dumping is only permitted if there are no practical opportunities for reuse or recycling and the effects of dumping on the marine environment will be minor.

Resource Management Act

The RMA addresses the discharge of pollutants into the marine area within the territorial sea. It includes provisions ,which:

- Prohibit the dumping of radioactive matter from any ship or offshore installation, and the storage of any radioactive waste in the sea (Section 15C)
- Require a resource consent to authorise the dumping of any waste from a ship or offshore installation or the incineration of any waste within the coastal marine area (section 15A)
- Require regulations, a rule in a plan or a resource consent to authorise the discharge of a harmful substance or a contaminant from a ship or offshore installation into water, land or air (section 15B)

The Resource Management (Marine Pollution) Regulations 1998 also address pollution from vessels and offshore installations within the territorial sea. They are designed to implement the provisions of the MARPOL and London conventions. The provisions include the following:

- The dumping of waste or other matter from a 'ship' (which is defined to include all boats and vessels used to travel over the sea) or offshore installation is a prohibited activity, unless it falls within specified categories, in which case it is a discretionary activity. The specified categories include:
 - Dredge material
 - Sewage sludge
 - Fish processing waste from an onshore facility, ships, platforms and other man-made marine structures
 - Bulky items consisting mainly of iron, steel and concrete
 - Organic material of natural origin
 - Inert inorganic geological material
- The incineration of any waste in a marine incineration facility is a prohibited activity
- The discharge of plastics, dunnage, lining and packaging materials from a ship is prohibited
- The discharge of garbage from any offshore installation is prohibited
- The discharge of other ship garbage is permitted so long as the particle size is 25 millimetres or less and the discharge occurs at least three nautical miles from low water mark and more than 0.27 nautical miles from an offshore installation
- Controls are placed over the discharge of sewage from a ship or offshore installation, and provide for minimum distances from land, marine farms, marine reserves and mātaihai reserves. Untreated sewage must not be discharged within 200 metres of marine reserve, within 500 metres of land, a marine farm or a mātaihai reserve or in water less than five metres deep.

A rule may only be included in a regional coastal plan for discharges included under the regulations if it increases the distances seaward or increases the depth specified for any area or increases the distances from a marine farm, marine reserve or mātaihai reserve. A rule in a regional coastal plan cannot relax the requirements in the regulation.

The New Zealand Coastal Policy Statement provides that the dumping of more than 50,000 cubic metres of material on the seabed is a restricted coastal activity and therefore requires the consent of the Minister of Conservation (clause S1.7).

Marine protection rules

Marine protection rules made by the Minister of Transport under the Maritime Transport Act 1994 provide for both marine safety and pollution prevention. They apply both within the coastal marine area and EEZ. They include:

- Requirements for the design and construction of oil tankers entering New Zealand waters which include segregated ballast tanks and double hulls for tankers delivered since July 1996 (Part 121A)
- Discharges of harmful substances from offshore installations (part 200)
- Requirements for regional councils to prepare regional oil spill contingency plans for spills occurring within the coastal marine area (part 130C)
- Controls on the discharge of garbage from ships and offshore installations within the EEZ (Parts 170 and 200)
- Mandatory ship routing to avoid the area surrounding the Poor Knight Islands and the Three Kings Islands (Part 190)

Monitoring and enforcement of discharges from ships and offshore installations within the territorial sea is undertaken by regional councils. Councils also consider resource consents for the dumping of waste at sea within this zone. Such discharges and dumping within the EEZ are controlled by Maritime New Zealand.

No waste may be dumped within the EEZ or continental shelf beyond without a permit issued by the Director of Maritime New Zealand (section 261, Maritime Transport Act 1994). Before issuing a permit, the Director must undertake consultation and also publicly notify the application. No less than 20 working days are to be allowed for public submissions to be made on the application. The Director then makes a decision and must have regard to the views submitted during the consultation and public notification procedures and the *New Zealand Guidelines for Sea Disposal of Waste*. The Director may also consider other relevant guidelines (Marine Protection Rule, Part 180).

Maritime New Zealand has issued a *National policy on the sea disposal of waste*, and this provides more detailed guidance on the conditions under which a permit is likely to be granted. The policy indicates a preference for the use of

‘designated’ sites for significant one-off or regular amounts of dredged material. Management plans are to be developed for these sites.

Marine biosecurity

Biosecurity New Zealand administers the requirements for the discharge in New Zealand of ships’ ballast water from overseas. This is under the umbrella of the *International convention for the control and management of ships’ ballast water and sediments*, which was adopted in 2004 but has yet to come into force.

An *Import Health Standard for ships’ ballast water from all countries* has been issued under the Biosecurity Act 1993. This provides that no ballast water can be discharged into New Zealand waters without the permission of an inspector. Permission will only be granted if the inspector is satisfied that the ballast water has been exchanged at sea or that the ballast water is fresh water. Provision is also made for the ballast water to be treated by an approved system or to be discharged into an approved onshore treatment system although neither of these options is currently available.

At present there is no control on the fouling of ships hulls and offshore installations, although at the time of writing Biosecurity New Zealand was in the process of developing an Import Health Standard to address this issue. A draft standard will be notified for written public comment before finalisation.

International guidelines for the control and management of ships’ bio-fouling to minimise the transfer of invasive aquatic species are currently being developed by the International Maritime Organisation.

Biosecurity New Zealand is also implementing a marine behaviour change programme for recreational boat owners encouraging them to keep their hulls clean, particularly before sailing to another region within New Zealand.

Elements of best practice

Potential environmental effects	Best practice response
Spills of oil and hazardous substances	The design and construction of ships and operational procedures should aim to prevent oil and other hazardous substances being discharged in into the marine environment The routes for vessels transporting oil and other hazardous substances should avoid marine areas with high ecological values including marine reserves The discharge of oily waste into the marine area in sensitive environments including estuaries, harbours and sheltered coastal areas should be avoided Up-to-date oil spill contingency plans should be in place
Sewage discharges	The discharge of untreated sewage in harbours and near marine farms and marine protected areas should be avoided
Marine mammals are injured or killed after being hit by ships	Where possible, redirect shipping routes away from marine mammal habitat Reduce ships speed in areas where marine mammals are likely to be present
Dumping of waste at sea	Minimise the production of waste Consider alternative disposal methods Ensure that the bulk of the waste remains within the disposal site Keep clear of areas of ecological, cultural, economic or recreational importance such as important benthic habitats, marine protected areas and fishing areas Instigate long-term monitoring programmes to measure the impacts of the waste material on the marine environment
Biosecurity risks	Exchange ballast water in mid ocean or operate a type approved ballast water management system Install an antifouling system that is designed for the ship's operating profile, to maintain the ship in a clean state throughout its docking cycle Regularly inspect and maintain the antifouling system of niche areas (such as sea chests, bow thrusters, propeller shafts, intake gratings) of a ship
Vessel wake	Reduce the speed of vessels in areas susceptible to wake-induced coastal erosion or ecological impacts

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- 1 Maritime New Zealand, 2009, iv
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 - 3 Van Roon M and S Knight, 2004, 114



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Catchment-based Activities

Overview

Various contaminants and substances from both natural and human influenced processes are discharged from the land and transferred to the coastal marine environment as part of the natural water cycle. Water flows in a constant cycle from the ocean to the sky and back again, often via a landmass, where it picks up other substances along the way back to the sea.

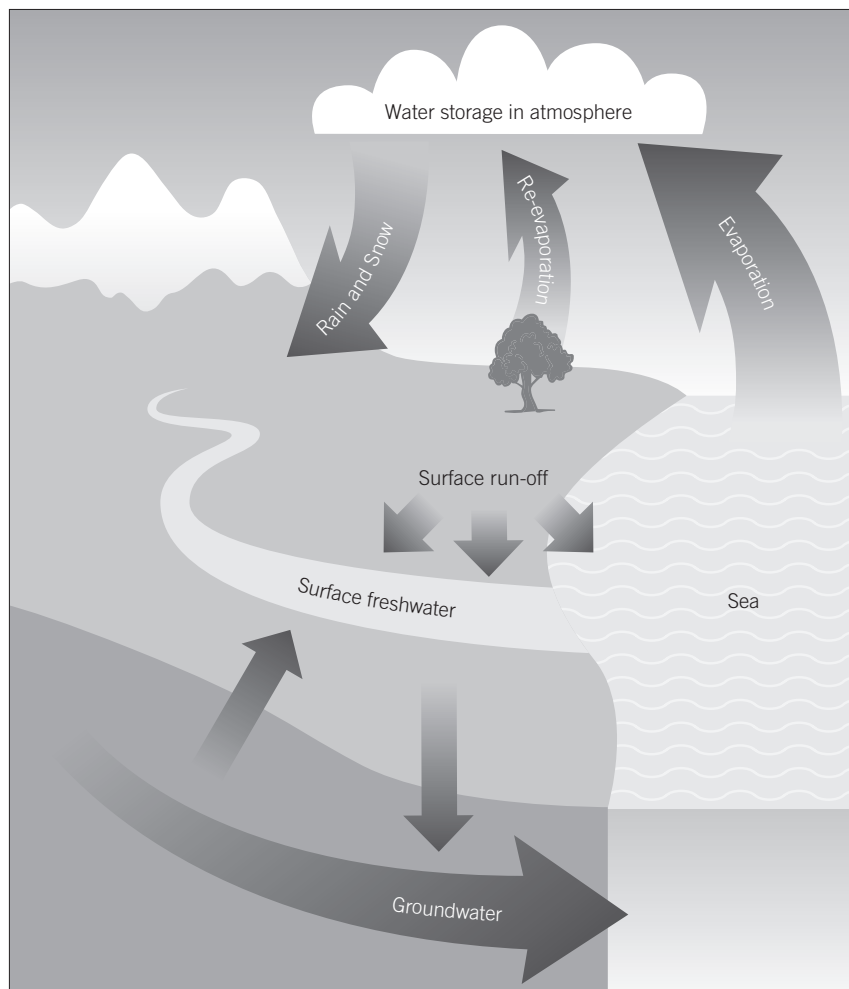


Figure 12.1 The Water cycle

The cycle starts when water evaporates from the oceans and is carried in air currents, which are forced up into cooler altitudes when they pass over a land mass. As the air cools the water droplets condense and become visible as clouds, and then become so heavy that they fall to the ground as rain (or snow).

Some rain is absorbed by plants and animals and re-evaporated back into the atmosphere, some flows back to the sea via streams, rivers, wetlands and estuaries (known generally as surface waters) taking with it sediment and dissolved material including nutrients, and some seeps down into underground rock and caves, (known as groundwaters) where it may either flow underground to the sea, flow back up to join rivers or streams or lie dormant underground.

Human activity has greatly affected the amounts and types of substances that are transferred from the land into the sea, with corresponding effects on the health of marine ecosystems. Human derived substances and excess amounts of natural substances pollute the marine environment by upsetting natural balances and introducing foreign or toxic elements to the ecosystem. Deforestation and devegetation has been a significant contributor to this problem because vegetation fulfils an important role of filtering contaminants from water and reducing overland flow.

It is generally accepted that pastoral agriculture causes the majority of the marine pollution in New Zealand because of its large extent and the large amounts of waste produced. In order of impact, the other land-sourced pollution in New Zealand's marine areas comes mainly from urban sewage, urban stormwater, industrial waste, agricultural waste, mining and forestry.¹

Non-point source pollution

Although pollution from point sources has declined in the last 30 years, pollution from non-point sources continues to be a major problem for the health of New Zealand's marine environment. Run-off from agricultural pasture is commonly thought to be the main cause of non-point source pollution of New Zealand's marine area. Run-off from urban areas is another significant contributor, including zinc, copper and lead washed off roads and iron roofs.

Agriculture

Agricultural pasture occupies more than a third of the land in New Zealand, and agricultural activity affects nearly all catchments.² Since European settlement, agriculture has changed the landscape of New Zealand dramatically, and it has a major impact on the marine environment. In particular, run-off

from agricultural activity causes increased sedimentation, abnormal nutrient enrichment (which can lead to eutrophication), microbiological contamination, and altered physical characteristics of surface water, which affect its appearance and sometimes its smell. These problems are dealt with separately below.

Sedimentation

The removal of hill and river plain forests for agricultural use increases the amount of rainfall running off the land into streams and rivers: forests act as sponges to soak up rainwater, as forest growth catches the rain as it falls and tree roots suck rainwater out of the soil. When forest is replaced by short-rooted pasture grass, increased amounts of rainfall can then run off the land and it runs off at a faster rate.

In the absence of tree roots to hold the soil together, greater amounts of sediment, made up of inorganic matter (rock and soil) and organic matter (mostly animal faeces), are washed from the land into groundwater, streams, rivers and wetlands and make their way to the sea. The effects of this are much worse, when riparian areas are stripped of growth, as well as the main pasture area. This was generally the practice, until awareness of the importance of riparian planting as a buffer zone to protect the catchment from land run-off, became more widespread.

Water run-off from pastoral land is not only greater, but it is faster, causing higher peak flows. In combination with increased sediment load, this destabilises stream and river channels often widening the channels and raising the level of the channel floors.

Pastoral catchments have a sediment load that can be two to five times greater than that of a forested catchment.³ Although most deforestation in New Zealand took place more than 80 years ago, the effects of these changes on the landscape continue to be felt, as the hills and riparian areas continue to be used for livestock production. In recent years, some councils have instigated programmes to provide incentives for farmers to replant riparian areas.⁴ In the past there have been assistance programmes for soil stabilisation but these are now less common.

Nitrogen-nitrate and phosphate pollution

Nitrogen, and to a slightly lesser extent, phosphorus, are among the most environmentally significant by-products of agricultural activity. Nitrogen and phosphorus are essential parts of amino acids and nucleic acids, and consequently they are essential building blocks for all life on earth. Unfortunately when they

appear in the marine environment in excess, they can be very harmful, because they disrupt natural processes of growth and decay.

Along with sediments, nitrogen-nitrates (nitrogen dissolved in water) and phosphorus-phosphates (phosphorus dissolved in water) produced by agricultural activity, find their way into the marine environment. These nutrients can come from a number of sources. Principal of these is animal waste. In 1995, faeces from cattle, sheep and farm goats in New Zealand, produced a waste load equivalent to a human population of 153 million people.⁵ The breakdown of organic nitrate and phosphate in faeces, and the oxidation of ammonia in urine, release large amounts of nutrients into the water.

Nitrogen and phosphorus pollution is also derived from fertilizers which are applied to agricultural pasture to promote grass growth. Furthermore, soil in agricultural pasture often has unusually high natural levels of nitrogen due to the presence of clover, which is sown by farmers as a method of fixing nitrogen into the soil, stimulated by the use of superphosphate. The use of nitrogenous fertilizers has become common to achieve high rates of pastureland production.

The Ministry for the Environment's 2007 State of the Environment report found that 39 per cent of the groundwaters in New Zealand that were monitored had higher than normal nitrate levels. Furthermore, nitrate levels were increasing, with those bodies of water that had previously suffered from the highest nutrient loads showing the greatest decline in quality. For example, the Piako, Waioa and Waihou rivers, which run through the dairy farming regions of the Waikato, are particularly polluted, delivering a total of more than 10 tonnes of nitrogen per day to the Firth of Thames.⁶

There has been a trend towards intensification of dairy farming and as farming becomes more intensive, more nitrogen fertiliser is used to increase yields, and consequently more nitrogen finds its way into the catchment.

In some areas, streams dependent on ground water for their summer flow have been depleted by the extraction of ground water, and nutrient concentrations in the remaining water are raised in the process.

Nitrogen is a significant pollutant in both freshwater and marine environments. Phosphorus is naturally more common in marine environments so systems there tend to be nitrogen limited. Phosphorous is often more of a concern in freshwater, but the addition of both substances to the marine environment can cause problems.

Forestry

Although the effects of forestry on a catchment are generally short-term and localised, they can be significant. Tree-felling and track construction can increase the amount of sediment and nutrients (in particular nitrogen, together

with potassium, phosphorus, magnesium and others) that are washed into the water because, without the trees and roots to provide a barrier, there is little to stop rainwater washing the soil into the waterways. The extent of this depends on the particular area, the type of soil and the amount of rain normally experienced. The worst effects are generally experienced for the first three to five years after felling.

The negative effects of forestry can be mitigated by the retention of riparian planting, which protects the waterways from the majority of the run-off. Pasture causes much more sedimentation and nutrient run-off than forestry in the longer term, so afforestation of an area previously used as pasture can return stream flows and sediment loads to normal levels within six to eight years of planting, and reduce run-off by 30 to 50 per cent within five to ten years.⁷

While water quality may benefit from afforestation, there is a potential downside to reduced sediment loads in rivers. Less sediment reaching the coast may result in accelerated in coastal erosion.

Urban stormwater

Approximately 86 per cent of New Zealand's population lives in urban centres of more than 1000 people.⁸ In these paved and developed areas, rainwater cannot be absorbed by vegetation and soil, as it would be in undeveloped areas. Therefore, to reduce the risk of flooding, towns and cities have stormwater systems that channel rainwater into gutters and pipes and eventually out into streams, lakes and coastal waters. Because a large number of New Zealand's urban centres are sited on estuaries and harbours, the stormwater many towns produce discharges directly to the coast.

Stormwater systems are often a significant source of pollutants into the catchment and coastal areas because the water that runs off into the pipe system picks up and carries with it a wide range of pollutants from the streets, industrial and construction sites and other surfaces. These pollutants include heavy metals (especially zinc, copper and lead – lead contamination has been falling since lead was removed from petrol but zinc contamination continues to rise⁹) hydrocarbons from vehicles (these come from the road surface itself, exhaust emissions, brake linings and tyre wear) sediment, and contaminated dust.¹⁰

Roof run-off is a major source of zinc contamination in catchments which have been urbanised for a long time, because of the large number of galvanized iron roofs in poor or unpainted condition. Zinc oxide used in tyres is another. In addition, earthworks have the potential to release large amounts of sediment into the catchment.¹¹

Point Source Pollution

Pollution of this type has generally declined over the last 20-30 years as treatment system technology has been upgraded and alternative methods of disposal have been developed. However, at specific sites in New Zealand it can still be a problem.

Agricultural and industrial enterprises

Meatworks, dairy factories, tanneries, fertiliser plants, stockyards, mills and many other types of agricultural and industrial enterprise all pose a risk of discharging pollutants into the catchment. In the early years of European settlement, there were many such enterprises scattered across the country, all discharging industrial or agricultural waste on a small scale. Some of these were connected to municipal treatment plants while others managed their own waste.

In the last 30 years, agricultural and industrial activity has changed, so that there are fewer, larger operations. For example, whereas in 1970 there were more than 200 dairy factories, there are now only about 30.¹² These bigger businesses tend to be better able to deal with waste effectively because of their relative financial strength and access to technology. Furthermore, the last 30 years have seen the implementation of stricter legal requirements and a shift in attitudes towards greater recognition of the importance of avoiding pollution, together with a greater interest in finding better technology to achieve this.

Agricultural enterprises generally now use two-pond treatment systems, often combined with land irrigation systems, to dispose of their effluent. This system passes the effluent through two ponds, each containing different naturally occurring types of bacteria, before discharging it into the catchment. It removes 95 per cent of the organic matter from the effluent. However it may not greatly affect nutrients levels, and may not reduce the amount of organic matter actually entering the water to safe levels, because of the sheer volume of waste.

Land irrigation disposes of waste water and nutrients to fields or forest rather than into the water. This is effective at protecting the catchment from some pollution, so long as the land is not overloaded – at this point nutrients will wash off into streams or leach into the groundwater, posing the same risk to marine ecosystems as they would if they had been discharged directly into the catchment.¹³ Only some nitrogen pollution will be removed by plants and only when they are growing. The remainder will pass into groundwater and eventually emerge in streams.

Constructed wetlands are an alternative treatment and disposal system. They can be effective at removing nutrients as well as organic material.

Mining

In the past, mining had a significant impact on water quality. Mines can discharge huge amounts of silt and sediment into the catchment, as well as large amounts of processing chemicals such as cyanide and heavy metals from the rock. Nowadays, mines have relatively effective treatment systems which minimise the amount of silt and sediment being transferred into the catchment (for example by holding waste water in pools to allow sediment to settle before releasing it into the river).

Gold mining gives rise to particular risks because, when the mineral ore is exposed to air and water, the metals can react chemically, dissolving and entering groundwater and streams. This can cause the catchment to become more acidic. The industry has methods to minimise the risk of this by transporting the rock to a treatment plant in a slurry, and then separating off the water which is then treated to remove the heavy metals before discharge.

Mining has often caused groundwater to flow through natural ground or mine waste, leaving water with high levels of heavy metals, often continuing long after the mining has ended.

Wastewater treatment plants

Approximately 1.5 billion litres of domestic wastewater is discharged into New Zealand's environment every day.¹⁴ Human effluent contains a number of substances that can be extremely damaging to the marine environment if released into coastal waters untreated. The most significant pollutants are rotting organic matter, disease-causing micro-organisms, excess nutrients, ammonia and suspended solids.

Most of the household sewage in New Zealand is piped through treatment plants to remove pollutants before it is discharged into the catchment: most towns of 5,000 or more have a reticulated sewage system which pumps everybody's waste water into a common treatment plant and discharge point. There are more than 250 treatment plants in New Zealand and although 24 of them discharge waste onto land, the rest discharge either into the ocean, rivers, streams or drains.

The plants vary in age and technological capability. They use a number of stages in the treatment process, depending on how advanced they are. 'Primary' treatment means that suspended solids are filtered out of the effluent before it is discharged into the catchment, and 'secondary' treatment refers to the use of

microorganisms to remove organic matter and some nutrients, and decompose them in process facilities or oxidation ponds. Tertiary treatment further reduces nutrients and organic matter using oxidation ponds and/or various types of organic or chemical treatment. Ultra-violet treatment, and various other technologies, are used by the most advanced plants in New Zealand to kill the pathogens that remain in the water.

Treatment plants are particularly designed for human wastes and effectively reduce organic pollution from this source. They are not generally designed for a broad range of industrial pollutants and may be quite ineffective against these. Industrial effluent can be much more difficult to manage in treatment plants because it can be highly variable in loading and volume.

The solid material from treatment plants (biosolids) needs careful management. It may be suitable for use as a soil additive (but this may add to pollution risks) or it may need to be disposed of in a landfill site.

All treatment plants in New Zealand practice at least primary treatment, and a large number also use secondary and tertiary treatment. Only a small number use ultraviolet or equivalent systems for the reduction of pathogens in the final effluent. The treatment level is largely related to the sensitivity of the receiving waters, but some legacy plants remain, awaiting replacement or improvement. The effectiveness of wastewater treatment systems consequently varies. Some wastewater treatment plants are ineffective because they are too small to cope with the volume of waste they receive. For example, some are put under pressure by new urban developments, or by the increased amounts of effluent generated by an influx of visitors during the holiday season.¹⁵

Septic tanks

Smaller settlements in New Zealand often do not have community wastewater treatment plants but rely on individual household septic tanks. Septic tanks are generally effective but can fail where the soil is unsuitable, or where the surrounding land area is not big enough for the soakage field. This is a particular problem in modern coastal developments, where large homes (rather than traditional baches) with modern conveniences requiring a large amount of water, are built on small sections very close together. Overloading of the soakage fields with resulting overland flow is the most common problem. Septic tanks can also fail when they are inadequately maintained. Effluent that seeps untreated from septic tanks may get into streams or groundwater, and from there into the marine environment. Even with properly operating systems, some nitrogen transfer to groundwater is inevitable.

Stormwater and sewage

A previous section discussed the treatment of wastewater. This section considers the environmental issues arising from the conveyance of wastewater to treatment plants.

Wastewater systems typically use gravity to move sewage discharged from houses and other premises along pipes towards the treatment works. When the sewage pipes reach the lowest point on land, gravity will no longer move the material, and pumping stations are required to pump the sewage up to the plant. Many pumping stations are located near the coast, as this is often the lowest point to which sewage can be moved by gravity.

Pumping stations will usually have some holding capacity for sewage in the event that the pump fails, such as when there is an interruption in the electricity supply. When the pump stops for any reason, the sewage will keep flowing as a result of gravity and it will build up in the holding tanks at the station. When the holding capacity of the station is exceeded, raw sewage typically flows into the sea or a nearby river.

Some regions use combined sewer and stormwater drains – this is common in old networks which have not yet been replaced with more modern systems that minimise the risk of pollution. In these areas pumping stations can overflow when there is heavy rain that quickly reduces their holding capacity. In other areas with separated sewers, stormwater infiltration through damaged pipes and poorly arranged or illegal connections can also result in inflows which exceed the capacity of the pumping stations. When this happens, raw sewage can also be discharged into the marine area, although it will be diluted to some extent by the large amount of stormwater.

Stormwater infiltration can also cause major problems in sewage treatment plants by increasing the volume of effluent that they have to deal with, reducing the effectiveness of treatment, and increasing the risk that treatment systems will flood and allow effluent to escape. This is quite common as a result of faulty pipes, illegal cross connections (where private properties illegally attach sewage pipes to the public stormwater drain), and old connections that have not been cut off.

Another way in which untreated urban sewage can enter the catchment is where sewage in the sewer system overflows into the stormwater network (exfiltration) and into the environment. This generally happens as a result of failures in the pipes such as cracking followed by tree root invasion. It can also occur as a result of heavy rainfall which causes overflows as well.

Pumped sewerage systems are becoming more common and are resistant to infiltration but failures can result in spills.

Potential environmental effects

The amount of pollutants that are transferred from land into the oceans, and the effect that they will have on coastal marine ecosystems, varies depending not only on the amount of waste actually released, and the type of waste, but also on conditions in the area. As one would expect, transfer rates are much worse at times of more extreme wet weather when more water is moving from the land to the sea.

Whether or not land-sourced pollutants remain in the coastal marine area depends on a number of factors, in particular the geographical makeup of the area and the winds and currents experienced by the locality. In harbours and estuaries, the amount of mixing that occurs and the time the inflowing water is retained in the area, are important. Shallow inlets are more prone to retain pollution than deeper ones, and estuaries and narrow inlets are more prone to retaining pollutants than wide exposed beaches, which are subject to stronger currents that can carry the pollutants out to sea. Many river estuaries have natural sea barriers broken only in high flows. Sediment and pollution retention behind these barriers is high.

New Zealand's particular geographic makeup means that there are large stretches of coastline that do tend to retain pollutants. As the Ministry for the Environment's 1997 State of the Environment report notes: 'Much of the coastline is made up of river-fed estuaries whose wide, shallow waters are permanently protected from ocean waves by sand or shingle bars or offshore islands.'¹⁶

Where there are high river flows, the fresh water may pass along an open coast, floating over seawater for some distance before it becomes mixed.

Sedimentation

An increase in the rate that sediment is transferred into the receiving environment can bring about changes in the marine ecosystem. An excess of sediment can affect water depths, water coverage, the type of sediment found on the sea floor (for example the size of the particles) and the turbidity of the water. When an excess of sediment enters the receiving environment it can be suspended in the water, reducing light by affecting water clarity.

Such environmental impacts can bring about changes in vegetation, be detrimental to fish habitats, degrade spawning areas and deplete invertebrate populations. As most sediment is deposited in intertidal zones, it is the species that live in these areas that are often most affected by excess sedimentation.¹⁷ This is particularly significant, because healthy intertidal zones support a

particularly high diversity of species, in comparison with other parts of the coastal area.

Organic material

Agricultural run-off, urban wastewater and industrial waste transfer organic material to the catchment. This can include faeces from humans and animals; proteins, vegetables and sugars from food preparation; and cleaning soaps. When effluent enters a healthy aquatic environment where there is plenty of oxygen dissolved in the water, aerobic (oxygen using) bacteria, which occur naturally in the water, will eat the organic material, using up oxygen in the water to do so. Consequently their numbers grow beyond what you would normally expect in a healthy aquatic ecosystem.

This use of oxygen can have very serious effects on the health of the ecosystem, as fish and plants need oxygen levels to be maintained in order to survive. Even an occasional reduction of the oxygen level can have the effect of removing the marine species that rely on a continuous supply at a high level. If all the oxygen is used up, anaerobic bacteria (non-oxygen-using bacteria) will take over, decomposing the waste material and producing gases such as methane hydrogen sulphide and carbon dioxide, making the water go septic.¹⁸

Thus, the potential harm caused by organic waste can be measured by its potential to remove oxygen from the water. There is a standard measure known as the biochemical oxygen demand (BOD) which is the amount of dissolved oxygen needed by aerobic bacteria to consume the waste (over a 5 day period where the water temperature is 20 degrees centigrade, measured in terms of the number of grams of organic material per cubic metre.) Thus, if waste deposited in the catchment has a BOD loading that is too high, too much oxygen will be removed from the water by the aerobic bacteria, jeopardising plant and animal life.

The surface sediment of the seabed is an important ecological zone and the home of some specialised forms of life. It is normally oxygenated. Excess sediments, and low oxygen in the water above, can cause it to become anoxic and unable to support its normal inhabitants.

Excess nutrients - nitrogen-nitrate/phosphorus/ammonia pollution

Nutrients, in particular nitrogen and phosphorus, occur naturally in aquatic ecosystems and are essential elements for growth. This is why they are used in large quantities in fertilizer to promote grass growth on agricultural pasture. However, where excess amounts of them reach the catchment from man-made sources, they can upset the balance of the ecosystem and cause serious problems.

When there are excess nutrients in the water, they serve to fertilise microscopic algae and water weeds, leading to abnormal increases in their growth and subsequent decay. Their decay stimulates the aerobic (oxygen consuming) bacteria, which in turn causes severe reductions in water quality and oxygen levels in the water, leading to the deaths of fish and other marine life. When the plants decay the nutrients are released back into the water, stimulating algal growth and allowing the unhealthy cycle to start again. This phenomenon is known as eutrophication.

Bacteria and viruses

Animal and human faecal matter contain large numbers of bacteria, which are produced by the gut. Although many are harmless, some of these bacteria can cause serious health problems in humans if they reach water used for recreation and shellfish gathering. For example, campylobacter is the most common source of food and waterborne illness in New Zealand. Typhoid is also easily transmitted via water contaminated by faeces.

Coliform bacteria are common in human and animal faecal waste – but can come from other sources. The amount of coliforms per hundred millilitres of water is used to indicate the likely presence of faecal matter in water. A more precise test for animal and human waste is that for a particular coliform, *E. coli*, which comes from warm blooded animals alone. Viruses can also be found in human and animal faecal matter – the main class is the enteric viruses, which cause illnesses such as polio, meningitis, and hepatitis.

Where levels of viruses or bacteria in coastal waters are relatively high, they can pose a risk to swimmers and other contact water users. They can also concentrate in shellfish, which eat by filtering food from the water, and can therefore be dangerous to humans who consume the shellfish.

Other dissolved pollutants

Many other pollutants find their way into the marine area, from industrial and agricultural activity and from domestic households. In particular, metals, chemicals, hormones, dioxins, organochlorines and PCBs can cause problems. The problem may not just be mortality of marine organisms, but disruption of growth, or reproduction, which may occur at much lower than lethal concentrations. Only a small proportion of these contaminants are dissolved in the water column; the majority remain attached to particles of sediment, which settle in the inner parts of harbours and estuaries. In these areas, where wave and current movements are likely to be too weak to move the polluted sediments around, they can build up to toxic levels.¹⁹

Marine organisms can be affected by the toxicity when they ingest the sediment or take up toxins that have dissolved into the water column. The pollutants can have effects higher up the food chain, as shellfish and other organisms, are eaten by fish and birds. The effects that such toxins have on marine life varies widely but can be very serious. For example, even small amounts of lead in the water are very toxic to marine life. Zinc is slightly less toxic, but can accumulate to very high concentrations in shellfish. Copper is toxic to some types of marine life – in particular algae are very sensitive to increased levels of copper in the environment. Polycyclic aromatic hydrocarbons have been linked with tumours in fish.

Decision-making framework

The RMA provides the framework for the control of activities that result in land-sourced pollution entering the marine area. The RMA adopts a catchment-based approach to environmental management, with the boundaries of regional councils largely coinciding with the boundaries of water catchments and including the coastal marine area. This is based on the understanding that many different activities within a catchment will cumulatively impact on the quality of lakes and rivers and ultimately on what is discharged into the sea. Effectively managing these impacts requires an integrated and catchment-focused approach.

Under section 15 of the RMA, the discharge of contaminants into water (or onto land where there is a possibility that the contaminants will run off into water) is prohibited unless the activity is expressly allowed by a national environmental standard, a rule in a regional plan as well as a rule in any proposed regional plan, a resource consent or regulations.

Section 15 Discharge of contaminants into environment

- (1) No person may discharge any –
 - (a) contaminant or water into water; or
 - (b) contaminant onto or into land in circumstances which may result in that contaminant (or any other contaminant emanating as a result of natural processes from that contaminant) entering water; or
 - (c) contaminant from any industrial or trade premises into air; or
 - (d) contaminant from any industrial or trade premises onto or into land –

unless the discharge is expressly allowed by a national environmental standard or other regulations, a rule in a regional plan as well as a rule in a proposed regional plan for the same region (if there is one), or a resource consent.

Regional councils have responsibility for controlling the use of land for the purpose of maintaining coastal water quality and coastal marine ecosystems (section 30(1)(c)) and controlling discharges into water bodies (section 30(1)(f)). This includes managing the negative effects of sedimentation and pollutants, which are derived from land-based sources and which end up being discharged into the marine area. Regional councils prepare regional policy statements and regional plans to address the issue. Some regional councils prepare separate plans, which specifically address issues such as sedimentation and discharges from dairy farms to help them effect their obligations under the RMA.²⁰ In practice it has proved easier to deal with point source pollution under the RMA than non-point source pollution.

There are two main approaches that regional plans take in relation to avoiding or minimising impacts of land use on the marine area. One method employed by a number of regional councils is to classify coastal receiving waters and apply appropriate water quality or sediment quality standards to those waters. This is a more “effects based” approach. It allows innovation on behalf of the consent holder as to what site management practices or controls could be implemented to meet the standards. Water quality standards which are linked to rules will have a more direct effect on outcomes than standards linked to policies.

Another method is to focus more on the land use activity itself and apply methods to control or regulate these types of activities. For example, if a rural catchment is identified as having a negative impact on the ecological health of the marine area because of the high levels of sedimentation, and the main causes are identified as forestry and farming activity, the regional plan might include rules to mitigate these effects, such as requiring stock to be kept out of waterways and preventing harvesting in riparian areas.

Where regional plans specify that particular types of discharge into water are allowed, the council must be satisfied that (as well as being consistent with the purposes of the RMA) after reasonable mixing with the water, the discharge will not result in (section 107(1)):

- The production of conspicuous oil or grease films, scums or foams, or floatable or suspended materials
- Any conspicuous change in the colour or visual clarity

- Any emission of objectionable odour
- The rendering of fresh water unsuitable for consumption by farm animals
- Any significant adverse effects on aquatic life

Most discharges into water will require a resource consent. When considering whether or not to grant an application for consent the authority must consider the factors set out in section 107(1) of the RMA (see above), and if the discharge would result in one or more of those effects, must not grant the consent. The exception to this rule is that the consent authority can grant the consent if it is satisfied that, as well as being consistent with the purposes of the RMA, exceptional circumstances justify the granting of the permit, the discharge is of a temporary nature, or the discharge is required for necessary maintenance work.

Non-point source pollution poses more of a challenge for regulators, because it is not caused by isolated and identifiable acts which are generally more easily controlled by the consent process. Regional councils are responsible for identifying the environmental effects of both urban and agricultural run-off, and advocating regional solutions and management. To discharge this duty they may produce technical guidelines for management of stormwater systems and sediment control programmes, set up incentive programmes and form community-based groups to encourage farmers, forestry companies and other users to alter the way they use their land to minimise the effects of run-off.

Under section 31 of the RMA territorial authorities are responsible for controlling the effects of land use outside the coastal marine area, and this includes downstream effects on the marine environment. Territorial authorities can prepare district plans to address these issues. Whereas regional councils tend to focus on catchment-wide issues, territorial authorities focus more on site development issues such as earthworks and the construction of impervious areas.

District plans can indicate the amount of impervious area (building or paving) that a property can have, and property owners who wish to exceed this limit must apply to the territorial authority for consent to do so. Territorial authorities and their local network operators also have responsibility for stormwater infrastructure in their area, such as maintenance and replacement of stormwater pipes, maintaining stormwater pumping systems and outfalls, and construction of silt traps and detention ponds.

Territorial authorities undertake monitoring and mapping to identify any issues associated with localized flooding and overland flow as well as any pollution-related issues. The Local Government Act 2002 requires them to

undertake an assessment of the water and sanitary services in their district, including wastewater and stormwater, to assess amongst other things whether they pose a danger to public health.

Reporting water pollution

Most regional and territorial authorities have a 24 hour pollution hotline, which can be called if pollution of the catchment occurs, and which can be found on council websites. Pollution response teams act on reports of pollution events, and aim to put an immediate halt to the discharge, ensure that polluted waters are cleaned up and collect evidence of the pollution incident.

When an incident occurs, the regional council can send a warning letter, setting out how the perpetrator has breached either the rules of a regional plan or the RMA. Then the council can issue an abatement notice requiring the perpetrator to stop the offending activity occurring or take some action to ensure compliance. The council can issue infringement fines of up to \$1000 or prosecute the offender. The maximum penalties for individual offenders are up to two years in prison or a fine of up to \$300,000, with a further \$10,000 for every day that the offence continues. Non-natural persons such as companies can be fined up to \$600,000. The Environment Court can issue an enforcement order requiring the offender to comply with the RMA.²¹

Elements of Best practice

Potential Environmental Effects	Elements of best practice
<i>Agriculture</i>	
Organic matter (animal faeces) enters the marine area, stimulating the activity of aerobic bacteria, which deplete oxygen levels in the water	Prevent stock from accessing waterways and the coastal edge Have adequate farm treatment systems to cope with the volume and strength of effluent produced Ensure land irrigation is not overused so that effluent sprayed onto fields runs straight off
Runoff from pasture causes increased sedimentation of the marine environment	Create riparian buffer zones using native flora along waterways to protect the catchment from sediment, organic material and nutrients washed from pasture by the rain Construct runoff management systems such as wetlands or pools along subsurface farm drains, where sediment and contaminants can settle Restore old wetlands to create natural obstacles to sediment runoff Construct farm tracks on sites where gradients are low Reduce stocking densities or retire land on steep or highly erodible slopes Develop and implement environmental farm plans
Nitrogen-nitrate and phosphate pollution derived from fertilisers used to promote pasture growth, encourage growth of algae and other marine vegetation, leading to eutrophication	Avoid using more fertiliser than can be taken up by the crop or pasture – soil nitrogen testing can indicate exactly how much nitrogen fertiliser a field requires, and therefore help avoid over-fertilising Ensure that catchments are not farmed over-intensively, to avoid generating more pollution than the marine environment can cope with
<i>Urban sewage</i>	
Wastewater treatment works fail to remove organic matter, nutrients and micro-organisms from the effluent before it is discharged into the marine area	Use the best available technology (tertiary with UV pathogen treatment or similar) in wastewater treatment plants to ensure that the water discharged into the coastal marine area contains minimal amounts of organic matter, micro-organisms and nutrients Plan the construction of wastewater treatment works so that they have the capacity to deal effectively with all the waste generated by the community, and with future or seasonal growth in the area.
Septic tanks result in effluent entering the marine area	Construct septic tanks only in areas where the soil is suitable Construct septic tanks only where there is an adequate amount of land surrounding the property to cope with the amount of effluent generated Adequately maintain septic tanks and soakage fields to ensure that they do not fail
<i>Industrial effluent</i>	
Industrial processes result in the discharge of effluent into the marine area	Use the best available technology to minimise pollution, and instigating programmes for the reduction of waste

Forestry

Felling of trees and construction of tracks results in increased runoff into the marine area

Undertake tree felling and track construction in plantation forests so that a riparian buffer and coastal margin is retained to protect the catchment from sedimentation

Install cut-off drains at regular intervals on tracks so that water runoff can be managed and directed away from waterways

Avoiding felling and track construction during high rainfall times

Use sediment control measures when harvesting trees on steep or erodible slopes

Burning to clear land for forestry or after harvesting, and mechanical clearing of the plantation, cause intensive soil disturbance leading to increased run-off into the catchment

Maintain riparian buffer zones along waterways and the coastal edge to prevent large amounts of sediment from entering the marine area

Avoid highly invasive mechanical and burning techniques to clear land with high erosion potential

Urban Stormwater

In periods of heavy rain stormwater increases the volume of sewage so that wastewater treatment plants can not treat it all

Replace combined sewage/stormwater systems with separate networks and adequately monitor and maintain sewage pipes to prevent leakage

Sewage enters the stormwater system and from there enters the marine area untreated

Prevent illegal connections

Minimise the amount of permeable paving in new developments and modifications to existing urban areas

Adequately maintain piping

Use swales and planted areas as buffers to limit and filter stormwater run-off

Restore old wetlands to filter and limit the flow of stormwater to the sea

Earthworks generate increased sediment

Prepare catchment management plans to protect natural channels and provide stormwater retention ponds

Require an erosion and sediment control plan as part of building consent applications and include strict conditions in resource consents to require adequate sediment detention methods to be used. These may include chemical sediment control (flocculation), silt fences, decanting earth bunds or sediment retention ponds.

Undertake major building works at times of the year when rainfall is unlikely to be heavy

Retain existing vegetation wherever possible and effect revegetation as soon as possible

Stabilise disturbed areas as soon as possible, and limit the area disturbed as much as possible

When considering urban zoning for plans, pay particular attention to the potential for the cumulative effects of earthworks in an area to cause significant sedimentation of the marine area

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Conclusions

New Zealand is extremely fortunate to have a large and very diverse marine area. It supports a wide array of marine life including seabirds, marine mammals, fish, invertebrates and plants. It is of great value to New Zealanders for both cultural and recreational reasons and because it supports many important economic activities. Human activities have had a major impact on marine ecosystems and it is important that these impacts are better managed to ensure the ongoing health and productivity of our seas.

The legal and policy framework for managing the marine environment is fragmented and complex. There are however, many opportunities for members of the public to become involved in seeking improved environmental outcomes for New Zealand's seas. This publication has identified these opportunities. It has also identified the potential environmental impacts of the raft of activities which impact on the marine area and provided guidance on how they might be better managed. EDS would urge all those with an interest in the future of New Zealand's marine area to become proactively involved in its ongoing management.



appendix one

Cases on Seabed Mining

Summary of cases

This appendix summarises six cases in the Planning Tribunal or Environment Court which considered resource consent applications for seabed mining. Five cases related to sand mining and one case related to gas. The Planning Tribunal was the former resource consent appeals body which was subsequently reconstituted as the Environment Court.

The cases on seabed mining are very fact specific with evidence playing a major role in the Court's determination. Where the evidence was not contested the Court held in favour of that evidence. The Court upheld expert evidence over lay evidence. In all the cases reviewed, the Court granted consent subject to conditions.

In most cases, the Court considered the precautionary approach, when assessing the effects of extraction. This was because of the uncertain scientific data presented in many of the cases. Two different approaches to the precautionary principle were expounded:

- Some cases took a hard line approach, and only made a decision based on the evidence put before the Court. In these cases the Court did not consider that caution, based on the possibility that its findings on the evidence may not accord with what in fact occurs, was relevant to the decision.
- In other cases, the Court found that the precautionary approach was appropriate, considering that the extractions were to occur within dynamic systems subject to constant change. However, these cases generally took the view that monitoring and other conditions could encompass this precaution.

The following factors were considered relevant by the Court in their decisions. However, it is important to note that the factors the Court consider relevant are limited by the appeal points. It is likely that there are other factors of relevance that have not been tested by the Court through an appeal.

- Ecology (marine mammals and shellfish)
- Māori cultural and spiritual values
- Impact of extraction on Treaty claims
- Consultation
- Demand for sand
- Alternative sand sources

- Natural character
- Amenity
- Quality of the environment
- Is the system open or closed?
- Cumulative effect of the proposed extraction when combined with previous extractions
- Noise
- Indirect effects, such as on shorebirds
- Recreation
- Economics of extraction
- Overall sustainability (largely dependent on findings on the above matters)

Oruawharo Marae Trust v Auckland RC A083/06

This interim decision of the Environment Court concerned appeals against the council's decision to grant replacement consents for sand extraction in the Kaipara Harbour. Oruawharo Marae Trust opposed the applications for replacement consents on the grounds that:

- Continued sand extraction would have a negative adverse effect on Māori cultural and spiritual values
- The extraction site was wāhi tapu
- The effects of previous sand extraction had not been properly assessed and the extraction had depleted shellfish which had a knock on effect up the food chain.
- There had been inadequate consultation

The Royal Forest and Bird Society did not oppose the granting of replacement consents but sought the imposition of a scientifically-credible monitoring regime to protect conservation values. This included an 'adaptive management regime' involving the undertaking of baseline studies, particularly in relation to Maui's dolphin, as a pre-condition of increases in extraction rates.

The Court held that the concerns raised by the Trust did not of themselves provide a basis for declining or recommending against the granting of consents and that monitoring and review conditions were satisfactory. These findings were bound up with findings about the sustainability of the sand resource, and aspects of the ecological evidence. For example, the Court accepted the applicant's ecological evidence that the effects on the tuatua population would be no more than minor.

The Court examined the evidence on the sustainability of sand extraction in depth. It found the monitoring conditions suitable, subject to attention to their detailed wording, so as to deal with potential shoreline effects. In doing so the Court was mindful of the challenge that the data may present for interested parties in seeking to differentiate cause and effect in a dynamic system subject to constant change. The Court held that whatever interpretative challenges the data may set, shoreline and banks monitoring is essential as part of a precautionary approach.

In relation to marine mammals, the Court concluded that there was no evidence that harm had arisen from the current high level of operations and that the risk of adverse effects was less than minor and probably negligible. The Court also rejected a sophisticated monitoring programme proposed by a submitter in relation to shorebirds on the basis that the effects would be less than minor.

The Court held that there was no cogent challenge to the applicant's evidence in relation to demand for the sand and accepted that the purpose of the Act would be served by granting the consents.

The Court did not consider that continued extraction of sand would compromise the natural character of the coastal marine area and concluded that there would be no significant adverse effects on either amenity values or on the quality of the environment. Consent was granted.

Sea-Tow Limited v Auckland Regional Council A066/06

This matter concerned appeals against the council's decision to refuse consent for the continuing extraction of sand from near-shore areas of Mangawhai-Pakiri Bay. Sea-Tow applied for consents to enable extraction of 27,000 cubic metres of sand per year for 20 years. McCallum Bros applied for consents to enable extraction of 49,000 cubic metres of sand per year for 20 years. Both extractions were to be continuations of previous expired consents. Both applications were for discretionary activities.

A large number of section 274 parties opposed the granting of consent including the Director-General of Conservation, the Northland Regional Council, the Kaipara District Council and the University of Auckland. These parties' concerns, and the council's, were that the continuing extraction of sand would create serious adverse environmental effects.

The section 274 parties variously claimed that extraction of sand from a closed system such as this would result in degradation of the natural character of the beach, disruption of natural processes, cumulative effects in light of previous extraction at Pakiri and coastal erosion. The applicants disputed that the sand resource in the embayment was a closed system and argued that there

had been no significant erosion or change to the coastline attributable to sand extraction, and none attributable to natural processes.

The Court summarised the issue as:

being whether or not the proposed sand extraction would have the potential effect of so reducing the quantity of sand in the system that, cumulative on the effects of past extraction, would contribute to erosion (or decreasing accretion) in the beach and foredunes.

After reviewing extensive evidence on whether the system was closed or not the Court held that, in relation to whether the extraction is sustainable:

We accept that the source of the quartz-feldspar Holocene sand is non-recurring. We are not persuaded that we should depart from our finding that the input to the system from deeper water, across the inner shelf, is on average about 12,000 cubic metres per year. Plainly that alone would be insufficient to render sustainable extraction at the rate of 76,000 cubic metres per year. But we have also made findings of additional net inputs from other sources amounting in total to 150,000 cubic metres per year –roughly twice the proposed rate of extraction. Allowing for variations year by year, and uncertainties in the methods, on that finding we do not accept that the proposed extraction is unsustainable.

The Court then examined whether there are indicators of significant change in the environment due to past sand extraction from the Mangawhai-Pakiri embayment. It heard from numerous witnesses, both lay and expert. With regard to the lay evidence, the Court did not give it much weight because of a lack of scientific proof that the changes identified were attributable to sand extraction and were not just part of natural processes. The Court agreed with some of the witnesses, that signs of shoreline retreat and erosion could not be attributed to past sand extraction, and that past extraction had had no detectable effect on the environment. Overall, the Court held that the sand system was open.

The Court then went on to examine the evidence on the direct adverse effects on the environment of the extraction process. The Court agreed with the uncontested evidence of the applicant's ecologist that the extraction would have no significant adverse effect on the shellfish and other marine life.

The Court also agreed with the uncontested evidence about noise at the nearest dwelling, and held that the noise would comply with the applicable limits set by the Auckland Regional Council, and would not be a direct physical adverse effect on the environment.

The Court heard no landscape evidence on the visual effects, although it did hear from several lay witnesses. The Court held that the Hauraki Gulf is not a wilderness waterway. It is frequented by pleasure boats, shipping and other craft. Although a barge and tug passing regularly may not be an attractive view to residents or leisure users of the beach, their presence at sea could not be deemed markedly out of place. It held there were no more than minor adverse visual effects.

Most of the concerns expressed on behalf of tangata whenua were based on their understanding that the sand extraction would have direct physical effects on the environment. The Court's review of the evidence led to it not accepting that such effects would occur.

When assessing the indirect physical effects of the extraction, the Court noted that effects on indigenous vegetation and indigenous habitats (for shorebirds) were related to dune erosion the destabilisation of natural systems predicated on the embayment being a closed system. This was not established by evidence, so the Court held that there were no adverse indirect effects.

The Court then assessed the indirect non-physical adverse effects. It held there would not be any significant adverse effect on natural character (largely related to the fact that the same resource would not be depleted because the system was not closed).

The Court considered that beliefs of effects of the extraction (e.g Māori belief of effect on tangata whenua values with respect to mauri, and to the relationship Te Uri o Hau have with the area) were not supported by the evidence, did not provide an appropriate basis for judicial findings of adverse effects on the environment, and should not influence the Court's judgment of whether or not the extraction would represent sustainable management of natural and physical resources.

The Court accepted evidence that changes may have occurred to the quality of the waves at the beach for surfing, and to the beach itself. However, it held that there was no evidence suitable to base a finding that this has been, or could be, caused by sand extraction activity. Therefore, the Court held that there was no effect on the safety of swimming and surfing.

In relation to loss of amenity value for recreation, the Court held that recreational use continues in the Pakiri area, and can be expected to expand in the future. It did not accept that there has been or would be any significant effect on amenity values caused by the extraction operation, despite the perceptions of submitters about the appearance and noise from the barge, and assumptions about the cause of perceived changes.

The Court held that the extent to which the proposal would have beneficial effects due to economic efficiency is relevant to the decision because of the

wording of section 5 of the RMA which includes ‘at a rate’, ‘provide for their economic well-being’ and ‘managing the use, development and protection of natural and physical resources’. Section 7(b) also refers to having particular regard to the ‘efficient use and development of natural and physical resources’.

The Court held that minimising transport costs in meeting the future demand for sand in the Auckland region is a consideration which relates to efficiency. The Court found that the proposal would be economically efficient through delivering sand to Auckland.

The Court then looked at alternative sources of sand, and after hearing extensive evidence on the point, it concluded that there were currently no other sources of sand suitable for ready-mixed concrete manufacture.

On the precautionary approach, the Court held:

In short, our judgement on whether the purpose of the Act would be better served by granting or refusing consent, and on the conditions to be imposed if consent is granted, should be made having regard to our findings on the evidence on actual and potential environmental effects, and on the relevant provisions of the Act and applicable statutory instruments. We do not accept that in the circumstances of this case our evaluative judgement should be influenced by caution based on the possibility that our findings on the evidence may not accord with what in fact occurs in future.

The Court heard from a number of submitters about previous decisions and their findings. However, in the end the Court held that it would not be influenced, in deciding these appeals, by the Commerce Commission decision, or by findings or judgements made on other applications under the Resource Management Act. The Court was also not interested to hear evidence on what other countries do, and stated it is entirely unhelpful for the Court.

In determining the term of consent to be granted, the Court held that, in the absence of a demonstrated need for longer terms to protect capital investment in plant, terms for as long as 20 years would not be justified. The Court considered that terms of 14 years from commencement would be appropriate in the circumstances.

McDonald v University of Auckland A204/2002

This decision involves an appeal by McDonald (owner of coastal Māori land at Pakiri) in relation to an application by Kaipara Excavators Limited to take up to two million cubic metres of sand, by way of suction dredging, from depths greater than 25 metres for a term of 35 years. The area proposed for the dredging was approximately 500 square kilometers in the north-west Hauraki Gulf in the vicinity of Pakiri and Little Barrier Island. Kaipara also sought

approval for the temporary, non-exclusive occupation of the coastal marine area during suction dredging. The proposal was a discretionary activity and a restricted coastal activity.

Kaipara wanted to take sand to meet the needs of the construction industry, including a number of large 'one-off' projects, as well as regional beach replenishments over the next 30 to 40 year planning period. It sought a 'maximum quantity' permit as opposed to an 'annual amount' permit to ensure flexibility to meet the needs of all the potential end users of the sand resource.

The issues of the case were:

1. *What was the extent of the effects on Pakiri beach, including the dune area?*

Various operators were allowed to take a maximum of 165,000m³ of sand per year from the near-shore area of Pakiri Beach. Those permits expired in 2004. The current application was for an area further offshore. The Court held that even if the sand in the extraction area was a finite resource, the consent sought was for approximately 1.3 percent of that resource. There would be no lasting effects on the seabed, as sand movement during wave events would remove evidence of dredging. Subject to later comments about a reduced term of consent, it was unlikely that there would be any adverse effects on the beach, including the dune areas.

2. *What was the extent of effects on marine life?*

The Court only heard from one expert (for the applicant) on the effects on marine life. It held that with the conditions proposed, adverse effects of any significance on marine life were unlikely. It accepted evidence that there were no species which consistently spawned only in the proposed extraction area. Dredging activity would be excluded from areas identified in the Pre-Dredging Assessment Report as having an average proportion of mud exceeding 20 percent by volume. Therefore the sand taken would be relatively 'clean', which would limit turbidity. There would be no effects on macro-algae. The Court considered that the monitoring programme provided for by the conditions was an appropriate method to assess any adverse effects from dredging.

3. *What were the cultural and spiritual effects and implications of the proposal?*

The sand was a treasured taonga and, as a matter of principle, tangata whenua's preference was that there should be no sand mining. The Court preferred the evidence on cultural and spiritual effects by senior kaumatua and rangatira of the hapū of Manuhiri. The Court held that mana whenua and the role of kaitiaki in relation to the beach and dunes would remain largely unaffected. The amount of sand taken would be small and the area within which extraction would actually occur over the term of the

consent, in the context of the total extraction area sought, was also small. The Court concluded that the conditions of consent provided a range of protections. The risk of human remains being found within the extraction area was remote. Proposed conditions enabled heritage areas, should any be discovered, to be avoided or for appropriate cultural protocols to be followed. In summary, the Court held that there would be minimal, if any, adverse cultural or spiritual effects on tangata whenua and those with mana whenua and mana moana.

4. *How did provisions from the relevant statutory instruments apply to the proposal?*

The Court examined the relevant provisions of the various planning instruments and the NZCPS. It did not apply a precautionary approach when assessing the location of the extraction activity, the maximum volume to be extracted, and the term, because there had been a significant amount of study of coastal processes in the Pakiri area and there was no scientific uncertainty. When dealing with cumulative effects, the Court was satisfied that taking sand from the near-shore and offshore compartments resulted in different effects, particularly in relation to the Pakiri dunes. It held that granting consent in this case, so that extraction occurs offshore, is positive in terms of limiting cumulative adverse effects. The Court accepted the evidence that there were no alternatives to the use of this sand. In conclusion, the Court held that granting consent on the basis of a 20-year term and the proposed conditions would be consistent with the provisions of the various statutory instruments. Because of findings about the lack of effects on the sand dunes at Pakiri, the Court concluded that it was not necessary to consider the District Plan provisions.

5. *Was there a need for the sand and is the size of the extraction area excessive?*

The Court disposed of this matter briefly. It relied on the unchallenged evidence about the need for additional sand, even having regard to quantities in existing consents, to meet anticipated market demands. The Court also relied on uncontested evidence that the environmental effects were acceptable.

6. *What was the appropriate term for the consent?*

Consent was sought for 35 years. Given that there was scientific uncertainty over the precise period over which the dune-beach-bar morphodynamic system might be coupled with the inner shelf system, the Court inclined to the view that the consent term should be restricted to 20 years. That would be sufficient time to enable sand mining to occur, and for at least two decades to pass. At that time, whether there had been any identifiable coupling of the systems over that period of time, and whether there were any potential adverse effects that might result from that, may be more readily ascertainable.

J W Paterson & Sons Ltd v Bay of Plenty Regional Council A135/2000

This case involved an application for a coastal permit to excavate and annually remove up to 27,000m³ of sand from Otamarakau Beach in the Bay of Plenty. The proposal was for ten years, with a review to take place after expiration of the first five years. The current application was to succeed a coastal permit issued in July 1995. Otamarakau Beach sand had special and rare qualities that rendered it outstandingly suitable for use in building aggregates.

A rule in the Proposed Regional Coastal Plan prohibited the commercial extraction of beach sand on the open coast three years after the Plan becomes operative. The Plan was expected to become operative within the next two years. The company had not lodged a submission in respect of that rule.

The regional council sought to curtail further mining because of considerable doubt that existed as to whether sufficient sand existed at Otamarakau for mining to occur in the quantities and for the term sought by the applicant without causing, or contributing to, erosion along the coastline.

The district council and the Ratepayer's Association contended that there was only a finite volume of sand available within the Otamarakau Beach system and that there was a state of delicate balance between sediment inputs and outputs.

The critical issue for assessment by the Environment Court was whether continued sand extraction at Otamarakau, at the level and for the term sought, was sustainable. The applicant emphasised the benefits derived from the sand extraction, both in terms of local employment by the company and in assisting with local and regional building industry needs, with corresponding benefit to end users or recipients such as would-be home owners. However, the Court held that the task was not one of setting off perceived economic benefits of the activity against perceived adverse environmental effects and arriving at a net judgment. Rather an environmental outcome was required that accorded with the first part of section 5(2) of the RMA while meeting the requirements of paragraphs (a) to (c) of the subsection.

Section 5(2)(a), by excluding minerals, effectively indicated that there was no duty to manage the use of minerals so as to sustain their potential per se to meet the reasonably foreseeable needs for future generations. Even so, it was clear that the use of minerals, especially their extraction, was to be managed sustainably in all other respects. In the present case, the sustainable management of natural and physical resources involved consideration of effects that stem or result from, or may stem or result from, the proposed sand extraction in relation to the coastal environment.

On weighing all the evidence, particularly the technical evidence, the Court felt that this is a case where the following comment made by this Court

(similarly constituted) in *Rotorua Bore Users Association Inc. v Rotorua District Council* A138/98 warranted mention:

Much depends on the particular circumstances, including the nature of the perceived risks to the environment, as to whether the outcome should be that which is dictated by prevailing caution. Obviously, good environmental decision-making under the Act involves careful and balanced judgment. In many cases such a judgment is reached without the need to call in aid any principle of precaution; but in others, particularly those where scientific uncertainty exists, the need for balance may only be adequately achieved by applying precaution to the ultimate judgment.

The Court recommended a consent period expiring on 31 August 2003, with Year 1 under the consent running during the year 2001 (quantity of sand excavated not to exceed 20,000m³), Year 2 during 2002 (quantity not to exceed 15,000m³) and Year 3 over the eight month period to 31 August 2003 (quantity not to exceed 10,000m³).

Tawa and Ngatai v Bay of Plenty Regional Council A018/95

This case related to an application for a coastal permit to extract sand from the foreshore at four sites at Otamarakau Beach in the Bay of Plenty. J W Paterson & Sons Limited applied for consent to mine beach sand from below mean high water mark springs, involving a maximum total volume of 36,000m³ annually, with no more than 9,000m³ to be taken from any individual locality in any one year. The case was heard by the Planning Tribunal.

The case for the appellants, the Otamarakau Marae Trustees, was based on five main grounds: inadequate consultation; a claim to the land under the Treaty of Waitangi; involvement of other tribes whose standing in the area was not recognised by Ngāti Makino; monitoring resources of the regional council; and the credibility of the applicant.

With respect to consultation, the Tribunal held that the applicant's approaches adequately reflected its respect for Ngāti Makino's claim to mana whenua in respect of the application area; and that the trustees' declining to call a meeting of the iwi, and their attitude ruling out mining of any description, effectively ended the process. The Tribunal held that the applicant made a genuine attempt at consultation, and could not reasonably have been expected to take the matter any further with Ngāti Makino.

The regional council recommended a condition giving kaumatua of all three local iwi opportunity before mining started to verify whether the area contained wāhi tapu, urupa, or other sites or values of special importance. In light of this, the Tribunal did not accept that the regional council or its

hearings committee failed to take into account the relationship of Māori and their culture and traditions with their ancestral land, water, sites, wāhi tapu and their taonga or kaitiakitanga.

Citing the Tribunal's comments in *Sea Tow* (A139/93) the Tribunal did not accept the trustees' assertion that, because of the claim under the Treaty of Waitangi, the regional council had no authority to hear the resource consent application.

The Tribunal held that it was not for the regional council to decide which of the competing tribes is entitled to mana whenua over the area of foreshore affected by the coastal permit application. On receiving the present coastal permit application, the regional council was evidently aware of the difference among the iwi as to which of them was entitled to status as mana whenua over the area, and that the difference had not been resolved by the competent Court. In those circumstances, the Tribunal held that the regional council acted correctly in avoiding making any preference among the iwi, and in involving all three iwi.

The Tribunal did not agree that the past provided a sound basis for a finding that the regional council would not have the ability or resources to monitor compliance with any conditions of a coastal permit to the extent that it is within its responsibility to do so.

The Tribunal accepted unchallenged evidence that the impact on the shore biology of the proposed mining would be minimal. Conditions were proposed that specified that the mining activities would be kept well away from the habitats of birds. The Tribunal held that, if carried on in compliance with the recommended conditions, the proposed mining operation would not directly adversely affect fish species.

The Court found that, while the proposal would restrict public access and recreational use of the beach, the extent of those effects would be temporary and confined to a small area of the beach at a time; and would be no more than the unavoidable minimum restriction on public access and recreation if the activity is to be carried on at all. The restrictions would be compensated for by proposed public access and picnic and parking areas.

When assessing the sustainability of the proposal, the Tribunal held that the extraction of the sand would enable people and the community to provide for their economic wellbeing, both in the end-use of the sand taken, and in the employment provided in the extraction, processing and transporting of it. The Tribunal accepted the unchallenged evidence that the proposed rate of mining would be equivalent to about a quarter of the new sand being deposited on the beach by natural processes.

The effect of the works would be localised and temporary, and would only be visible from the beach itself or from the sea. Even so, the activity would involve some disturbance to the natural character of the coastal environment. However, the Tribunal held that with the proposed restrictions and controls to mitigate its effects, the activity was not inappropriate. The Tribunal recommended that consent be granted.

Sea Tow Ltd v Auckland Regional Council A129/93

This case has a complicated history involving five different applications for coastal permits to extract sand from the Mangawhai-Pakiri sand system. The applications were:

- Sea-Tow to take up to 25,000m³ per annum offshore at Pakiri and also up to 100,000m³ per annum outside the Mangawhai Harbour
- McCallum Brothers to take up to 45,000m³ offshore at Pakiri
- Kaipara Excavators to take up to 55,000m³ per annum off shore at Pakiri (the regional council recommended that only 45,000m³ per annum be extracted);
- MP Wilkinson to take up to 48,000m³ in the Mangawhai Harbour channel entrance (the regional council recommended that only 25,000m³ per annum be extracted)

All applications were dealt with as restricted coastal activities. From the recommendations of the regional council, largely granting consent for ten years subject to conditions which would be reviewed on completion of a proposed sand study, nine notices were lodged requiring inquiries. All parties other than Haddon and Brown had reached agreement on a basis for settlement of the issues raised in the notices.

The outstanding issues related to the relationship between the Treaty of Waitangi and consideration of applications for resource consents and the role of the tangata whenua in making decisions.

The Planning Tribunal examined the background of sand mining and held that the Mangawhai-Pakiri sand system was considered by the ready-mix concrete industry to be an important source of sand for the Auckland Region. Waikato River deposits were not available for any additional extraction operations and existing extraction there would eventually cease. Research undertaken by one of the applicants suggested that potential on-shore sources in the Auckland region were severely limited because of land use conflicts, unsuitable physical characteristics, or because of the presence of deleterious material. Increasing demand would favour offshore sands, particularly sand from the Mangawhai-Pakiri system.

The Tribunal acknowledged that the Pakiri-Mangawhai coastline had many features of regional, national and/or international significance (wildlife, vegetation and landform associations). The Mangawhai harbour was an important link in the chain of largely natural estuarine ecosystems around New Zealand. The area also had some important historical and cultural values.

The particulars of the Māori issues were addressed in detail by the Tribunal. In summary, Haddon and Brown argued that the Crown's presumptive ownership of the sand under the RMA was in breach of the Treaty and that the Crown should not have sole responsibility to make decisions which impact on their rangatiratanga and kaitiaki role. They also contended that consultation by the Auckland Regional Council had been inadequate.

The Tribunal held that it was not an appropriate forum in which to determine whether the vesting of the seabed in the Crown was a breach of the Treaty of Waitangi. Similarly, if the iwi of Ngātiwai had grievances about past actions by the Crown affecting their relationship with their ancestral taonga, those could be addressed in the claim lodged with the Waitangi Tribunal on their behalf. Granting the permits sought would not be permanent denial of the rights claimed by the Trust Board, nor would it necessarily have an adverse effect on the settlement of the claim if successful. If the sand was ultimately held to have been the property of the claimants, any injustice arising from extraction in the interim, might be able to be redressed by accounting for royalties collected from the exercise of the permits.

In relation to the claim that neither the Crown nor the Auckland Regional Council had the sole right to make decisions about extraction of sand, that being the customary authority of Ngātiwai, the Tribunal reiterated that it must apply the law as it is. The regional council had the sole function and duty to hear the applications and to make recommendations to the Minister. It was free to delegate its duty only to a committee set up as prescribed under the RMA (as it did), but it could not lawfully delegate the function to the tangata whenua.

The tangata whenua had been involved in the decision-making process to the extent provided for by the RMA, and as the law did not give them a greater part, there was no basis for an adverse report on the regional council's recommendations in that respect.

The Planning Tribunal held that, a proposal that its decision be deferred pending the outcome of the Trust Board's claim to the Waitangi Tribunal concerning ownership of the sand was not appropriate in this case, as it was unlikely that there would be a report from the Waitangi Tribunal within the period for which it would be reasonable to expect decisions on the present applications to be delayed.

The Tribunal found that the applicants and the regional council all demonstrated proper respect for the interests of the tangata whenua in the applications and consulted fully with them. The tangata whenua's mana, role and kaitiaki are recognised by the inclusion of their representative in the working party for the sand study. They would benefit from the results of that study, without being required to contribute to the cost. If the permits were granted, the relationships could also be recognised by involvement in monitoring the effects of sand extraction, and perhaps even by making inspections and observations to ensure compliance by the consent holders.

The Tribunal held that a condition relating to the use of the sand was not necessary, even though it acknowledged that it was important for tangata whenua to consider the use to which the resource would be put.

The Tribunal was satisfied that the sand resource was so large that extraction at the proposed rates for up to 10 years could be sustained. The Tribunal was also satisfied that the natural character of the coastline would be preserved and protected, as would the natural features of the Pakiri sand dunes. No plausible basis was raised for concern that indigenous vegetation or fauna would be affected, and that could be confirmed in the proposed sand study. The Tribunal recommended that coastal permits be issued subject to conditions.

Taranaki Energy Watch Incorporated v Taranaki Regional Council and New Plymouth District Council W039/03

This case involved the Pohokura Gas Field, lying off the shore of North Taranaki at a depth of about 3,500 metres under the seabed. The field was New Zealand's only significant undeveloped gas resource. Gas from that source would become a valuable source of energy in place of gas from the Maui Field which would become depleted over the next few years.

The eight appeals related to the conditions that should be imposed on the resource consents needed for the facilities to exploit the Pohokura Field. The proposed facilities were up to three offshore wellhead platforms, two onshore drilling sites, and an onshore production station.

In the event, all but one of the eight appeals were settled by agreement being reached to propose to the Court amendments to the consent conditions. The one appeal that was not settled (by Te Ohu o Pohokura) gave rise to questions about:

- The Court's jurisdiction to impose conditions requiring financial contributions for solar water heaters
- Whether a separate consent was needed to use energy and minerals
- Requiring financial contributions for reforestation
- Requiring efficient end-use of gas products

The Council submitted that the condition requiring a financial contribution for solar water heaters would not conform with provisions of the Air Quality Plan in that it would require payment to a Trust (which would have to be formed); and that there was no evidence that the proposed discharge would have any adverse effect on public amenities, or places having heritage value or importance to tangata whenua.

Under section 108(10) of the RMA, a consent condition requiring payment of money can only be imposed if it is made for purposes specified in the plan, and if it does not exceed the maximum amount specified in, or determined in accordance with, the plan. The Court held that the proposed condition would not comply with s108(9), so it lacked jurisdiction to consider it, and lacked the power to impose it.

The Court found that a proposition that a separate resource consent was required for use of the energy potential of the gas extracted and produced was not even arguable. The Court held that the environmental effects, of the end-use of the energy potential of the gas to be produced, was too remote to be taken into account. The end-users' activities would either be within the permitted baseline, or if larger in scale, would require separate resource consents.

The Court held that conditions related to the efficiency of the end-use of gas products would not fairly and reasonably be related to emissions from the on-shore wellsite and the production station, and therefore would not be valid.

The effects of discharge of greenhouse gases by end users of gas is a policy issue for the New Zealand Government, not a matter to be taken into account in deciding a resource consent application. The Court found that the carbon dioxide content of the discharges was not likely to contribute to significant adverse effects on a public amenity, or on places having heritage value or importance to tangata whenua. Consequently the circumstances prescribed by the Air Quality Plan for requiring financial contributions would not be present, and it would not be lawful to impose the afforestation condition.

Te Ohu's appeal was disallowed. The Court recommended that the resource consents be granted subject to the conditions outlined.



appendix two

Cases on Aquaculture

Summary of cases

This appendix summarises eight cases where the Environment Court considered resource consent applications for marine farms.

One of the main factors raised in opposition to applications for new marine farms is that the proposed farms will have adverse effects on the landscape, natural character and amenity values of the area. The Court usually adopts a holistic approach when assessing a landscape, taking into account the water, the land-water interface and the surrounding land. If the broader coastal environment has already been modified, then it is likely that the Court will decide that any adverse effects from the marine farm will be minor. The Court considers whether the proposal can be ‘absorbed’ into the existing landscape and whether it is appropriate in the context. If the surrounding environment consists of pastoral land, then the Court will usually conclude that the presence of a marine farm is consistent with this environment.

Related to issues of amenity is the concern about cumulative effects. Although individual marine farms may have minor adverse effects, when there are several of them within the same area, then the combined effect can be significant. This issue is often raised in respect of amenity values and ecological effects.

Despite the fact that navigational issues are raised in the majority of cases, this concern is unlikely to carry much weight unless it can be established that the proposed marine farm is to be constructed on a direct navigational route. The Court will usually assess what are the most common routes used by vessels in the area, how often the proposed area for a marine farm is used, and whether the proposed site can safely and easily be avoided by navigators.

Ecological effects are usually a factor in cases involving marine farms, in particular the impacts on benthic fauna and marine mammals. The Court places great emphasis on the necessity for scientific evidence to adequately assess the likely ecological effects of a marine farm. If the evidence is unclear, or the potential effects have not been established, then the Court will often recommend an adaptive management approach. The application of this approach usually involves staged development and significant monitoring requirements.

The adaptive management approach is increasingly being proposed by the applicants themselves as conditions of consent. In some instances, the Court has required baseline monitoring for several years before any development can be undertaken. Adaptive management has involved subsurfacing of the marine farm and a proposal to remove the structures during the periods when marine mammals mostly frequent the area proposed for the marine farm.

When reading aquaculture cases it is important to note that the law in this area has changed significantly. Prior to amendments to the RMA which came into force on 1 January 2005, marine farms required two permits, a coastal permit under the RMA and a second permit under the Fisheries Act 1996. Since 1 January 2005, all consenting has been undertaken under the RMA.

Factors considered by the Court in deciding whether or not to grant consent:

- Natural character and landscape effects
- Visual and amenity effects
- Navigational and public access issues
- Recreational issues
- Ecological effects
- Impact on benthic fauna
- Impact on marine mammals
- Māori concerns
- Cumulative effects
- Beneficial effects
- New Zealand Coastal Policy Statement
- Regional policy statement
- Relevant district plan provisions
- Lack or availability of scientific information
- Baseline monitoring
- Adaptive management measures

Pigeon Bay Aquaculture Limited v Canterbury Regional Council C32/99

This case concerned an application for two marine farms in Pigeon Bay, Banks Peninsula. The marine farms were to be used for mussel farming and to grow dredge oysters and scallops. The Canterbury Regional Council granted resource consent for one of the marine farms and declined consent for the other.

The application was opposed on the basis that the adverse effects of the proposal were more than minor and were unacceptable. The main alleged adverse effects were on the natural character of the coastal environment and on amenity values.

This case is important as it consolidated and clarified the criteria to be applied when assessing whether a landscape is significant. These are as follows:

- The natural science factors – the geological, topographical and dynamic aspects of the landscape
- Its aesthetic values including memorability and naturalness

- Its expressiveness – how obviously the landscape demonstrates the formative processes leading to it
- Transient values – occasional presence of wildlife; or its values at certain times of the day or of the year
- Whether the values are shared and recognised
- Its values to tangata whenua
- Its historical associations

The Court concluded that, in terms of the policies in the regional policy statement, there would not be substantial modification to the naturalness of Pigeon Bay, the effects would be short-term and localised and the ecological state of the area would not be affected except for the proposed introduction of scallops and dredge oyster species to the area.

The Court heard evidence about the adverse effects of the proposal on navigation. The establishment of the two marine farms was not considered to create a navigation hazard because they were both out of the normal courses of transit in and out of the bay. One of the experts also noted that in his experience of the Marlborough Sounds, where there are approximately 500 marine farms, there is only one known incident of a vessel colliding with a marine farm.

The Court discussed the issue of whether the granting of one resource consent application creates a precedent which means that a second application should also be granted. The Court held that ‘granting a resource consent even for a non-complying activity may create a substantive precedent.’ This comment was made on the basis that justice required that similar cases should be treated equally. The Court noted that this was especially applicable where adverse ecological effects were not of major concern but amenity issues were. However, in this case, the Court considered that the granting of the consent would not create any precedent that could be usefully relied on because there were no other areas suitable for development within the Banks Peninsula area.

In determining the appeals the Court concluded that the resource consents for the marine farms should both be granted with conditions, but only mussels were permitted to be grown because the Court had not heard evidence about the effects of the proposal on dredge oysters and scallops in the bays of Banks Peninsula.

Ngati Kahu Ki Whangaroa Co-Operative Society Limited v Northland Regional Council A95/2000

This case is an appeal against a decision made by the Northland Regional Council to grant a coastal permit for structures that enable oysters to be farmed in the Whangaroa Harbour. The appellants asserted that the grant should be

cancelled and resource consent refused on the basis that the proposed site is in part of the harbour which has traditional and historical links with the community and is also used for recreational purposes.

The Court rejected some of the grounds of appeal, and excluded the associated evidence, on the basis that some of the alleged adverse effects related to matters to be considered under the Fisheries Act. This included those grounds relating to the effects on other types of fishers and existing marine farms.

The Court commented that when the structures of the proposed marine farm protruded above the water surface this would detract from the natural character of the coastal environment. However because the environment already included similar structures, and the proposed regional coastal plan provides that marine farming is generally suitable, the Court concluded that the proposal would not be inappropriate development.

One of the arguments made by the appellants was that the marine farm would have adverse effects on water quality. This was because 'the filter-feeding activity of bivalve shellfish increases the rate of biodeposition, because finer particles of suspended sediment and organic material are concentrated in faeces and pseudofaeces which then settle out of the water column,' and this would have a negative impact on water quality. The Court accepted that it was likely that there would be some adverse effect on water quality from the proposed marine farm yet these were not quantified and there was no evidence that such a deterioration in water quality would raise the total contamination of the harbour to a significant threshold.

In relation to Māori cultural effects, a kaumatua of Ngāti Kahu Ki Whangaroa gave evidence of the value placed by local people on seafood from Whangaroa harbor. The evidence asserted that oyster farms were 'exterminating' the natural and physical resources valued by Māori people and that the proposed oyster farm was in direct contrast with their tikanga.

The Court noted that under section 6(e) of the RMA it is a matter of national importance that consent authorities are to recognise and provide for the relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wāhi tapu and other taonga. Based on the evidence, the Court concluded that the proposed marine farm would not preclude Māori from collecting shellfish or other seafood and there were no actual or potential effects on Māori cultural values.

The Court held that the proposed marine farm would have positive effects of enabling people to provide for their economic and social wellbeing through the establishment of a small business which would employ local people. The proposal would result in some adverse effects on the natural character of the environment and limit recreational opportunities. On balance the Court

determined that the proposal did satisfy the purpose of sustainable management in the RMA. The application was granted and the respondent's decision confirmed subject to minor amendments.

Clifford Bay Marine Farms Limited v Marlborough District Council
C113/2004

This case related to applications for resource consents to establish a large open water mussel farm in Clifford Bay, Marlborough Sounds. The Marlborough District Council granted consent for one of the marine farms and the Director-General of Conservation appealed the decision. The central issue in the case was what effect the proposed marine farm would have on Hector's dolphins that allegedly frequented the area.

In the proposed Wairau-Awatere resource management plan the Cloudy and Clifford Bay area was identified as an area of national importance for Hector's dolphins. The Director-General of Conservation asserted that there were potential detrimental impacts on Hector's dolphins which are an indigenous species that is endangered.

The applicant contended that there was no scientific evidence to support the assertion that the proposed farm could cause harm to Hector's dolphins. However to address the concerns, the applicant proposed to stage the development, and to monitor the effect of each stage before proceeding with further development. Any adverse effects could be remedied by an adaptive management response.

The Court placed great weight on the provisions in the New Zealand Coastal Policy Statement relating to the national priorities of avoiding areas important to the continued survival of any indigenous species and to protect the functioning of the coastal environment in terms of natural biodiversity. Both of these national priorities were identified as being relevant to the assessment of effects on Hector's dolphins.

The Court observed that the proposed marine farm could potentially result in adverse effects on the habitat of indigenous fauna, especially Hector's dolphins. However the Court noted that there was minimal certainty about the potential effects. The Court can consider effects of low probability with high potential impact but it is somewhat unclear from the case law whether the Court can consider accumulating effects of low probability as an 'actual and potential effect' as set out in section 104(1)(a) of the RMA. In this case the Court avoided the issue by dealing with such effects as 'other relevant matters' under section 104(1)(i) of the RMA.

The Court discounted any risk that the marine farm would cause death to a Hector's dolphin. It accepted the following findings: there would be a reduction

in the available habitat but the impact would be low, it was uncertain what effect the marine farm would have on breeding, it was unlikely that the marine farm would reduce the food supply of the dolphin, and the proposal would have benefits to Hector's dolphins because there would be more comprehensive monitoring undertaken which would result in greater understanding.

Overall the Court held that the proposed marine farm was not likely to have any significant adverse accumulative effect on Hector's dolphins as long as the development was staged, involved monitoring and adaptive management. The Court upheld the resource consent and recommended that the Minister of Conservation grant a coastal permit for occupation of the site.

Southern Seas Marine Farms Limited v Canterbury Regional Council C119/2005

This case related to an application for resource consents to establish a marine farm at Port Levy, Banks Peninsula. The proposal involved the establishment of two blocks on the eastern side of Port Levy with a total area of 20.68 hectares. The applicant sought occupation of the marine area for a standard long line facility for the purposes of farming green lipped mussels, blue mussels and spat gathering. The Canterbury Regional Council had declined to grant consent and the applicants subsequently appealed against the decision.

For the purposes of the application the activity was discretionary, and the Court noted that this meant it was generally appropriate within the area, but not on every site and not at every scale. The regional council had since amended the proposed regional coastal environment plan to make marine farming a non-complying activity within certain bays including Port Levy.

The major issues in the case related to consideration of the potential effects of the proposed marine farm on landscape and amenity values, recreation, benthic habitat, and cumulative effects.

Three landscape architects provided evidence to the Court, and all of them agreed that the overall landscape of Banks Peninsula could be regarded as an outstanding natural landscape, as could the headlands of Port Levy. They agreed that a marine farm on the western side of the Port would be inconsistent with landscape values. However in relation to the eastern side of the Port they reached different conclusions. The essential point of difference related to what scale should be used to assess the adverse effects on natural character. Port Levy itself demonstrates pastoral and other uses of the landscape, where at the shoreline the natural elements are dominant.

The Court adopted the approach that when assessing the natural character of an area 'the water, land-water interface and land above, together with the coastal environment' must all be taken into account. This approach was

consistent with that used by tangata whenua. The Court concluded that the marine farm was sufficiently distant from features of high natural character and that the proposal could be 'absorbed' along the coastline because the context of Port Levy was one of a pastoral use of the land.

In relation to recreation and matters of public access and navigation, the Court accepted that the shoreline of both the northern and southern sites is from time to time accessed by boats to allow both diving and reef fishing. The Court was satisfied that a separation between the northern and southern sites of a minimum of 60 metres would allow adequate room for vessels to access the coastline and the shore. Related to this was the issue of cumulative effects. Concerns were raised that the number of marine farms within the Banks Peninsula area created cumulative adverse amenity effects. However this factor was ultimately discounted on the basis that it was not a significant effect.

The Court concluded that there was no evidence of potential adverse effects on benthic species and that the monitoring proposed by the applicant would appropriately identify any phytoplankton depletion or other benthic effects.

The Court considered it had addressed Part II matters within the general criteria that it had applied to the application. It held that the potential adverse effects would be minimal and that any effects could be sufficiently dealt with by the terms and conditions of the proposed consent. The Court granted the consents.

East Bay Conservation Society Incorporated v Marlborough District Council W106/2006

This appeal concerned Marlborough District Council's decision to grant consent for a marine farm immediately east of Parea Point in East Bay, Marlborough Sounds. The three main issues were the possible adverse effects on the natural character, landscape and amenity of the area, navigation and benthic issues.

The Court held that the proposed farm is not sited on a 'recognised navigational route'. Although it was possible to choose a route through the proposed farm, the Court concluded that most people would choose a more direct route that bypasses the farm, or would choose to take a route further offshore for safety reasons.

In relation to natural character issues, Policy 1.1 of the New Zealand Coastal Policy Statement states that it is a national priority to preserve the natural character of the coastal development and avoid sprawling, or sporadic subdivision, use or development. The Court held that the proposed marine farm would not result in sprawling, because it would form part of an existing cluster of marine farm activity.

The Court agreed with the expert evidence that Puriri Point was an outstanding natural landscape. However the Court accepted that the proposed marine farm would be viewed in the context of the existing salmon farm and the production forest and the adverse effects on the natural character of Puriri Point would be minor.

This demonstrates that the Court will often consider that there are minor adverse visual effects on the natural character if a similar development already exists near the proposed marine farm. There was minimal discussion by the Court of the cumulative adverse effects that a cluster of marine farms may have on the natural character, except for stating that ‘the addition of the proposed farm would not tip the balance in favour of too much marine farm activity’.

The benthic habitat that was a concern to the appellants was the ‘fine sand and hard substrate partly underlying and along the western boundary of the proposed site ... in this area are giant lampshells, burrowing anemone and common lampshells.’ In response to a recommendation from an ecological consultant the applicants altered the proposed location for the marine farm.

However concerns were raised by other consultants that a farm on the proposed site would deposit several hundred tonnes of sediment each year into the surrounding environment and this could have a detrimental effect on sensitive inshore areas. The Court concluded that an adaptive management response was most appropriate and only three lines should be installed coupled with a baseline study and ongoing monitoring. The Court concluded that the proposal met the requirements of sustainable management and upheld the council’s consent.

Biomarine Limited v Auckland Regional Council A14/2007

This case concerned an appeal by the applicant of the Auckland Regional Council’s decision to decline an application for consents to construct and operate an oyster farm in the Kaipara Harbour, comprising four 22.5 hectare blocks connected by a central corridor. These activities were classified as discretionary under the proposed regional plan. The regional council also refused to recommend to the Minister of Conservation that the applicant should be permitted to occupy the coastal marine area.

The Kaipara harbour is considered to be the largest estuary in New Zealand. It is said to have ‘very extensive tidal flats, relatively deep channels, very strong tidal flows and is connected to the open coast by an entrance some two kilometres wide.’ There is a diversity of habitats within the harbour including some unique species.

The case focused on three main potential issues from the proposed marine farm; the effect on *Zostera* (sub-tidal seagrass) on the site, the effect on benthic

fauna and effects on natural character. Deposition was not an issue in this case due to the particularly strong tidal movements.

The applicant proposed to pursue a staged development so that monitoring and adaptive management processes could be undertaken which would ensure that there were no adverse effects from the marine farm. The construction of further stages of the development would be contingent on the results of the previous stages.

One issue which arose out of the applicant's proposal was how to determine what qualified as an adverse effect, and how it could be accurately measured, when there were natural variations in the flora and fauna of the area. The evidence presented to the Court was that monitoring could identify effects of between two and five per cent change and that the *Zostera* habitat and species relying on this habitat should be the specific focus of monitoring. The Court adopted this approach and proceeded on the basis that an adverse effect was a scientifically significant adverse effect.

In respect of *Zostera*, the Court found that the particular species present in New Zealand was indigenous and performed a number of ecological functions in the marine environment. The particular *Zostera* found in the Kaipara harbour was subtidal and this type was nationally rare. The Court emphasised the importance of the *Zostera* to the overall ecosystem. Significantly the Court noted that a decrease in *Zostera* would result in 'a reduction in estuarine productivity and decline in the abundance and diversity of invertebrate and fish species.' *Zostera* is sensitive to environmental changes and there was a possibility that it would be slow to recover after a disturbance.

The Court stated that:

The possible effects of an oyster farm on *Zostera* have been identified as including enrichment of sediments by organic matter, physical damage during construction and by farm management operations, smothering by material deposition, shading by structures, and the combination of these factors.

Despite the potential adverse effects on *Zostera*, the Court concluded that these could largely be addressed through the monitoring and adaptive management measures as outlined above.

It was undisputed there was benthic fauna on the site including a unique tube-building worm. The point of contention was whether the benthic fauna was indigenous or exotic. This is relevant in determining whether section 6(c) of the Resource Management Act 1991 applies. The Court proceeded on the basis that there may be indigenous fauna but that any adverse effects could be

adequately mitigated through conditions preventing trampling and propeller scour.

The Court held that the natural character of Kaipara harbour was:

dominated by the large-scale visual catchment, and the natural processes, winds, tides, seasons which dominate it. Any built structures, such as the oyster farm, will be of such a small scale that they will have little, if any, effect on the overall natural character already dominated by modified rural landscapes.

The Court observed that there was a continuum of effect on natural character, from an extremely modified setting to an untouched environment. In the case of the proposed oyster farm it would be situated within a coastal environment that already included pastoral land surrounding the harbour.

In exercising its discretion under the RMA, the Court held that the consents should be granted for Blocks 1, 2 and 3, as 70-75 hectares would be adequate for the activity while providing for protection of indigenous ecology. The consents were granted for 20 years as opposed to 30 years in order to balance the need for commercial certainty for investment with the need to adopt a cautious approach. However in relation to Block 4, the Court declined to grant consent because the potential adverse effects on *Zostera* were estimated to be the most severe.

CRA 2 Rock Lobster Management Company Ltd v Bay of Plenty Regional Council A103/2008

This case concerned a proposed major mussel farm to be located approximately 9.3 kilometres offshore of Opotiki. The mussel farm required four different types of consent under the RMA. Three of these had been granted by the Bay of Plenty Council and were confirmed on appeal to the Environment Court, subject to conditions in consent orders.

This case relates to the fourth resource consent sought to allow the occupation of the area proposed for the marine farm. The extent of proposed occupation of the coastal marine area was a restricted coastal activity and required a permit which could only be granted by the Minister of Conservation.

During the processing of the application, the law surrounding marine aquaculture was substantially altered due to the enactment of the Māori Commercial Aquaculture Claims Settlement Act 2004, the Aquaculture Reform (Repeals and Transitional Provisions) Act 2004 and the amendment of both the Fisheries Act 1996 and the RMA.

However the transitional provisions of the legislative amendments meant the application was to be considered under the previous legislative framework.

The Court did note that the proposed marine farm would satisfy an obligation of the Crown under the Māori Commercial Aquaculture Claims Settlement Act 2004 to provide water space to iwi in respect of aquaculture.

The Court observed that the consent process for the proposed marine farm had been a lengthy one, but the extensive scientific investigation which was undertaken along with the thorough consultation, had been of great assistance to the final outcome. The regional council had conducted an offshore science project and mapping exercise which demonstrated that the Opotiki site had the least potentially conflicting uses and values in the Bay of Plenty. The findings of the project also highlighted that the cumulative effects of the proposed mussel farm and a proposed farm at Otamarakau were sustainable.

Further research included the construction and testing of a prototype offshore mussel farming line in the Marlborough Sounds and then offshore from Napier. This established that it is feasible to anchor a mussel line in an exposed environment and generate a crop.

The Court recommended that the Minister of Conservation grant consent to a restricted coastal activity so as to allow the proposed marine farm to occupy part of the coastal marine area. This was on the basis of the inquiry it conducted and the fact that all the parties indicated that they wanted the Court to make such a recommendation subject to amended conditions as agreed between the parties.

Marlborough Aquaculture Limited v The Marlborough District Council W027/2009

This recent interim decision concerned an application to establish a marine farm of 24 hectares and an application to extend an existing marine farm in Admiralty Bay in the Marlborough Sounds.

The Court had issued a previous interim decision (W36/2006) in relation to this matter where it held that Admiralty Bay was a significant habitat for the dusky dolphins, particularly over winter. The existing marine farms had displaced the dolphins from that area, but this had not resulted in any harm to the 1000 plus dusky dolphins which go there. However the Court accepted the evidence that there will be a point at which the extension of the marine farms in Admiralty Bay will adversely affect the dolphin habitat. Significantly the Court noted that ‘the preservation and protection required by s6 ... become increasingly dominant factors in the balancing exercise. In other words, the precaution inherent in the RMA will be increasingly likely to outweigh factors such as, for instance, economic wellbeing.’

The applicants attempted to address the above concerns by proposing mitigation measures. One proposed measure was to remove the surface structures and long lines from the sites over the winter period when the dolphins used

the Bay the most and to perhaps subsurface the farm. One of the applicants suggested that a further mitigation measure could involve baseline monitoring over a three year period to more precisely establish the use of various parts of the Bay by the dolphins.

The Court observed that establishing baseline information and removing the farm structures would adequately avoid or mitigate possible adverse effects on the dolphin habitat. However these were indications and not conclusive determinations. The Court decided not to grant consents for the marine farms, until the results of the three year study were known, because it could not properly assess the affects on the dolphin habitat without such information. In addition, the possible adverse effects on the natural character of Admiralty Bay and on the safe navigation on the waters of the Bay, could not be adequately assessed until the details of the subsurface proposal were developed and explained.

This decision demonstrates that the Court is likely to adopt a precautionary approach to marine farms when the marine environment is close to the point of saturation. It will be hesitant to grant consent when there is a lack of robust information about the potential adverse effects that a proposal may have on marine mammals.



appendix three

New Zealand Coastal Policy Statement

This appendix reproduces the 1994 NZCPS, with the exclusion of introductory material.

General Principles for the Sustainable Management of New Zealand's Coastal Environment

... In addition to the foregoing [RMA purpose and principles], to provide for the special context of the coastal environment, regard shall be had to the following general principles:

1. Some uses and developments which depend upon the use of natural and physical resources in the coastal environment are important to 'the social, economic and cultural well-being' of 'people and communities'. Functionally, certain activities can only be located on the coast or in the coastal marine area.
2. The protection of the values of the coastal environment need not preclude appropriate use and development in appropriate places.
3. The proportion of the coastal marine area under formal protection is very small and therefore management under the Act is an important means by which the natural resources of the coastal marine area can be protected.
4. Expectations differ over the appropriate allocation of resources and space in the coastal environment and the processes of the Act are to be used to make the appropriate allocations and to determine priorities.
5. People and communities expect that lands of the Crown in the coastal marine area shall generally be available for free public use and enjoyment.
6. The protection of habitats of living marine resources contributes to the social, economic and cultural wellbeing of people and communities.
7. The coastal environment is particularly susceptible to the effects of natural hazards.
8. Cultural, historical, spiritual, amenity and intrinsic values are the heritage of future generations and damage to these values is often irreversible.
9. The tangata whenua are the kaitiaki of the coastal environment.

10. It is important to maintain biological and physical processes in the coastal environment in as natural a condition as possible, and to recognise their dynamic, complex and interdependent nature.
11. It is important to protect representative or significant natural ecosystems and sites of biological importance, and to maintain the diversity of New Zealand's indigenous coastal flora and fauna.
12. The ability to manage activities in the coastal environment sustainably is hindered by the lack of understanding about coastal processes and the effects of activities. Therefore, an approach which is precautionary but responsive to increased knowledge is required for coastal management.
13. A function of sustainable management of the coastal environment is to identify the parameters within which persons and communities are free to exercise choices.
14. The potential for adverse effects of activities to spread beyond regional boundaries may be significant in the coastal marine area.

CHAPTER 1 - National Priorities for the Preservation of the Natural Character of the Coastal Environment Including Protection from Inappropriate Subdivision, Use and Development

Policy 1.1.1

It is a national priority to preserve the natural character of the coastal environment by:

- a. encouraging appropriate subdivision, use or development in areas where the natural character has already been compromised and avoiding sprawling or sporadic subdivision, use or development in the coastal environment;
- b. taking into account the potential effects of subdivision, use, or development on the values relating to the natural character of the coastal environment, both within and outside the immediate location; and
- c. avoiding cumulative adverse effects of subdivision, use and development in the coastal environment.

Policy 1.1.2

It is a national priority for the preservation of the natural character of the coastal environment to protect areas of significant indigenous vegetation and significant habitats of indigenous fauna in that environment by:

- a. avoiding any actual or potential adverse effects of activities on the following areas or habitats:
 - i. areas and habitats important to the continued survival of any indigenous species; and
 - ii. areas containing nationally vulnerable species or nationally outstanding examples of indigenous community types;
- b. avoiding or remedying any actual or potential adverse effects of activities on the following areas:

- i. outstanding or rare indigenous community types within an ecological region or ecological district;
- ii. habitat important to regionally endangered or nationally rare species and ecological corridors connecting such areas; and
- iii. areas important to migratory species, and to vulnerable stages of common indigenous species, in particular wetlands and estuaries;
- c. protecting ecosystems which are unique to the coastal environment and vulnerable to modification including estuaries, coastal wetlands, mangroves and dunes and their margins; and
- d. recognising that any other areas of predominantly indigenous vegetation or habitats of significant indigenous fauna should be disturbed only to the extent reasonably necessary to carry out approved activities.

Policy 1.1.3

It is a national priority to protect the following features, which in themselves or in combination, are essential or important elements of the natural character of the coastal environment:

- a. landscapes, seascapes and landforms including:
 - i. significant representative examples of each landform which provide the variety of each region;
 - ii. visually or scientifically significant geological features; and
 - iii. the collective characteristics which give the coastal environment its natural character including wild and scenic areas;
- b. characteristics of special spiritual, historical or cultural significance to Māori identified in accordance with tikanga Māori; and
- c. significant places or areas of historic or cultural significance.

Policy 1.1.4

It is a national priority for the preservation of natural character of the coastal environment to protect the integrity, functioning, and resilience of the coastal environment in terms of:

- a. the dynamic processes and features arising from the natural movement of sediments, water and air;
- b. natural movement of biota;
- c. natural substrate composition;
- d. natural water and air quality;
- e. natural biodiversity, productivity and biotic patterns; and
- f. intrinsic values of ecosystems.

Policy 1.1.5

It is a national priority to restore and rehabilitate the natural character of the coastal environment where appropriate.

CHAPTER 2 - The Protection of the Characteristics of the Coastal Environment of Special Value to the Tangata Whenua Including Waahi Tapu, Tauranga Waka, Mahinga aataitai, and Taonga Raranga

Policy 2.1.1

Provision should be made for the identification of the characteristics of the coastal environment of special value to the tangata whenua in accordance with tikanga Māori. This includes the right of the tangata whenua to choose not to identify all or any of them.

Policy 2.1.2

Protection of the characteristics of the coastal environment of special value to the tangata whenua should be carried out in accordance with tikanga Māori. Provision should be made to determine, in accordance with tikanga Māori, the means whereby the characteristics are to be protected.

Policy 2.1.3

Where characteristics have been identified as being of special value to tangata whenua, the local authority should consider:

- a. The transfer of its functions, powers and duties to iwi authorities in relation to the management of those characteristics of the coastal environment in terms of Section 33 of the Resource Management Act 1991; and/or
- b. The delegation of its functions, powers and duties to a committee of the local authority representing and comprising representatives of the relevant tangata whenua, in relation to the management of those characteristics of the coastal environment in terms of Section 34 of the Resource Management Act 1991.

CHAPTER 3 - Activities Involving the Subdivision, Use or Development of Areas of the Coastal Environment

3.1 Maintenance and Enhancement of Amenity Values

Policy 3.1.1

Use of the coast by the public should not be allowed to have significant adverse effects on the coastal environment, amenity values, nor on the safety of the public nor on the enjoyment of the coast by the public.

Policy 3.1.2

Policy statements and plans should identify (in the coastal environment) those scenic, recreational and historic areas, areas of spiritual or cultural significance, and those scientific and landscape features, which are important to the region or district and which should therefore be given special protection; and that policy statements and plans should give them appropriate protection.

Policy 3.1.3

Policy statements and plans should recognise the contribution that open space makes to the amenity values found in the coastal environment, and should seek to maintain and enhance those values by giving appropriate protection to areas of open space.

3.2 Providing for the Appropriate Subdivision, Use and Development of the Coastal Environment

Policy 3.2.1

Policy statements and plans should define what form of subdivision, use and development would be appropriate in the coastal environment, and where it would be appropriate.

Policy 3.2.2

Adverse effects of subdivision, use or development in the coastal environment should as far as practicable be avoided. Where complete avoidance is not practicable, the adverse effects should be mitigated and provision made for remedying those effects, to the extent practicable.

Policy 3.2.3

Policy statements and plans should recognise the powers conferred by Section 108 to obtain environmental benefits which will (to a degree) offset environmental damage, by specifying purposes in their plans for which 'financial contributions' can be sought, in cases where there will be unavoidable adverse effects from subdivision, use or development in the coastal environment.

Policy 3.2.4

Provision should be made to ensure that the cumulative effects of activities, collectively, in the coastal environment are not adverse to a significant degree.

Policy 3.2.5

Subdivision, use and development in the coastal environment should be conditional on the provision of adequate services (particularly the disposal of wastes), and the adverse effects of providing those services should be taken into account when preparing policy statements and plans and when considering applications for resource consents.

Policy 3.2.6

Policy statements and plans should make provision for papakainga housing and marae developments in appropriate places in the coastal environment. 'Papakainga housing' means residential occupancy on any ancestral land owned by Māori.

Policy 3.2.7

Policy statements and plans should identify any practicable ways whereby the quality of water in the coastal environment can be improved by altered land management practices, and should encourage the adoption of those practices.

Policy 3.2.8

Provision should be made for the protection of the habitats (in the coastal marine area) of species which are important for commercial, recreational, traditional or cultural purposes.

Policy 3.2.9

Policy statements and plans should contain a requirement that the Maritime Safety Authority and the Hydrographic Office of the Royal New Zealand Navy are to be notified of new structures and works in the coastal marine area at the time permission is given for their construction.

Policy 3.2.10

Policy statements and plans should indicate that when restoration plantings are carried out, preference should be given to the use of indigenous species, with a further preference for the use of local genetic stock.

3.3 Adoption of a Precautionary Approach to Activities with Unknown but Potentially Significant Adverse Effects

Policy 3.3.1

Because there is a relative lack of understanding about coastal processes and the effects of activities on coastal processes, a precautionary approach should be adopted towards proposed activities, particularly those whose effects are as yet unknown or little understood. The provisions of the Act which authorise the classification of activities into those that are permitted, controlled, discretionary, noncomplying or prohibited allow for that approach.

Policy 3.3.2

Local authorities should share information and knowledge gained by them about the coastal environment, particularly where it relates to coastal processes and/or to activities with previously unknown or little known effects.

3.4 Recognition of Natural Hazards and Provision for Avoiding or Mitigating Their Effects

Policy 3.4.1

Local authority policy statements and plans should identify areas in the coastal environment where natural hazards exist.

Policy 3.4.2

Policy statements and plans should recognise the possibility of a rise in sea level, and should identify areas which would as a consequence be subject to erosion or inundation. Natural systems which are a natural defence to erosion and/or inundation should be identified and their integrity protected.

Policy 3.4.3

The ability of natural features such as beaches, sand dunes, mangroves, wetlands and barrier islands, to protect subdivision, use, or development should be recognised and maintained, and where appropriate, steps should be required to enhance that ability.

Policy 3.4.4

In relation to future subdivision, use and development, policy statements and plans should recognise that some natural features may migrate inland as the result of dynamic coastal processes (including sea level rise).

Policy 3.4.5

New subdivision, use and development should be so located and designed that the need for hazard protection works is avoided.

Policy 3.4.6

Where existing subdivision, use or development is threatened by a coastal hazard, coastal protection works should be permitted only where they are the best practicable option for the future. The abandonment or relocation of existing structures should be considered among the options. Where coastal protection works are the best practicable option, they should be located and designed so as to avoid adverse environmental effects to the extent practicable.

3.5 Maintenance and Enhancement of Public Access to and Along the Coastal Marine Area

Policy 3.5.1

In order to recognise the national importance of maintaining public access to and along the coastal marine area, a restriction depriving the public of such access should only be imposed where such a restriction is necessary :

- a. to protect areas of significant indigenous vegetation and/or significant habitats of indigenous fauna;
- b. to protect Māori cultural values;
- c. to protect public health or safety;
- d. to ensure a level of security consistent with the purpose of a resource consent; or
- e. in other exceptional circumstances sufficient to justify the restriction notwithstanding the national importance of maintaining that access.

Policy 3.5.2

In order to recognise the national importance of enhancing public access to and along the coastal marine area, provision should be made to identify, as far as practicable:

- i. the location and extent of places where the public have the right of access to and along the coastal marine area;
- ii. those places where it is desirable that physical access to and along the coastal marine area by the public should be enhanced; and
- iii. those places where it is desirable that access to the coastal marine area useable by people with disabilities be provided.

Policy 3.5.3

In order to recognise and provide for the enhancement of public access to and along the coastal marine areas as a matter of national importance, policy statements and plans should make provision for the creation of esplanade reserves, esplanade strips or access strips where they do not already exist, except where there is a specific reason making public access undesirable.

Policy 3.5.4

Policy statements and plans should as far as practicable identify the access which Māori people have to sites of cultural value to them, according to tikanga Māori.

CHAPTER 4 - The Crown's Interests in Land of the Crown in the Coastal Marine Area

4.1 Maintenance of the Crown's Interest in Land of the Crown in the Coastal Marine Area

Policy 4.1.1

Regional coastal plans should identify land and areas under the Conservation Act 1987 and other land and areas administered by the Department of Conservation so that their status will be taken into account in deciding resource consents.

Policy 4.1.2

If an application for a resource consent affects an area proposed for protection under a statute administered by the Department of Conservation then the publicly notified purpose of the proposal should be taken into account when deciding the application.

Policy 4.1.3

In respect of the erection of any structures in the coastal marine area, in cases where the provisions of the Building Act 1992 do not apply, provision should be made to ensure that, where appropriate, the consent holder will remove any abandoned or redundant structure that the consent holder erected or took responsibility for, or any structure that is not in active use and is not likely to be used in the future.

Policy 4.1.4

Provision should be made to ensure that material used to create and form a reclamation, or material sited on a reclamation, in land of the Crown in the coastal marine area, does not include contaminants which are likely to, or have the potential to, adversely affect the coastal marine area.

Policy 4.1.5

Regional coastal plans should make provision for use of the coastal marine area for Defence Purposes. Defence Purposes are those in accordance with the Defence Act 1990.

Policy 4.1.6

Policy statements and plans should require that on applications for coastal permits for the following in relation to lands of the Crown in the coastal marine area:

- a. reclamations;
- b. the removal of sand, shingle, shell or other natural materials for commercial purposes; and
- c. rights to occupy;

regard shall be had to any available alternatives to what the applicant seeks to do, and the applicant's reasons for making the proposed choice.

4.2 Taking Into Account the Principles of the Treaty of Waitangi (Te Tiriti o Waitangi) in Land of the Crown in the Coastal Marine Area

Policy 4.2.1

All persons exercising functions and powers under the Act in relation to land of the Crown in the coastal marine area shall recognise and facilitate the special relationship between the Crown and the tangata whenua as established by the Treaty of Waitangi (Te Tiriti o Waitangi).

Policy 4.2.2

All persons exercising functions and powers under the Act in relation to land of the Crown in the coastal marine area should follow these general guidelines:

- a. take into account the principles of the Treaty of Waitangi;
- b. make provision for consultation with tangata whenua which is early, meaningful and on-going, and which is as far as practicable in accordance with tikanga Māori;
- c. have regard to any relevant planning document recognised by the appropriate iwi authority;

- d. where appropriate, involve iwi authorities and tangata whenua in the preparation of plans and policy statements, in recognition of the relationship of Māori and their culture and traditions with their ancestral lands; and
- e. where practicable, and with the consent of the tangata whenua, incorporate in policy statements and plans and in the consideration of applications for resource consents, Māori customary knowledge about the coastal environment, in accordance with tikanga Māori.

CHAPTER 5 - The matters to be included in any or all Regional Coastal Plans in regard to the preservation of the Natural Character of the Coastal Environment, including the specific circumstances in which the Minister of Conservation will decide Resource Consents

5.1 Maintenance and Enhancement of Water Quality

Policy 5.1.1

Rules should be made as soon as possible with the object of enhancing water quality in the coastal environment (including aquifers) where that is desirable to assist in achieving the purpose of the Act, and in particular where:

- a. there is a high public interest in, or use of the water;
- b. there is a particular tangata whenua interest in the water;
- c. there is a particular value to be maintained or enhanced; or
- d. there is a direct discharge containing human sewage.

Policy 5.1.2

Those rules should provide that a discharge of human sewage direct into water, without passing through land, may occur only where:

- a. it better meets the purpose of the Act than disposal onto land;

- b. there has been consultation with the tangata whenua in accordance with tikanga Māori and due weight has been given to Sections 6, 7 and 8 of the Act; and
- c. there has been consultation with the community generally.

Policy 5.1.3

Those rules should also provide that, after reasonable mixing, no discharge (either by itself or in combination with other discharges) may give rise to any significant adverse effects on habitats, feeding grounds or ecosystems.

Policy 5.1.4

Policy statements and plans should provide:

- a. that once such rules have been made, a review of all permits to discharge a contaminant into water in the coastal environment will be undertaken; and
- b. that where the standards set by the rules are not being met, and it is desirable that those standards be met, steps will be taken pursuant to Section 128 to review the conditions of those permits.

Policy 5.1.5

Consideration should be given to reducing any contamination of natural water in the coastal environment caused by the discharge of trade wastes.

Policy 5.1.6

Consideration should be given to reducing contamination of natural water in the coastal environment from non-point sources.

Policy 5.1.7

Provision should be made to ensure that the public is adequately warned when the degradation of water in the coastal environment has rendered the water unsafe for swimming, shell-fish gathering or other activities.

5.2 Limiting of Adverse Environmental Effects From Vessel Waste Disposal or Maintenance

Policy 5.2.1

Provision should be made to require adequate and convenient rubbish disposal facilities in ports, marinas and other such busy areas, and for the provision of facilities for the collection and appropriate disposal of the residues from vessel maintenance.

Policy 5.2.2

Provision should be made to require in all new ports and marinas adequate and convenient facilities to collect sewage from boats.

Policy 5.2.3

Provision should be made to encourage those in charge of vessels to discharge sewage and rubbish into collection facilities.

Policy 5.2.4

In the preparation of regional coastal policies and plans, consideration should be given to specifying a minimum distance off-shore within which sewage should not be discharged from vessels, and to the prohibition of the discharge of non-biodegradable rubbish into the sea.

Policy 5.2.5

Provision should be made to allow vessels to take and use sea water for normal operational and fire-fighting purposes in the coastal marine area.

5.3 Defining the Specific Circumstances in Which the Minister of Conservation Will Decide on Resource Consent Applications

Policy 5.3.1

The types of activities which have or are likely to have a significant or irreversible adverse effect on the coastal marine area and for which therefore the Minister of Conservation will decide resource consent applications are those defined in Schedule 1.

CHAPTER 6 - The Implementation of New Zealand's International Obligations Affecting the Coastal Environment

Statement

Where the government has accepted international obligations which affect the coastal environment, the intention is that guidelines shall be issued from time to time by the government outlining the manner in which these obligations can best be carried out and implemented.

CHAPTER 7 - The Procedures and Methods to be Used to Review the Policies and to Monitor their Effectiveness

7.1 Monitoring the Effectiveness of the New Zealand Coastal Policy Statement

Policy 7.1.1

The effectiveness of the New Zealand Coastal Policy Statement will be reviewed by a person or persons independent of the Minister no later than 9 years after its gazettal, and the Minister shall then consider the desirability of reviewing, changing or revoking the Statement.

Policy 7.1.2

The Minister of Conservation shall monitor the effectiveness of the New Zealand Coastal Policy Statement in achieving the purpose of the Act by:

- a. assessing the effect of the statement on all subordinate regulatory planning instruments; and
- b. working with regional councils and with all other interested bodies willing to co-operate to establish a national state of the coastal environment monitoring programme.

Policy 7.1.3

In order to assist in the establishing of a national state of the coastal environment monitoring programme, local authority policy statements and plans should identify the procedures and methods which the local authority intends to use to gather information and monitor the state of their coastal environment.

SCHEDULE 1 - The Circumstances in which Activities that have a Significant or Irreversible Adverse Effect on the Coastal Marine Area will be made Restricted Coastal Activities

S1.1 Reclamations

- a. Any activity reclaiming foreshore or seabed which is less than 1 hectare or extends less than 100 metres in all directions, including incremental reclamations connected to or part of another reclamation which was commenced or received a resource consent after the 5 May 1994, and the sum of the existing and proposed reclamations do not exceed these dimensions is **not** a restricted coastal activity.
- b. Any activity reclaiming foreshore or seabed which:
 - i. is equal to or greater than 1 hectare but less than or equal to 2 hectares, or extends 100 or more metres up to or equalling 300 metres linear in any direction (or which is an incremental reclamation connected to, or part of, another reclamation which was commenced or received a resource consent after 5 May 1994, and the sum of the existing and proposed reclamations are within these dimensions);
 - ii. which is specified in the relevant operative or proposed regional coastal plan as a discretionary activity; and
 - iii. the plan contains the criteria for, or the permissible locations of, reclamations, and the permissible adverse effects and the usages of any areas so reclaimed;is **not** a restricted coastal activity.

- c. Except as in S1.1(a) or (b) above, any activity reclaiming foreshore or seabed which:
 - i. equals or exceeds 1 hectare;
 - ii. extends 100 or more metres in any direction; or
 - iii. is an incremental reclamation connected to, or part of, another reclamation which;
 - was commenced or received a resource consent after 5 May 1994, and
 - the sum of the existing and proposed reclamations are equal to or exceed the dimensions in (i) and (ii); and
- is a restricted coastal activity.

S1.2 Structures which impound or effectively contain the coastal marine area

- a. Any activity involving the erection of a structure or structures which:
 - i. will impound or effectively contain less than 4 hectares of the coastal marine area; or
 - ii. is floating or open pile structure which can be demonstrated to not impede water flows;

is **not** a restricted coastal activity.
- b. Any activity involving the erection of a structure or structures which will:
 - i. impound or effectively contain less than 8 hectares, of the coastal marine area;
 - ii. which is specified in the relevant operative or proposed regional coastal plan as a discretionary activity; and
 - iii. the plan defines or provides the criteria for determining:
 - where it would be permissible to locate any such structure or structures; and
 - the materials to be used in the construction of any structure or structures; and

- the activities for which such structure or structures can be used; and

iv. the plan:

- requires consideration of the likely adverse effects of the structure or structures; and
- defines, or provides the criteria for determining, the limits on likely adverse effects of the structure or structures;

is **not** a restricted coastal activity.

- c. Except as provided for in S1.2(a) and (b) above, any activity involving the erection of a structure or structures which will impound or effectively contain 4 hectares or more of the coastal marine area is a restricted coastal activity.

S1.3 Structures in the coastal marine area more or less parallel to mean high water springs

- a. Any activity involving the erection of a structure or structures:

- which is solid (or presents a significant barrier to water or sediment movement), and when established on the foreshore or seabed extends less than 300 metres in length more or less parallel to the line of mean high water springs (including separate structures which total less than 300 metres contiguous length);
- which is a submarine or sub-aqueous cable, or
- which is a floating or open pile structure which can be demonstrated not to have adverse effects;

is **not** a restricted coastal activity.

- b. Any activity involving the erection of a structure or structures:

- which are solid (or presents a significant barrier to water or sediment movement);
- when established on the foreshore or seabed would extend 300 metres or more, but not more than 1000 metres, in length more or less parallel to the line of mean high water springs (including separate structures which incrementally total at least 300 metres and up to 1000 metres, contiguously);

- iii. which is specified in the relevant operative or proposed regional coastal plan as a discretionary activity; and
- iv. for which the plan defines, or provides the criteria for determining:
 - where it would be permissible to locate any such structure or structures;
 - the materials to be used in the construction of any structure or structures;
 - the activities for which such structure or structures can be used; and
- v. for which the plan:
 - requires consideration of the likely adverse effects of the structure or structures; and
 - defines, or provides, the criteria for determining, the limits on likely adverse effects of the structure or structures;

is **not** a restricted coastal activity.

- c. Except as provided for in S1.3(a) and (b) above, any activity involving the erection of a structure or structures which:
 - i. are solid (or present a significant barrier to water or sediment movement); and
 - ii. when established on the foreshore or seabed would extend 300 metres or more in length more or less parallel to the line of mean high water springs (including separate structures which total 300 metres or more contiguous);

is a restricted coastal activity.

S1.4 Structures in the coastal marine area oblique or perpendicular to mean high water springs

- a. Any activity involving the erection of a structure or structures:
 - i. which is solid (or present a significant barrier to water or sediment movement), and is sited obliquely or perpendicular in horizontal projection to the line of mean high water springs in the coastal

marine area, and is in horizontal projection less than 100 metres;
or

ii. which is a submarine or sub-aqueous cable;

is **not** a restricted coastal activity.

b. Any activity involving the erection of a structure or structures:

i. which is solid (or presents a significant barrier to water or sediment movement);

ii. which is sited obliquely or perpendicular to the line of mean high water springs in the coastal marine area;

iii. which in horizontal projection is not more than 1000 metres in length;

iv. which is specified in the relevant operative or proposed regional coastal plan as a discretionary activity; and

v. for which the plan defines, or provides the criteria for determining:

- where it would be permissible to locate any such structure or structures;

- the materials to be used in their construction; and

- specifies the activities for which such structure or structures can be used; and

vi. for which the plan:

- requires consideration of the likely adverse effects of the structure or structures; and

- defines, or provides the criteria for determining, the limits on adverse effects of the structure or structures;

is **not** a restricted coastal activity.

c. Except as provided for in S1.4(a) and (b) above, any activity which includes erecting a structure or structures in the coastal marine area which is:

i. solid (or presents a significant barrier to water or sediment movement);

- ii. is sited obliquely or perpendicular in horizontal projection to the line of mean high water springs in the coastal marine area; and
 - iii. is in horizontal projection 100 metres or more in length;
- is a restricted coastal activity.

S1.5 Structures in the coastal marine area used in the petroleum and chemical industry

- a. Any activity involving the erection of structures for the storage or containment of any petroleum, petroleum products, or contaminants in quantities less than or equal to 50,000 litres is not a restricted coastal activity.
- b. Any activity involving the erection of structures for the storage or containment of any petroleum, petroleum products, or contaminants in quantities less than 100,000 litres and more than 50,000 litres and the relevant operative or proposed regional coastal plan specifies that the activity is a discretionary activity and defines, or provides criteria determining:
 - i. where it would be permissible to locate any such structures; and
 - ii. the effects of the structure;
 is **not** a restricted coastal activity.
- c. Except as provided for in S1.5(a) and (b) above, any activity involving the erection of structures for the storage or containment of any petroleum, petroleum products, or contaminants, in quantities greater than 50,000 litres is a restricted coastal activity.

S1.6 Disturbance of foreshore and seabed (excavate, drill, move, tunnel etc) including any removal of sand, shell or shingle

- a. Any activity involving, in any 12-month period, disturbance of foreshore and seabed for specific purposes, including any removal of sand, shell or shingle or other material which is either:
 - i. maintenance dredging;

- ii. in volumes less than or equal to 50,000 cubic metres; and extracted from areas less than 4 hectares; and extending less than 1000 metres over foreshore and seabed; or
- iii. in volumes less than 300,000 cubic metres; and extracted from areas of less than 10 hectares; and extending less than 10,000 metres over foreshore and seabed; and
 - is specified in an operative or proposed regional coastal plan as a discretionary activity; and
 - the plan defines or provides the criteria which specifies the location where the removal can be carried out; and
 - specifies when the activity can be undertaken; and
 - specifies conditions to control any adverse effects of any removal or disturbance;

is **not** a restricted coastal activity.

- b. Except as in S1.6(a) above any activity involving, in any 12 month period, disturbance of foreshore and seabed for specific purposes, including any removal of sand, shell or shingle:
 - i. in volumes greater than 50,000 cubic metres;
 - ii. extracted from areas equal to or greater than 4 hectares; or
 - iii. extending 1000 metres or more over foreshore and seabed;

is a restricted coastal activity.

For the purposes of this clause, **maintenance dredging** means any dredging of the bed of the sea necessary to maintain water depths to previously approved levels, for the safe and convenient navigation of vessels, in navigation channels and at berthing and mooring facilities, including marina developments.

S1.7 Depositing substances in the coastal marine area

- a. Any activity involving the depositing of any material on the foreshore and seabed which involves quantities less than or equal to 50,000 cubic metres at a site in the coastal marine area in any 12 month period is not a restricted coastal activity.

- b. Any activity involving the depositing of any material on the foreshore and seabed:
 - i. which is specified in an operative or proposed regional coastal plan as a discretionary activity;
 - ii. for which the plan defines or provides the criteria for determining:
 - the location where the activity can be carried out;
 - the time during which the activity can be carried out; and
 - iii. for which the plan:
 - requires consideration of the likely adverse effects of the depositing of the material; and
 - defines, or provides the criteria for determining, the limits on the likely adverse effects of the depositing of the material;
- is **not** a restricted coastal activity.

Except as provided for in S1.7(a) and (b) above, any activity involving the depositing of any material on the foreshore or seabed in quantities greater than 50,000 cubic metres in any 12 month period in the coastal marine area is a restricted coastal activity.

S1.8 Exotic plants in the coastal marine area

Any activity involving the introduction of any exotic plant species to the coastal marine area is a restricted coastal activity, except where that plant is already present in an area and an operative or proposed regional coastal plan specifies that the planting of it is a discretionary activity.

S1.9 Exclusive occupation of the coastal marine area

Any activity involving occupation of the coastal marine area which:

- a. would exclude or effectively exclude public access from areas of the coastal marine area over 10 hectares (except where such exclusion is required in commercial port areas for reasons of public safety or security);
- b. would exclude or effectively exclude the public from more than 316 metres along the length of the foreshore; or

- c. would involve occupation or use of areas greater than 50 hectares of the coastal marine area and such occupation or use would restrict public access to or through such areas;

is a restricted coastal activity.

S1.10 Discharges to the coastal marine area

- a. Any discharge of human sewage to the coastal marine area, except from vessels, which has not passed through soil or wetland, shall be a restricted coastal activity.
- b. Any discharge to the coastal marine area in respect of which the applicant may desire to rely on section 107(2)(a) shall be a restricted coastal activity.



appendix four

Sections 7 and 8,
Hauraki Gulf Marine
Park Act 2000

7. Recognition of national significance of Hauraki Gulf –

- (1) The interrelationship between the Hauraki Gulf, its islands, and catchments and the ability of that interrelationship to sustain the life-supporting capacity of the environment of the Hauraki Gulf and its islands are matters of national significance.
- (2) The life-supporting capacity of the environment of the Gulf and its islands includes the capacity –
 - (a) to provide for –
 - (i) the historic, traditional, cultural, and spiritual relationship of the tangata whenua of the Gulf with the Gulf and its islands; and
 - (ii) the social, economic, recreational, and cultural well-being of people and communities:
 - (b) to use the resources of the Gulf by the people and communities of the Gulf and New Zealand for economic activities and recreation:
 - (c) to maintain the soil, air, water, and ecosystems of the Gulf

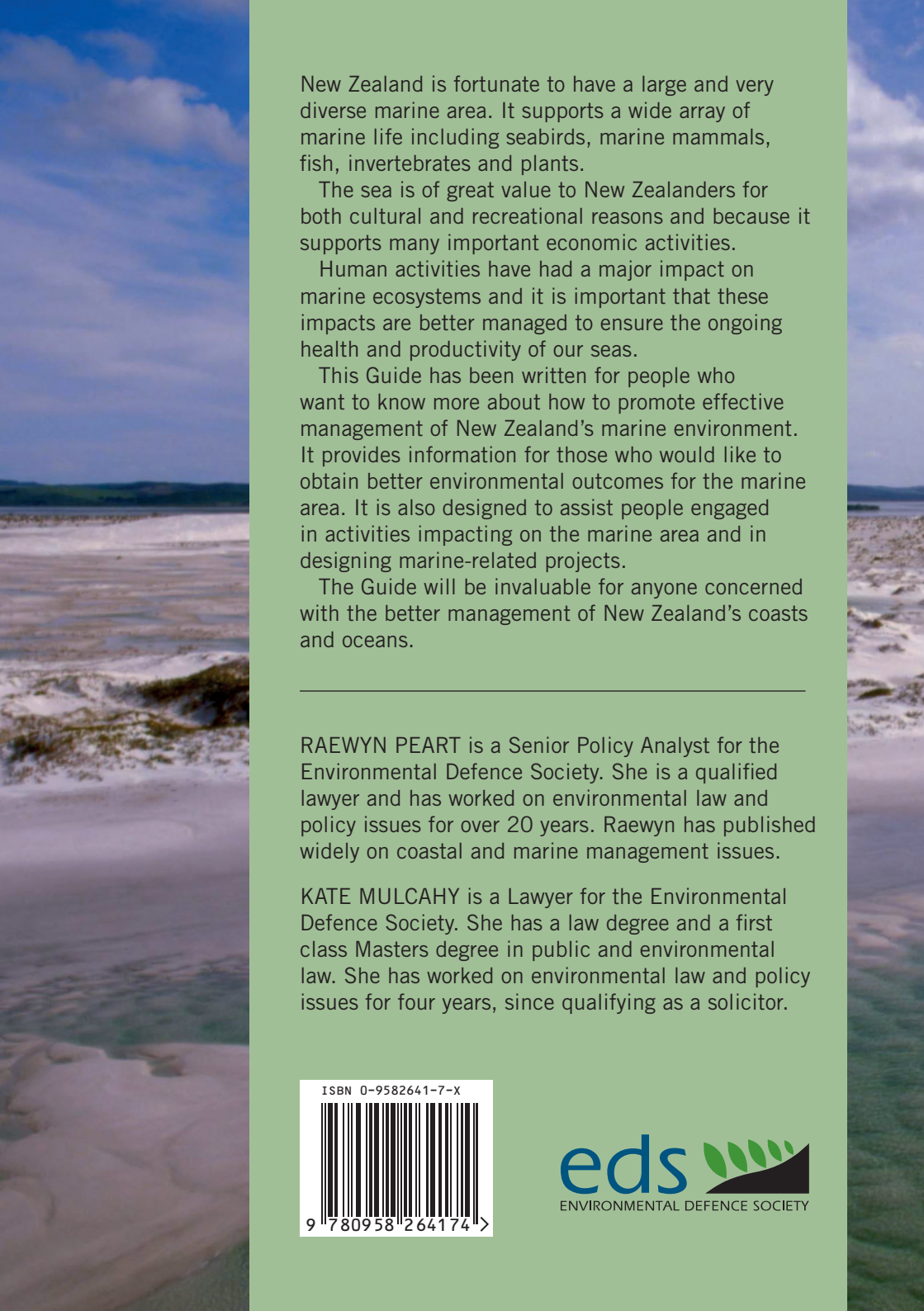
8. Management of Hauraki Gulf –

To recognise the national significance of the Hauraki Gulf, its islands, and catchments, the objectives of the management of the Hauraki Gulf, its islands, and catchments are –

- (a) the protection and, where appropriate, the enhancement of the life-supporting capacity of the environment of the Hauraki Gulf, its islands, and catchments:
- (b) the protection and, where appropriate, the enhancement of the natural, historic, and physical resources of the Hauraki Gulf, its islands, and catchments:
- (c) the protection and, where appropriate, the enhancement of those natural, historic, and physical resources (including kaimoana) of the Hauraki Gulf, its islands, and catchments with which tangata

whenua have an historic, traditional, cultural, and spiritual relationship:

- (d) the protection of the cultural and historic associations of people and communities in and around the Hauraki Gulf with its natural, historic, and physical resources:
- (e) the maintenance and, where appropriate, the enhancement of the contribution of the natural, historic, and physical resources of the Hauraki Gulf, its islands, and catchments to the social and economic well-being of the people and communities of the Hauraki Gulf and New Zealand:
- (f) the maintenance and, where appropriate, the enhancement of the natural, historic, and physical resources of the Hauraki Gulf, its islands, and catchments, which contribute to the recreation and enjoyment of the Hauraki Gulf for the people and communities of the Hauraki Gulf and New Zealand.



New Zealand is fortunate to have a large and very diverse marine area. It supports a wide array of marine life including seabirds, marine mammals, fish, invertebrates and plants.

The sea is of great value to New Zealanders for both cultural and recreational reasons and because it supports many important economic activities.

Human activities have had a major impact on marine ecosystems and it is important that these impacts are better managed to ensure the ongoing health and productivity of our seas.

This Guide has been written for people who want to know more about how to promote effective management of New Zealand's marine environment. It provides information for those who would like to obtain better environmental outcomes for the marine area. It is also designed to assist people engaged in activities impacting on the marine area and in designing marine-related projects.

The Guide will be invaluable for anyone concerned with the better management of New Zealand's coasts and oceans.

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